



APPROVED

By John Ozoga III at 9:49 am, Apr 30, 2024

March 28, 2024

Michigan Department of Environment, Great Lakes and Energy
Materials Management Division
Cadillac District Office
120 W. Chapin Street
Cadillac, Michigan 49601
Submitted via email

Attn: John Ozoga, III, District Supervisor/Engineer
ozogaj@michigan.gov

Subject: **PART 115 Landfill Gas Monitoring and Management Plan**
Wexford County Landfill
Manton, Michigan

Dear Mr. Ozoga:

Attached please find the above referenced PART 115 Landfill Gas Monitoring and Management Plan for the Wexford County Landfill. The Plan was prepared by Tetra Tech in response to the March 22, 2023, "Landfill Gas Amendments Advisory Letter" from the Michigan Department of Environment, Great, Lakes, and Energy (EGLE) to GFL Environmental USA Inc.

Should you have any questions regarding the attached Plan, please contact me at (770) 575-7610.

Sincerely,

A handwritten signature in blue ink, appearing to read 'Tami Craig'.

Tami Craig
Regional Landfill Gas Program Manager

attached: PART 115 Landfill Gas Monitoring and Management Plan

cc: C. Gee, cgee@gflenv.com
C. Oleniacz, doleniacz@eilllc.com
T.Unseld, unseldt@michigan.com

**PART 115 Landfill Gas Monitoring and Management Plan
for
Wexford County Landfill**

Manton, Michigan



March 28, 2024

Prepared By:

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1.0 Purpose

This document serves as a comprehensive PART 115 Landfill Gas Monitoring and Management Plan (“the Statute 11512 Plan”) for Wexford County Landfill (“the Facility”) in accordance with the requirements of Part 115, Subpart 2, Section 11512. The effective date of this statute was March 29, 2023.

The Statute 11512 Plan, submitted on or before March 29, 2024, fulfills this requirement.

2.0 Applicability

Section 11512f(1)(a) requires submittal of revised engineering plans and reports within 90 days after the effective date of the amendatory act that added this section for landfills that have an active gas collection and control system but were not previously subject to monthly wellhead monitoring pursuant to the new source performance standards under 40 CFR Part 60, Subpart WWW (later clarified by EGLE to be Subpart XXX) or under any other federal air program. These landfills must submit plans within one year of the effective date of the regulations; i.e. by March 29, 2024. This includes the Wexford County Landfill, which is subject to the Federal Plan (40 CFR 62 Subpart OOO) but has demonstrated via periodic Tier 2 testing that NMOC emissions are less than 34 Mg/year.

3.0 Introduction

The State of Michigan amended 1994 PA 451 (Solid Waste) via Enrolled House Bill 4456. The amendments included addition of the following new sections to Part 115, Subpart 2:

- 11512b “Active gas collection and control system; prevention of the migration of explosive gases; system requirements; compliance with gas migration monitoring plan; alternative gas venting”;
- 11512d “Installation of monitoring port; sampling of gas extractions for effectiveness; surface monitoring design plan; submission and retention of records”;
- 11512f “Type II landfill; revised engineering plans; compliance requirements; report”; and
- 11512h “Schedule for operating and monitoring an active gas collection and control system; surface emission scans”.

These newly added sections to the solid waste statute prescribed several mandates for landfill gas management and monitoring, including the preparation of the following plans:

- Engineering Design Plans (revised or new)
- Gas Migration Monitoring Plan (Note – EGLE required submittal of this plan for all landfills on March 29, 2023 per an Advisory Letter sent to facilities on March 22, 2023)
- Gas Collection and Control System Sampling and Monitoring Plan
- Surface Monitoring Design Plan

The document presented herein incorporates all required plans, with the exception of the already-submitted Landfill Gas Migration Monitoring Plan, into one comprehensive submittal. The Engineering Design Plan is provided in Section 4. Variances from the prescriptive 11512b design requirements for future unconstructed portions of the GCCS will be clearly highlighted for Agency approval.

Section 5 includes the prescriptive 11512d monitoring and sampling requirements for the landfill gas collection and control system components.

Section 6 provides the required Surface Monitoring Design Plan as described in 11512d.

Appendices A through D include engineering documentation and drawings to support the plans.

4.0 Gas Management System Engineering Design Plan – Section 11512f(1)

For elements of the Engineering GCCS Design Plan which do not conform with the prescriptive Section 11512b(2) requirements, variances and demonstrations are presented as allowed by the Statute.

Please note that the GCCS Design Plan submitted herein has been developed for the site at final grade conditions. Note that this Monitoring and Management Plan is only required under Part 115 and not by Federal air regulation 40 CFR Part 62, Subpart OOO.

The Wexford County Landfill is currently at interim grades. The evolution of the collection and control system as the landfill is filled will ultimately produce the design specified in this Engineering Design Plan. Until the landfill has attained its permitted final grades, the collection and control of landfill gas pursuant to the Michigan Part 115 regulations may be accomplished using methods not specifically included as part of the final design (i.e. horizontal gas collection trenches, collection from leachate sumps and risers, etc.). However, once the facility has reached the final permitted grades, the collection and control system will meet state criteria.

Additionally, Part 115 now requires annual submittal of construction drawings to EGLE prior to onset of planned GCCS expansions and as-builts of GCCS construction after it is completed. These drawings will detail the methods the site will use to control gas at interim waste grades, and represent an opportunity for EGLE to maintain records of GCCS construction at the facility as it evolves over time. These records can be considered periodic supplements to this Part 115 engineering design plan. The facility will begin submittal of the construction documents after this PART 115 Landfill Gas Monitoring and Management Plan has been approved by EGLE.

4.1 Vertical Gas Extraction Well Design - Section 11512b(2)(a)

If a landfill gas collection system includes vertical gas extraction wells, Section 11512b(2)(a) lists the following requirements for these types of wells for any unconstructed portion of the GCCS that is not already in place at the site as of March 29, 2023 (unless as otherwise indicated in Section 11512f(1)(a)):

- (i) Wells will have a maximum radius of influence (ROI) of 150 feet per well and lesser radii for wells located near the perimeter of the landfill. The radii of influence of adjacent wells shall overlap. Alternate well spacings may be used for portions of a site or the entire site if approved by the department after a site-specific demonstration.*
- (ii) Wells have target depths of at least 75% of the waste depth at the well location. However, the wells should not extend closer than 10 feet above the leachate collection system.*
- (iii) Wells are constructed of pipe that meets all of the following requirements:*
 - (A) Is at least 6 inches in diameter.*
 - (B) Is manufactured from polyvinylchloride, high-density polyethylene, chlorinated polyvinyl chloride, or an alternate material approved by the department.*
 - (C) Is designed to convey projected amounts of gas; withstand installation, static, and settlement forces; and withstand planned overburden and traffic loads.*
 - (D) When constructed, is slotted or otherwise perforated and is screened in the lower 2/3 to 3/4 of its length in the borehole. The department may approve alternative perforated screened length requirements based on waste thickness or other factors.*
- (iv) Has boreholes that meet all of the following requirements:*

*PART 115 Monitoring and Management Plan
Wexford County Landfill*

- (A) Are 36 inches in diameter. The department may approve alternate diameter boreholes as part of a design prepared by a licensed professional engineer and approved by the department.*
- (B) Are backfilled around the perforated pipe with 3/4- to 3- inch washed stone or an alternate material if approved by the department after a site-specific demonstration.*
- (C) The top 10 feet are sealed in a manner approved by the department.*

Well Spacing

For the remaining unconstructed areas of the gas collection system as of March 29, 2023, the facility's conceptual GCCS design utilized a maximum ROI of 150 feet for the interior wells and 125 feet for the perimeter wells, which meets the requirements of Section 11512b(2)(a). Therefore, no department approval of a variance is required. Remaining components of the gas collection system will be installed in accordance with 11512b(2)(a) and the conceptual GCCS design plan provided in Appendix C. A table showing the calculated ROI's for the remaining unconstructed wells is provided in Appendix A. The well placement of existing wells as of March 29, 2023, is shown on the drawing in Appendix B. Well placement for remaining unconstructed wells for the landfill at final permitted grades is shown on the drawing in Appendix C.

Well Depth

Vertical wells are now required to have target depths of at least 75% of the waste depth at the well location. The wells must not extend closer than 10 feet above the leachate collection system. The facility will comply with these requirements for future unconstructed wells, unless:

- the wells are sited in shallow areas of waste and cannot meet both of the requirements above simultaneously.
- the wells are sited in very deep areas of waste where due to limitations or the drilling equipment the well depth equal to or greater than minimum 75% waste depth target cannot be met.
- the borehole collapses during drilling and varies more than 25% of the proposed depth.
- Drilling equipment is not able to penetrate the waste mass due to drilling refusal.

If any of the above cases occur the requirements of Sec. 11512f(4) would apply: The owner or operator shall submit to the department 1 of the following:

- (a) A statement from a licensed professional engineer that the gas wells installed will provide adequate control of landfill gas emissions and meet the intent of the design.
- (b) A schedule for installing additional gas collectors to meet the design requirements included with the approved engineering plans.

Well Construction

Conceptual details included in the GCCS Design Plan for vertical gas extraction wells demonstrate compliance with the following Section 11512b(2)(a) requirements:

- Minimum 6 inch diameter well casing;

- Well casing material type (polyvinylchloride, high-density polyethylene, chlorinated polyvinyl chloride);
- 36 inch diameter borehole;
- 3/4- to 3- inch washed stone; and
- Surface seal in top 10 feet of well.

Therefore, the conceptual details for vertical wells in the GCCS Design Plan are incorporated by reference into this Gas Management System Engineering Design Plan.

For the following Section 11512b(2)(a) requirements:

- Slotting/perforations/screens in the lower 2/3 to 3/4 of its length in the borehole

The facility will comply with these requirements for future unconstructed wells, unless:

- the wells are sited in shallow areas of waste and cannot meet the requirement without providing adequate surface seal.

In case the above occurs the requirements of Sec. 11512f(4) would apply: The owner or operator shall submit to the department 1 of the following:

- (a) A statement from a licensed professional engineer that the gas wells installed will provide adequate control of landfill gas emissions and meet the intent of the design.
- (b) A schedule for installing additional gas collectors to meet the design requirements included with the approved engineering plans.

4.2 Horizontal Collector Design - Section 11512b(2)(b)

Section 11512b(2)(b) states the following with respect to horizontal collectors:

Horizontal gas extraction wells that are properly sloped to drain accumulated liquids and designed to withstand expected overburden pressures.

The facility plans to install additional horizontal collectors in unconstructed portions of the GCCS after March 29, 2023, for interim and/or permanent gas collection, as needed. The collectors will be designed to meet the requirements of Section 11512b(2)(b); specifically, they will be properly sloped to drain accumulated liquids and will be designed to withstand expected overburden pressures.

Interim horizontal collectors will not require piping that is as robust as a permanent horizontal collector since these locations are “sacrificial” and will ultimately be supplemented by a permanent vertical gas collection well once an area reaches final grades.

4.3 Flow Control Valve/Sampling Port - Section 11512b(2)(c)

Section 11512b(2)(c) requires the following:

A flow control valve and sampling access port on each gas extraction well.

Each future vertical or horizontal gas collector at the landfill will be equipped with a flow control valve and a sampling access port. All the existing extraction wells at the site contain this equipment as well.

4.4 Gas Header System - Section 11512b(2)(d)

Section 11512b(2)(d) lists the following requirements for any unconstructed portion of the header system that is not already in place at the site as of March 29, 2023 (unless as otherwise indicated in Section 11512f(1)(a)):

- (i) The entire gas header system is designed with a loop to allow alternative flow paths for the gas as soon as practicable during both the interim and final development phases of construction.*
- (ii) The slope on the header pipe over the waste mass is at least 2% wherever possible. The slope outside of the waste mass shall allow efficient removal of condensate and prevents sags.*
- (iii) The header and lateral pipes meet both of the following requirements:*
 - (A) Are manufactured from polyethylene or another material approved by the department.*
 - (B) Are designed to convey projected amounts of gas and liquids; withstand installation, static, and settlement forces; and withstand planned overburden and traffic loads.*

The conceptual GCCS design includes a looped header system that allows alternative flow paths when the site is at closure conditions. The site is currently at interim grades. As future header components are installed after March 29, 2023, the facility will, when field conditions allow, incorporate header piping and valves when practicable to allow for collection of gas to continue along alternate flow paths, should the main header system be shut down for construction or repair.

The minimum header slope in waste for future unconstructed portions of the header system will be 2%, which meets the minimum 2% requirement of 11512b(2)(d). The minimum header slopes outside the waste mass will be minimum 0.5% to allow efficient removal of condensate and prevent sags.

Header and lateral piping is and will be constructed of high density polyethylene (HDPE), as required. HDPE piping in landfill gas collection systems has historically been demonstrated to have the ability to withstand typical installation, static and settlement forces, planned overburden and traffic loads. Should a specific section of header line fail, the site will initiate repairs to restore vacuum as soon as practicable. Header and lateral piping alignment for the remaining unconstructed piping for the landfill at final permitted grades is shown on the drawing in Appendix C.

4.5 10-Inch Vacuum Requirement - Section 11512b(2)(e)

Section 11512b(2)(e) requires the following:

A blower, header, and laterals designed so that a vacuum of at least 10 inches of water column is available at the well located furthest from the blower. An available header vacuum of less than 10 inches of water column at the well located furthest from the blower complies with this subdivision if the owner or operator of the landfill demonstrates to the department that the available vacuum is adequate to meet performance criteria.

A KYPipe model was prepared for this Engineering Plan to demonstrate that the proposed GCCS design will be able to provide minimum vacuum of at least 10 inches of water column at the well located furthest from the blower, when the site is at final development. Output from the KYPipe™ model is provided in Appendix A.

4.6 Blower Dripleg Requirement - Section 11512b(2)(f)

Condensate must be managed before the blower (or other gas mover equipment) as follows:

A drip leg or equivalent installed immediately before the blower to separate condensate from gas while preserving the suction at the wells when under maximum operating vacuum.

The landfill has installed a condensate knockout immediately prior to the blower station.

4.7 Secondary Containment for Liquid Piping Outside Waste Limits – Section 11512b(2)(g)

Section 11512b(2)(g) requires the following:

An approved secondary containment method for condensate and liquid transfer piping if the piping is located outside of the limits of the waste and installed after the effective date of the amendatory act that added this section.

All piping for condensate and liquid transfer that are located outside of the limits of waste, installed after March 29, 2023 will be provided with secondary containment in accordance with a department-approved method.

4.8 Condensate and Liquid Management/Gas Sampling – Section 11512b(2)(h)

Section 11512b(2)(h) of the statute states that active systems must have:

The ability to collect and manage all condensate, measure volumes of liquid removed from the gas extraction wells, and collect samples of landfill gas.

Condensate from the gas collection system is co-mingled with the landfill leachate for disposal. The facility disposes of landfill liquids at an offsite deep injection well.

Existing and future liquid removal pumps are and will continue to be equipped with equipment that is capable of estimating volumes of liquids removed from gas wells. Note that current pump technology such as pump cycle counters or run hour meters can only approximate liquid volumes removed from gas wells. Section 11512b(2)(h) does not state that measured volumes must be recorded; it only imposes a measurement requirement.

The GCCS is designed so that samples of landfill gas can either be collected from individual wellheads, or, from a central location at the blower/flare station.

4.9 Control Device – Section 11512b(2)(i)

Pursuant to Section 11512b(2)(i), collected gas must be sent to:

A control device to which collected landfill gas is routed that meets all of the following requirements:

- (i) Operates at all times gas is routed to it.*
- (ii) Is designed and operated to meet the requirements of part 55 or the new source performance standards under 40 CFR part 60.*
- (iii) Operates backup blower or control equipment required under subdivision (j).*

The collected gas is sent to an open flare. This device meets the requirements of the air quality regulations of Part 55.

See Section 4.10 for a discussion of backup equipment at the landfill.

4.10 Backup Equipment – Section 11512b(2)(j)

Section 11512b(2)(j) of the statute states that active systems must have:

Available backup equipment to effectively control landfill gas emissions during an equipment breakdown.

In order to minimize equipment downtime during a breakdown of the flare or blower, the facility has implemented the following:

- Inventory of critical spare parts for blower repair,
- Access to a back-up company or supplier-owned skid-mounted flare, which can immediately be transported to the site,
- An emergency generator to supply power during utility downtimes OR Rental agreements for emergency generators during power outages,

4.11 Five Day Downtimes – Section 11512b(2)(k)

Pursuant to Section 11512b(2)(k):

The active gas collection and control system shall not be inoperable or unable to maintain a vacuum required by subdivision (e) for more than 5 consecutive days.

The facility will make every effort to minimize downtimes of the GCCS. However, in the event of extended power outages from ice storms, terrorism, Acts of God, etc., it is not practicable to guarantee that electrical power for devices applying system vacuum will always be restored within five days.

5.0 Gas Collection and Control System Sampling and Monitoring Plan – Section 11512d

The revised solid waste statute imposes monitoring requirements for landfills that operate active landfill gas collection and control systems. Some of the requirements are already being conducted at the facility in accordance with applicable air regulations. This includes monthly monitoring of gas extraction wells for oxygen or nitrogen, pressure and temperature, and monitoring gas flow to the flare. The site will continue to follow these federal air requirements. Section 11512(d) also requires additional monitoring as described in the following sections. Surface emissions monitoring required by Section 11512(d)(5) is discussed separately in Section 6 of this Statute 11512 Plan.

Pursuant to Section 11512h(1)(a), monitoring for landfills described in section 11512f(1)(a) shall commence within 90 days after EGLE’s approval of this Statute 11512 Plan. Note that Part 115 monitoring will only apply to areas of the site that are/were subject to Part 115 requirements, and not to portions of the landfill that were subject to and final closed under older solid waste rules, such as Act 641 or Act 87.

5.1 Gas Extraction Well and Control Device Sampling – Section 11512d(2) & (3)

Pursuant to Section 11512d(2) & (3), the facility will conduct the following monitoring on a monthly basis:

Gas Extraction Wells:

- Gas quality parameters of nitrogen or oxygen, and methane;
- Temperature
- Pressure
- Gas Flow (if existing wellheads allow)

Control Device(s):

- Gas Flow
- Methane Content

Liquid level measurements at each gas extraction well are also required, but the frequency will vary depending on the length of well screen not covered by liquids. This is described in more detail in the next section.

5.2 Liquid Level Sampling and Corrective Action Triggers – Section 11512d(3)

Section 11512d(3) contains a variable schedule for monitoring liquid levels in wells based on how much of the screened length of the well is covered by liquids. In addition to the description of monitoring frequencies below, a flow chart is also provided for better illustration. For purposes of this section, the screen interval is defined as the as the interval between the measured depth to current bottom and the top of the screen:

- Liquid levels in each gas extraction well shall be monitored semiannually except as otherwise provided.
- If for two consecutive monitoring events the liquid level in a well exceeds 50% but does not exceed 75% of the screened interval length, the owner or operator shall submit to the department for review a liquids removal evaluation and corrective action report for the well, unless the well has a functional, operated liquid pump.
- If the liquid level in a well exceeds 75% of the screened interval length during a monitoring event, then the frequency of liquid level measurements for that well shall be increased to quarterly.

PART 115 Monitoring and Management Plan
Wexford County Landfill

- If the liquid level in a well exceeds 75% of the screened interval length for two consecutive quarterly monitoring events, then the owner or operator shall install a liquids pump, unless the department approves an alternative corrective action plan.
- If the liquid level in a well did not exceed 50% for the immediately preceding two consecutive monitoring events, the owner or operator may petition the department for a decreased monitoring frequency. However, decreased monitoring shall be conducted at least annually.

The statute does not clearly state what monitoring frequency an extraction well defaults to once a liquids removal pump is installed. Therefore, the facility is requesting approval of the following liquid level monitoring schedule, irrespective of screened interval length liquids occlusion:

Once a functional pump is installed and operating within a gas extraction well, liquid level monitoring frequency shall revert to semiannual until the well is eligible for a decreased monitoring frequency petition.

6.0 Surface Emissions Monitoring Design Plan – Section 11512d(6)

Section 11512d(4) states that the owner or operator of a landfill with an active landfill gas collection and control system shall operate the system so that the methane concentration is 500 parts per million or less above background at the surface of the landfill. This Surface Emissions Monitoring Design Plan presented herein establishes protocol for this monitoring consistent with Section 11512d requirements. Note that the facility is not currently required to conduct surface emissions monitoring (SEM) since it has demonstrated that NMOC emissions rates are below thresholds requiring controls and subsequent monitoring. The procedures for conducting surface emissions testing will be in accordance with EPA Method 21.

6.1 SEM Frequency

Although not explicitly stated in Section 11512d(5), the frequency of SEM monitoring is presumed to be quarterly for active sites and annually for portions of sites that have been final closed under the Part 115 rules. The definition of “final closed” is that the area has had certified final cover placed. Any portions of the site that were final closed under prior solid waste regulations (such as Act 641 or Act 87) are NOT subject to Part 115 SEM monitoring at all, and these areas (if any) will be shown as exempt on the surface scan map.

For the areas of the site that have been final closed under Part 115, once three quarters of exceedance-free SEM monitoring have been demonstrated, the facility will revert to an annual schedule in these areas. Should an exceedance be detected in a closed area during an annual SEM event, SEM monitoring will revert back to quarterly until three subsequent quarters of SEM monitoring without an exceedance are demonstrated. At that point, the closed area can revert back to an annual SEM monitoring frequency. SEM monitoring conducted under Part 115 will be discontinued altogether once active waste receipt ceases and the facility is permanently closed.

6.2 Initiation of SEM in New Waste Areas – Section 11512d(5)

Not later than 180 days after initial waste receipt in a portion of a landfill, surface emissions monitoring for methane will be commenced at all of the following locations:

- (a) *Where visual observations, such as of distressed vegetation or cracks or seeps in the cover, indicate elevated concentrations of landfill gas.*
- (b) *At each penetration of daily, interim, or final landfill cover.*
- (c) *Around the perimeter of the active gas collection and control system.*
- (d) *Along a pattern that traverses the landfill at no more than 30-meter intervals, unless the owner or operator establishes an alternative traversing pattern that is approved by the department after a site-specific demonstration. NOTE: The facility is not proposing to establish an alternative traversing pattern.*

The definition of cover penetration that will be used by the site for SEM monitoring is the definition provided in the Landfill NESHAP (40 CFR 63 Subpart AAAA - §63.1990):

Cover penetration means a wellhead, a part of a landfill gas collection or operations system, and/or any other object that completely passes through the landfill cover. The landfill cover includes that portion which covers the waste, as well as the portion which borders the waste

extended to the point where it is sealed with the landfill liner or the surrounding land mass. Examples of what is not a penetration for purposes of this subpart include but are not limited to: Survey stakes, fencing including litter fences, flags, signs, utility posts, and trees so long as these items do not pass through the landfill cover.

6.3 Monitoring Route and Exclusions – Section 11512d(6)

The proposed surface monitoring topographical route map for the facility is provided in Appendix D. The route shown is for the facility at closure/final grade conditions. The path followed while the landfill is actively filling will vary depending on the footprint of in-place waste at the time of the survey. However, the route will meet the requirements of Section 11512d(5) with respect to route pattern, spacing, cover penetrations, in-place waste perimeter and visual observation results of potential elevated landfill gas concentrations. Areas excluded from each survey due to dangerous conditions will be documented on the route map for each quarterly survey. Dangerous conditions will be determined by the monitoring technician at the time of each SEM event. Dangerous conditions are defined as those conditions that may present a substantial risk of harm or potential loss of life to the individual conducting the monitoring, and can include steep interim waste slopes, icy or slippery conditions, truck traffic from construction or cover placement, the active working face, etc.

Appendix A

Supporting Documentation for Gas Collection and Control System (GCCS) Design

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
DESCRIPTION: Landfill Gas Modeling Projections
Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 4241336
SHEET: 1
OF: 3
DATE: 3/29/2024

Given:

Landfill gas generation projections have been made utilizing the USEPA's Landfill Gas Emission Model (LandGEM) V3.03. LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of land filled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 k L_o \left[\frac{M_i}{10} \right] (e^{-k t_{ij}})$$

Where:

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year (*decimal years*, e.g., 3.2 years)

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate.

Site Conditions:

Waste conditions at the Wexford County Landfill have been developed based upon recorded documentation and reports from site operations personnel. Degradable waste mass (MSW and other degradable waste streams) has been utilized in projecting LFG generation rates. Fill placement activities at the Wexford County Landfill began in 1997. Given the projected site life, the landfill will close in 2073.

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
 DESCRIPTION: Landfill Gas Modeling Projections

Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 4241336
 SHEET: 2
 OF: 3
 DATE: 3/29/2024

Year	East Hill Total Waste Intake (ton/year)	West Hill Total Waste Intake (ton/year)	Total Waste In Place (ton/year)	East Hill LFG Generation Rate (scfm)	West Hill LFG Generation Rate (scfm)	Total LFG Generation Rate (scfm)
1977	83,828		83,828	0		0
1978	64,400		148,228	40		40
1979	138,340		286,568	70		70
1980	114,859		401,427	133		133
1981	105,238		506,665	183		183
1982	96,642		603,307	226		226
1983	47,834		651,141	264		264
1984	63,988		715,129	277		277
1985	59,485		774,614	296		296
1986	59,596		834,210	313		313
1987	39,908		874,118	330		330
1988	47,512		921,630	336		336
1989	59,566		981,196	346		346
1990	55,117		1,036,313	361		361
1991	85,811		1,122,124	373		373
1992	78,653		1,200,777	399		399
1993	98,544		1,299,321	422		422
1994	122,263		1,421,584	452		452
1995	97,664		1,519,248	493		493
1996	95,009		1,614,257	521		521
1997	89,444		1,703,701	546		546
1998	100,193		1,803,894	568		568
1999	86,259		1,890,153	593		593
2000	146,993		2,037,146	611		611
2001	128,262		2,165,408	658		658
2002	129,132		2,294,540	694		694
2003	129,895		2,424,435	729		729
2004	120,611		2,545,046	762		762
2005	106,394		2,651,440	790		790
2006	36,141		2,687,581	810		810
2007	70,828		2,758,409	796		796
2008	57,179		2,815,588	799		799
2009	47,158		2,862,746	795		795
2010	65,490		2,928,236	786		786
2011	55,595		2,983,831	787		787
2012	119,892		3,103,723	783		783
2013	134,224		3,237,947	810		810
2014	153,860		3,391,807	842		842
2015	162,147		3,553,954	883		883
2016	177,011		3,730,965	926		926
2017	155,881		3,886,846	975		975
2018	187,078		4,073,924	1,012		1,012
2019	199,092		4,273,016	1,062		1,062
2020	226,039		4,499,055	1,116		1,116
2021	202,941		4,701,996	1,180		1,180
2022	201,109		4,903,105	1,231		1,231

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
 DESCRIPTION: Landfill Gas Modeling Projections

Prepared By: JPS Checked By: JPS Approved By: KM

PROJECT NO: 4241336
 SHEET: 3
 OF: 3
 DATE: 3/29/2024

Year	East Hill Total Waste Intake (ton/year)	West Hill Total Waste Intake (ton/year)	Total Waste In Place (ton/year)	East Hill LFG Generation Rate (scfm)	West Hill LFG Generation Rate (scfm)	Total LFG Generation Rate (scfm)
2023	338,298		5,241,403	1,280		1,280
2024	328,424		5,569,827	1,392		1,392
2025	328,424		5,898,252	1,495		1,495
2026	328,424		6,226,676	1,594		1,594
2027	328,424		6,555,100	1,689		1,689
2028	328,424		6,883,524	1,781		1,781
2029	328,424		7,211,948	1,868		1,868
2030	328,424		7,540,372	1,953		1,953
2031	328,424		7,868,797	2,034		2,034
2032	328,424		8,197,221	2,112		2,112
2033	328,424		8,525,645	2,187		2,187
2034		311,905	8,837,550	2,258	0	2,258
2035		311,905	9,149,455	2,170	150	2,320
2036		311,905	9,461,359	2,085	294	2,378
2037		311,905	9,773,264	2,003	432	2,435
2038		311,905	10,085,169	1,924	565	2,489
2039		311,905	10,397,074	1,849	692	2,541
2040		311,905	10,708,979	1,777	815	2,591
2041		311,905	11,020,884	1,707	932	2,639
2042		311,905	11,332,788	1,640	1,046	2,685
2043		311,905	11,644,693	1,576	1,154	2,730
2044		311,905	11,956,598	1,514	1,259	2,773
2045		311,905	12,268,503	1,454	1,359	2,814
2046		311,905	12,580,408	1,397	1,455	2,853
2047		311,905	12,892,312	1,343	1,548	2,891
2048		311,905	13,204,217	1,290	1,637	2,927
2049		311,905	13,516,122	1,239	1,723	2,962
2050		311,905	13,828,027	1,191	1,805	2,996
2051		311,905	14,139,932	1,144	1,884	3,028
2052		311,905	14,451,837	1,099	1,960	3,059
2053		311,905	14,763,741	1,056	2,032	3,089
2054		311,905	15,075,646	1,015	2,102	3,117
2055		311,905	15,387,551	975	2,170	3,145
2056		311,905	15,699,456	937	2,234	3,171
2057		311,905	16,011,361	900	2,296	3,196
2058		311,905	16,323,266	865	2,356	3,221
2059		311,905	16,635,170	831	2,413	3,244
2060		311,905	16,947,075	798	2,468	3,267
2061		311,905	17,258,980	767	2,521	3,288
2062		311,905	17,570,885	737	2,572	3,309
2063		311,905	17,882,790	708	2,621	3,329
2064		311,905	18,194,694	680	2,668	3,348
2065		311,905	18,506,599	654	2,713	3,367
2066		311,905	18,818,504	628	2,756	3,384
2067		311,905	19,130,409	603	2,798	3,401
2068		311,905	19,442,314	580	2,838	3,418

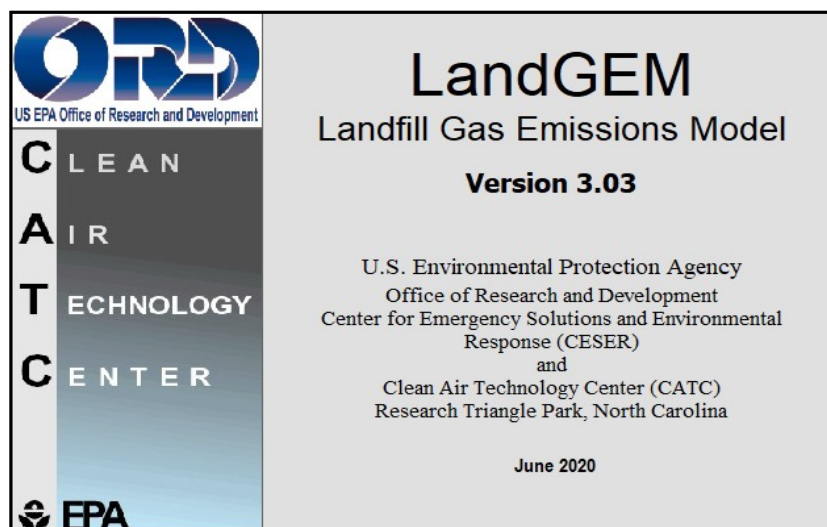
COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill - Gas Model Update
DESCRIPTION: Landfill Gas Modeling Projections

PROJECT NO: 4241336
SHEET: 3
OF: 3
DATE: 3/29/2024

Prepared By: JPS Checked By: JPS Approved By: KM

Year	East Hill Total Waste Intake (ton/year)	West Hill Total Waste Intake (ton/year)	Total Waste In Place (ton/year)	East Hill LFG Generation Rate (scfm)	West Hill LFG Generation Rate (scfm)	Total LFG Generation Rate (scfm)
2069		311,905	19,754,219	557	2,876	3,433
2070		311,905	20,066,124	535	2,913	3,448
2071		311,905	20,378,029	514	2,949	3,463
2072		311,905	20,689,933	494	2,983	3,477
2073		311,905	21,001,838	475	3,016	3,490
2074			21,001,838	456	3,047	3,503
2075			21,001,838	438	2,928	3,366
2076			21,001,838	421	2,813	3,234
2077			21,001,838	404	2,703	3,107
2078			21,001,838	389	2,597	2,985
2079			21,001,838	373	2,495	2,868
2080			21,001,838	359	2,397	2,756
2081			21,001,838	345	2,303	2,648
2082			21,001,838	331	2,213	2,544
2083			21,001,838	318	2,126	2,444
2084			21,001,838	306	2,043	2,348
2085			21,001,838	294	1,962	2,256
2086			21,001,838	282	1,886	2,168
2087			21,001,838	271	1,812	2,083
2088			21,001,838	260	1,741	2,001
2089			21,001,838	250	1,672	1,923
2090			21,001,838	240	1,607	1,847
2091			21,001,838	231	1,544	1,775
2092			21,001,838	222	1,483	1,705
2093			21,001,838	213	1,425	1,638
2094			21,001,838	205	1,369	1,574
2095			21,001,838	197	1,315	1,512
2096			21,001,838	189	1,264	1,453
2097			21,001,838	182	1,214	1,396
2098			21,001,838	175	1,167	1,341
2099			21,001,838	168	1,121	1,289
2100			21,001,838	161	1,077	1,238
2101			21,001,838	155	1,035	1,190
2102			21,001,838	149	994	1,143
2103			21,001,838	143	955	1,098
2104			21,001,838	137	918	1,055
2105			21,001,838	132	882	1,014
2106			21,001,838	127	847	974
2107			21,001,838	122	814	936
2108			21,001,838	117	782	899
2109			21,001,838	112	751	864
2110			21,001,838	108	722	830
2111			21,001,838	104	694	797
2112			21,001,838	100	666	766
2113			21,001,838	96	640	736
2114			21,001,838	92	615	707



Summary Report

Landfill Name or Identifier: Wexford East Landfill

Date: Wednesday, March 27, 2024

Description/Comments:

About LandGEM:

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 k L_o \left(\frac{M_i}{10} \right) e^{-k t_{ij}}$$

Where,

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year (*decimal years*, e.g., 3.2 years)

LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of landfilled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate. Defaults for estimating emissions for this type of operation are being developed to include in LandGEM along with defaults for conventional landfills (no leachate or liquid additions) for developing emission inventories and determining CAA applicability. Refer to the Web site identified above for future updates.

Input Review

LANDFILL CHARACTERISTICS

Landfill Open Year	1977	
Landfill Closure Year (with 80-year limit)	2033	
Actual Closure Year (without limit)	2033	
Have Model Calculate Closure Year?	No	
Waste Design Capacity	8,525,645	<i>short tons</i>

MODEL PARAMETERS

Methane Generation Rate, k	0.040	<i>year⁻¹</i>
Potential Methane Generation Capacity, L ₀	100	<i>m³/Mg</i>
NMOC Concentration	354	<i>ppmv as hexane</i>
Methane Content	50	<i>% by volume</i>

GASES / POLLUTANTS SELECTED

Gas / Pollutant #1:	Total landfill gas
Gas / Pollutant #2:	Methane
Gas / Pollutant #3:	Carbon dioxide
Gas / Pollutant #4:	NMOC

WASTE ACCEPTANCE RATES

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
1977	76,207	83,828	0	0
1978	58,545	64,400	76,207	83,828
1979	125,764	138,340	134,753	148,228
1980	104,417	114,859	260,516	286,568
1981	95,671	105,238	364,934	401,427
1982	87,856	96,642	460,605	506,665
1983	43,485	47,834	548,461	603,307
1984	58,171	63,988	591,946	651,141
1985	54,077	59,485	650,117	715,129
1986	54,178	59,596	704,195	774,614
1987	36,280	39,908	758,373	834,210
1988	43,193	47,512	794,653	874,118
1989	54,151	59,566	837,845	921,630
1990	50,106	55,117	891,996	981,196
1991	78,010	85,811	942,103	1,036,313
1992	71,503	78,653	1,020,113	1,122,124
1993	89,585	98,544	1,091,615	1,200,777
1994	111,148	122,263	1,181,201	1,299,321
1995	88,785	97,664	1,292,349	1,421,584
1996	86,372	95,009	1,381,135	1,519,248
1997	81,313	89,444	1,467,506	1,614,257
1998	91,085	100,193	1,548,819	1,703,701
1999	78,417	86,259	1,639,904	1,803,894
2000	133,630	146,993	1,718,321	1,890,153
2001	116,602	128,262	1,851,951	2,037,146
2002	117,393	129,132	1,968,553	2,165,408
2003	118,086	129,895	2,085,945	2,294,540
2004	109,646	120,611	2,204,032	2,424,435
2005	96,722	106,394	2,313,678	2,545,046
2006	32,855	36,141	2,410,400	2,651,440
2007	64,389	70,828	2,443,255	2,687,581
2008	51,981	57,179	2,507,645	2,758,409
2009	42,871	47,158	2,559,625	2,815,588
2010	59,536	65,490	2,602,496	2,862,746
2011	50,541	55,595	2,662,033	2,928,236
2012	108,993	119,892	2,712,574	2,983,831
2013	122,022	134,224	2,821,566	3,103,723
2014	139,873	153,860	2,943,588	3,237,947
2015	147,406	162,147	3,083,461	3,391,807
2016	160,919	177,011	3,230,867	3,553,954

WASTE ACCEPTANCE RATES (Continued)

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2017	141,710	155,881	3,391,786	3,730,965
2018	170,071	187,078	3,533,496	3,886,846
2019	180,993	199,092	3,703,567	4,073,924
2020	205,490	226,039	3,884,560	4,273,016
2021	184,492	202,941	4,090,050	4,499,055
2022	182,827	201,109	4,274,542	4,701,996
2023	307,544	338,298	4,457,368	4,903,105
2024	298,567	328,424	4,764,912	5,241,403
2025	298,567	328,424	5,063,479	5,569,827
2026	298,567	328,424	5,362,047	5,898,252
2027	298,567	328,424	5,660,614	6,226,676
2028	298,567	328,424	5,959,182	6,555,100
2029	298,567	328,424	6,257,749	6,883,524
2030	298,567	328,424	6,556,317	7,211,948
2031	298,567	328,424	6,854,884	7,540,372
2032	298,567	328,424	7,153,451	7,868,797
2033	298,567	328,424	7,452,019	8,197,221
2034	0	0	7,750,586	8,525,645
2035	0	0	7,750,586	8,525,645
2036	0	0	7,750,586	8,525,645
2037	0	0	7,750,586	8,525,645
2038	0	0	7,750,586	8,525,645
2039	0	0	7,750,586	8,525,645
2040	0	0	7,750,586	8,525,645
2041	0	0	7,750,586	8,525,645
2042	0	0	7,750,586	8,525,645
2043	0	0	7,750,586	8,525,645
2044	0	0	7,750,586	8,525,645
2045	0	0	7,750,586	8,525,645
2046	0	0	7,750,586	8,525,645
2047	0	0	7,750,586	8,525,645
2048	0	0	7,750,586	8,525,645
2049	0	0	7,750,586	8,525,645
2050	0	0	7,750,586	8,525,645
2051	0	0	7,750,586	8,525,645
2052	0	0	7,750,586	8,525,645
2053	0	0	7,750,586	8,525,645
2054	0	0	7,750,586	8,525,645
2055	0	0	7,750,586	8,525,645
2056	0	0	7,750,586	8,525,645

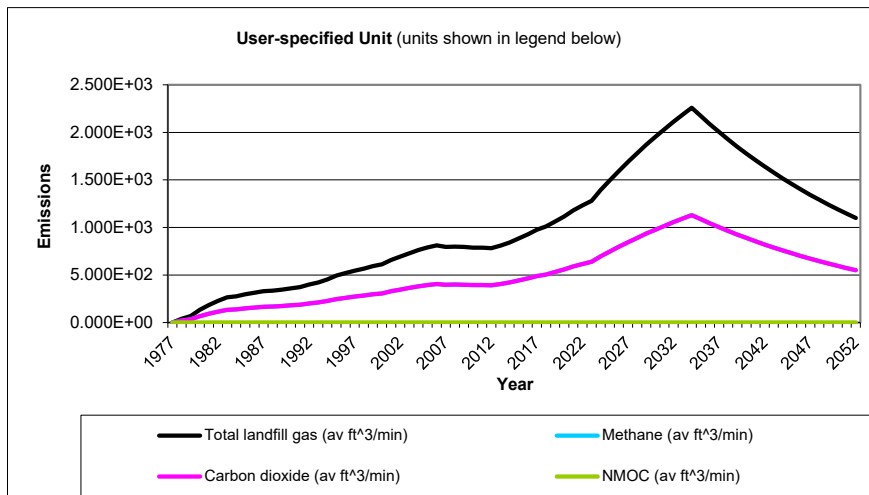
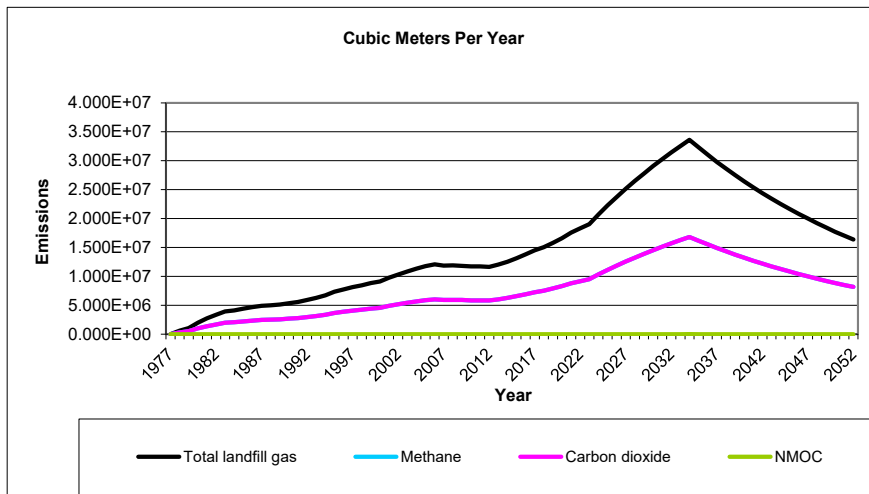
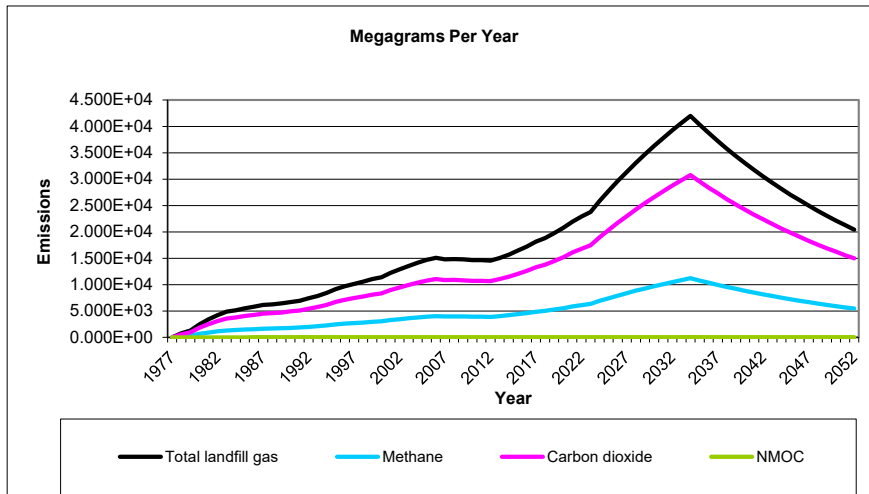
Pollutant Parameters

Gas / Pollutant Default Parameters:				User-specified Pollutant Parameters:	
	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Gases	Total landfill gas		0.00		
	Methane		16.04		
	Carbon dioxide		44.01		
	NMOC		86.18		
Pollutants	1,1,1-Trichloroethane (methyl chloroform) - HAP	0.48	133.41		
	1,1,2,2- Tetrachloroethane - HAP/VOC	1.1	167.85		
	1,1-Dichloroethane (ethylidene dichloride) - HAP/VOC	2.4	98.97		
	1,1-Dichloroethene (vinylidene chloride) - HAP/VOC	0.20	96.94		
	1,2-Dichloroethane (ethylene dichloride) - HAP/VOC	0.41	98.96		
	1,2-Dichloropropane (propylene dichloride) - HAP/VOC	0.18	112.99		
	2-Propanol (isopropyl alcohol) - VOC	50	60.11		
	Acetone	7.0	58.08		
	Acrylonitrile - HAP/VOC	6.3	53.06		
	Benzene - No or Unknown Co-disposal - HAP/VOC	1.9	78.11		
	Benzene - Co-disposal - HAP/VOC	11	78.11		
	Bromodichloromethane - VOC	3.1	163.83		
	Butane - VOC	5.0	58.12		
	Carbon disulfide - HAP/VOC	0.58	76.13		
	Carbon monoxide	140	28.01		
	Carbon tetrachloride - HAP/VOC	4.0E-03	153.84		
	Carbonyl sulfide - HAP/VOC	0.49	60.07		
	Chlorobenzene - HAP/VOC	0.25	112.56		
	Chlorodifluoromethane	1.3	86.47		
	Chloroethane (ethyl chloride) - HAP/VOC	1.3	64.52		
	Chloroform - HAP/VOC	0.03	119.39		
	Chloromethane - VOC	1.2	50.49		
	Dichlorobenzene - (HAP for para isomer/VOC)	0.21	147		
	Dichlorodifluoromethane	16	120.91		
	Dichlorofluoromethane - VOC	2.6	102.92		
	Dichloromethane (methylene chloride) - HAP	14	84.94		
	Dimethyl sulfide (methyl sulfide) - VOC	7.8	62.13		
	Ethane	890	30.07		
	Ethanol - VOC	27	46.08		

Pollutant Parameters (Continued)

[illegible]

Graphs



Results

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1977	0	0	0	0	0	0
1978	7.478E+02	5.988E+05	4.023E+01	1.998E+02	2.994E+05	2.012E+01
1979	1.293E+03	1.035E+06	6.957E+01	3.454E+02	5.177E+05	3.478E+01
1980	2.476E+03	1.983E+06	1.332E+02	6.615E+02	9.915E+05	6.662E+01
1981	3.404E+03	2.726E+06	1.831E+02	9.092E+02	1.363E+06	9.157E+01
1982	4.209E+03	3.371E+06	2.265E+02	1.124E+03	1.685E+06	1.132E+02
1983	4.906E+03	3.929E+06	2.640E+02	1.311E+03	1.964E+06	1.320E+02
1984	5.141E+03	4.116E+06	2.766E+02	1.373E+03	2.058E+06	1.383E+02
1985	5.510E+03	4.412E+06	2.965E+02	1.472E+03	2.206E+06	1.482E+02
1986	5.825E+03	4.664E+06	3.134E+02	1.556E+03	2.332E+06	1.567E+02
1987	6.128E+03	4.907E+06	3.297E+02	1.637E+03	2.453E+06	1.648E+02
1988	6.244E+03	5.000E+06	3.359E+02	1.668E+03	2.500E+06	1.680E+02
1989	6.423E+03	5.143E+06	3.456E+02	1.716E+03	2.571E+06	1.728E+02
1990	6.702E+03	5.367E+06	3.606E+02	1.790E+03	2.683E+06	1.803E+02
1991	6.931E+03	5.550E+06	3.729E+02	1.851E+03	2.775E+06	1.865E+02
1992	7.425E+03	5.945E+06	3.995E+02	1.983E+03	2.973E+06	1.997E+02
1993	7.835E+03	6.274E+06	4.216E+02	2.093E+03	3.137E+06	2.108E+02
1994	8.407E+03	6.732E+06	4.523E+02	2.246E+03	3.366E+06	2.262E+02
1995	9.168E+03	7.342E+06	4.933E+02	2.449E+03	3.671E+06	2.466E+02
1996	9.680E+03	7.751E+06	5.208E+02	2.586E+03	3.876E+06	2.604E+02
1997	1.015E+04	8.126E+06	5.460E+02	2.711E+03	4.063E+06	2.730E+02
1998	1.055E+04	8.446E+06	5.675E+02	2.818E+03	4.223E+06	2.838E+02
1999	1.103E+04	8.831E+06	5.933E+02	2.946E+03	4.415E+06	2.967E+02
2000	1.137E+04	9.101E+06	6.115E+02	3.036E+03	4.550E+06	3.057E+02
2001	1.223E+04	9.794E+06	6.581E+02	3.267E+03	4.897E+06	3.290E+02
2002	1.290E+04	1.033E+07	6.938E+02	3.445E+03	5.163E+06	3.469E+02
2003	1.354E+04	1.084E+07	7.286E+02	3.617E+03	5.422E+06	3.643E+02
2004	1.417E+04	1.135E+07	7.624E+02	3.785E+03	5.673E+06	3.812E+02
2005	1.469E+04	1.176E+07	7.904E+02	3.924E+03	5.882E+06	3.952E+02
2006	1.506E+04	1.206E+07	8.104E+02	4.024E+03	6.031E+06	4.052E+02
2007	1.480E+04	1.185E+07	7.960E+02	3.952E+03	5.924E+06	3.980E+02
2008	1.485E+04	1.189E+07	7.988E+02	3.966E+03	5.944E+06	3.994E+02
2009	1.477E+04	1.183E+07	7.949E+02	3.946E+03	5.915E+06	3.975E+02
2010	1.462E+04	1.170E+07	7.864E+02	3.904E+03	5.852E+06	3.932E+02
2011	1.463E+04	1.171E+07	7.870E+02	3.907E+03	5.856E+06	3.935E+02
2012	1.455E+04	1.165E+07	7.828E+02	3.886E+03	5.825E+06	3.914E+02
2013	1.505E+04	1.205E+07	8.097E+02	4.020E+03	6.025E+06	4.048E+02
2014	1.566E+04	1.254E+07	8.423E+02	4.182E+03	6.268E+06	4.212E+02
2015	1.641E+04	1.314E+07	8.832E+02	4.385E+03	6.572E+06	4.416E+02
2016	1.722E+04	1.379E+07	9.263E+02	4.599E+03	6.894E+06	4.632E+02
2017	1.812E+04	1.451E+07	9.750E+02	4.840E+03	7.255E+06	4.875E+02
2018	1.880E+04	1.506E+07	1.012E+03	5.022E+03	7.528E+06	5.058E+02
2019	1.973E+04	1.580E+07	1.062E+03	5.271E+03	7.901E+06	5.309E+02
2020	2.074E+04	1.660E+07	1.116E+03	5.539E+03	8.302E+06	5.578E+02
2021	2.194E+04	1.757E+07	1.180E+03	5.860E+03	8.784E+06	5.902E+02
2022	2.289E+04	1.833E+07	1.231E+03	6.114E+03	9.164E+06	6.157E+02
2023	2.379E+04	1.905E+07	1.280E+03	6.353E+03	9.523E+06	6.399E+02
2024	2.587E+04	2.072E+07	1.392E+03	6.910E+03	1.036E+07	6.960E+02
2025	2.779E+04	2.225E+07	1.495E+03	7.422E+03	1.113E+07	7.475E+02
2026	2.963E+04	2.372E+07	1.594E+03	7.914E+03	1.186E+07	7.970E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2027	3.139E+04	2.514E+07	1.689E+03	8.386E+03	1.257E+07	8.446E+02
2028	3.309E+04	2.650E+07	1.781E+03	8.840E+03	1.325E+07	8.903E+02
2029	3.473E+04	2.781E+07	1.868E+03	9.276E+03	1.390E+07	9.342E+02
2030	3.629E+04	2.906E+07	1.953E+03	9.695E+03	1.453E+07	9.764E+02
2031	3.780E+04	3.027E+07	2.034E+03	1.010E+04	1.513E+07	1.017E+03
2032	3.925E+04	3.143E+07	2.112E+03	1.048E+04	1.571E+07	1.056E+03
2033	4.064E+04	3.254E+07	2.187E+03	1.086E+04	1.627E+07	1.093E+03
2034	4.198E+04	3.361E+07	2.258E+03	1.121E+04	1.681E+07	1.129E+03
2035	4.033E+04	3.229E+07	2.170E+03	1.077E+04	1.615E+07	1.085E+03
2036	3.875E+04	3.103E+07	2.085E+03	1.035E+04	1.551E+07	1.042E+03
2037	3.723E+04	2.981E+07	2.003E+03	9.944E+03	1.491E+07	1.002E+03
2038	3.577E+04	2.864E+07	1.924E+03	9.554E+03	1.432E+07	9.622E+02
2039	3.437E+04	2.752E+07	1.849E+03	9.180E+03	1.376E+07	9.245E+02
2040	3.302E+04	2.644E+07	1.777E+03	8.820E+03	1.322E+07	8.883E+02
2041	3.172E+04	2.540E+07	1.707E+03	8.474E+03	1.270E+07	8.534E+02
2042	3.048E+04	2.441E+07	1.640E+03	8.142E+03	1.220E+07	8.200E+02
2043	2.929E+04	2.345E+07	1.576E+03	7.822E+03	1.173E+07	7.878E+02
2044	2.814E+04	2.253E+07	1.514E+03	7.516E+03	1.127E+07	7.569E+02
2045	2.703E+04	2.165E+07	1.454E+03	7.221E+03	1.082E+07	7.272E+02
2046	2.597E+04	2.080E+07	1.397E+03	6.938E+03	1.040E+07	6.987E+02
2047	2.496E+04	1.998E+07	1.343E+03	6.666E+03	9.992E+06	6.713E+02
2048	2.398E+04	1.920E+07	1.290E+03	6.405E+03	9.600E+06	6.450E+02
2049	2.304E+04	1.845E+07	1.239E+03	6.153E+03	9.223E+06	6.197E+02
2050	2.213E+04	1.772E+07	1.191E+03	5.912E+03	8.862E+06	5.954E+02
2051	2.127E+04	1.703E+07	1.144E+03	5.680E+03	8.514E+06	5.721E+02
2052	2.043E+04	1.636E+07	1.099E+03	5.458E+03	8.180E+06	5.496E+02
2053	1.963E+04	1.572E+07	1.056E+03	5.244E+03	7.860E+06	5.281E+02
2054	1.886E+04	1.510E+07	1.015E+03	5.038E+03	7.551E+06	5.074E+02
2055	1.812E+04	1.451E+07	9.750E+02	4.840E+03	7.255E+06	4.875E+02
2056	1.741E+04	1.394E+07	9.367E+02	4.651E+03	6.971E+06	4.684E+02
2057	1.673E+04	1.340E+07	9.000E+02	4.468E+03	6.698E+06	4.500E+02
2058	1.607E+04	1.287E+07	8.647E+02	4.293E+03	6.435E+06	4.324E+02
2059	1.544E+04	1.237E+07	8.308E+02	4.125E+03	6.183E+06	4.154E+02
2060	1.484E+04	1.188E+07	7.982E+02	3.963E+03	5.940E+06	3.991E+02
2061	1.425E+04	1.141E+07	7.669E+02	3.808E+03	5.707E+06	3.835E+02
2062	1.370E+04	1.097E+07	7.369E+02	3.658E+03	5.484E+06	3.684E+02
2063	1.316E+04	1.054E+07	7.080E+02	3.515E+03	5.269E+06	3.540E+02
2064	1.264E+04	1.012E+07	6.802E+02	3.377E+03	5.062E+06	3.401E+02
2065	1.215E+04	9.727E+06	6.535E+02	3.245E+03	4.863E+06	3.268E+02
2066	1.167E+04	9.345E+06	6.279E+02	3.117E+03	4.673E+06	3.140E+02
2067	1.121E+04	8.979E+06	6.033E+02	2.995E+03	4.490E+06	3.017E+02
2068	1.077E+04	8.627E+06	5.796E+02	2.878E+03	4.313E+06	2.898E+02
2069	1.035E+04	8.289E+06	5.569E+02	2.765E+03	4.144E+06	2.785E+02
2070	9.945E+03	7.964E+06	5.351E+02	2.656E+03	3.982E+06	2.675E+02
2071	9.555E+03	7.651E+06	5.141E+02	2.552E+03	3.826E+06	2.570E+02
2072	9.181E+03	7.351E+06	4.939E+02	2.452E+03	3.676E+06	2.470E+02
2073	8.821E+03	7.063E+06	4.746E+02	2.356E+03	3.532E+06	2.373E+02
2074	8.475E+03	6.786E+06	4.560E+02	2.264E+03	3.393E+06	2.280E+02
2075	8.142E+03	6.520E+06	4.381E+02	2.175E+03	3.260E+06	2.190E+02
2076	7.823E+03	6.264E+06	4.209E+02	2.090E+03	3.132E+06	2.105E+02
2077	7.516E+03	6.019E+06	4.044E+02	2.008E+03	3.009E+06	2.022E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2078	7.222E+03	5.783E+06	3.885E+02	1.929E+03	2.891E+06	1.943E+02
2079	6.939E+03	5.556E+06	3.733E+02	1.853E+03	2.778E+06	1.867E+02
2080	6.666E+03	5.338E+06	3.587E+02	1.781E+03	2.669E+06	1.793E+02
2081	6.405E+03	5.129E+06	3.446E+02	1.711E+03	2.564E+06	1.723E+02
2082	6.154E+03	4.928E+06	3.311E+02	1.644E+03	2.464E+06	1.655E+02
2083	5.913E+03	4.735E+06	3.181E+02	1.579E+03	2.367E+06	1.591E+02
2084	5.681E+03	4.549E+06	3.056E+02	1.517E+03	2.274E+06	1.528E+02
2085	5.458E+03	4.371E+06	2.937E+02	1.458E+03	2.185E+06	1.468E+02
2086	5.244E+03	4.199E+06	2.821E+02	1.401E+03	2.100E+06	1.411E+02
2087	5.038E+03	4.035E+06	2.711E+02	1.346E+03	2.017E+06	1.355E+02
2088	4.841E+03	3.876E+06	2.605E+02	1.293E+03	1.938E+06	1.302E+02
2089	4.651E+03	3.724E+06	2.502E+02	1.242E+03	1.862E+06	1.251E+02
2090	4.469E+03	3.578E+06	2.404E+02	1.194E+03	1.789E+06	1.202E+02
2091	4.293E+03	3.438E+06	2.310E+02	1.147E+03	1.719E+06	1.155E+02
2092	4.125E+03	3.303E+06	2.219E+02	1.102E+03	1.652E+06	1.110E+02
2093	3.963E+03	3.174E+06	2.132E+02	1.059E+03	1.587E+06	1.066E+02
2094	3.808E+03	3.049E+06	2.049E+02	1.017E+03	1.525E+06	1.024E+02
2095	3.659E+03	2.930E+06	1.968E+02	9.773E+02	1.465E+06	9.842E+01
2096	3.515E+03	2.815E+06	1.891E+02	9.389E+02	1.407E+06	9.456E+01
2097	3.377E+03	2.704E+06	1.817E+02	9.021E+02	1.352E+06	9.086E+01
2098	3.245E+03	2.598E+06	1.746E+02	8.668E+02	1.299E+06	8.729E+01
2099	3.118E+03	2.497E+06	1.677E+02	8.328E+02	1.248E+06	8.387E+01
2100	2.995E+03	2.399E+06	1.612E+02	8.001E+02	1.199E+06	8.058E+01
2101	2.878E+03	2.305E+06	1.548E+02	7.687E+02	1.152E+06	7.742E+01
2102	2.765E+03	2.214E+06	1.488E+02	7.386E+02	1.107E+06	7.439E+01
2103	2.657E+03	2.127E+06	1.429E+02	7.096E+02	1.064E+06	7.147E+01
2104	2.553E+03	2.044E+06	1.373E+02	6.818E+02	1.022E+06	6.867E+01
2105	2.452E+03	1.964E+06	1.319E+02	6.551E+02	9.819E+05	6.597E+01
2106	2.356E+03	1.887E+06	1.268E+02	6.294E+02	9.434E+05	6.339E+01
2107	2.264E+03	1.813E+06	1.218E+02	6.047E+02	9.064E+05	6.090E+01
2108	2.175E+03	1.742E+06	1.170E+02	5.810E+02	8.709E+05	5.851E+01
2109	2.090E+03	1.673E+06	1.124E+02	5.582E+02	8.367E+05	5.622E+01
2110	2.008E+03	1.608E+06	1.080E+02	5.363E+02	8.039E+05	5.402E+01
2111	1.929E+03	1.545E+06	1.038E+02	5.153E+02	7.724E+05	5.190E+01
2112	1.854E+03	1.484E+06	9.972E+01	4.951E+02	7.421E+05	4.986E+01
2113	1.781E+03	1.426E+06	9.581E+01	4.757E+02	7.130E+05	4.791E+01
2114	1.711E+03	1.370E+06	9.206E+01	4.570E+02	6.851E+05	4.603E+01
2115	1.644E+03	1.316E+06	8.845E+01	4.391E+02	6.582E+05	4.422E+01
2116	1.579E+03	1.265E+06	8.498E+01	4.219E+02	6.324E+05	4.249E+01
2117	1.518E+03	1.215E+06	8.165E+01	4.054E+02	6.076E+05	4.082E+01

Results (Continued)

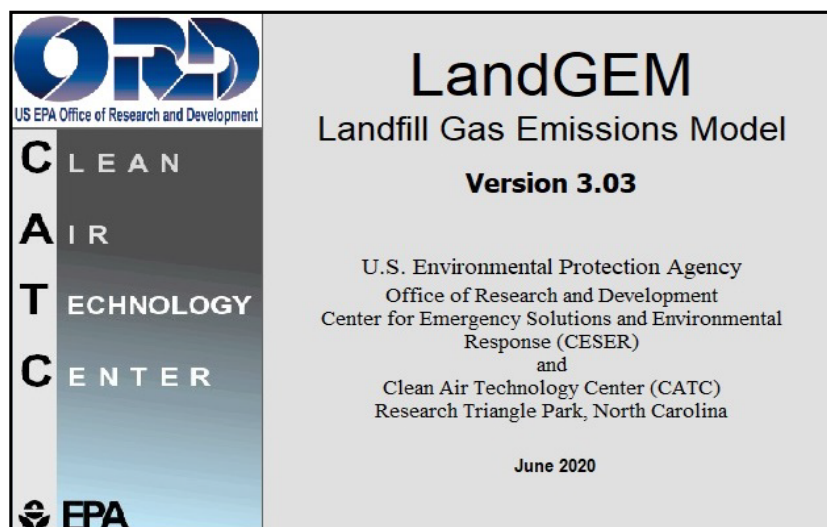
Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
1977	0	0	0	0	0	0
1978	5.481E+02	2.994E+05	2.012E+01	7.598E-01	2.120E+02	1.424E-02
1979	9.476E+02	5.177E+05	3.478E+01	1.314E+00	3.665E+02	2.463E-02
1980	1.815E+03	9.915E+05	6.662E+01	2.516E+00	7.020E+02	4.717E-02
1981	2.495E+03	1.363E+06	9.157E+01	3.459E+00	9.649E+02	6.483E-02
1982	3.085E+03	1.685E+06	1.132E+02	4.277E+00	1.193E+03	8.017E-02
1983	3.596E+03	1.964E+06	1.320E+02	4.985E+00	1.391E+03	9.345E-02
1984	3.768E+03	2.058E+06	1.383E+02	5.223E+00	1.457E+03	9.791E-02
1985	4.038E+03	2.206E+06	1.482E+02	5.599E+00	1.562E+03	1.049E-01
1986	4.269E+03	2.332E+06	1.567E+02	5.918E+00	1.651E+03	1.109E-01
1987	4.491E+03	2.453E+06	1.648E+02	6.226E+00	1.737E+03	1.167E-01
1988	4.576E+03	2.500E+06	1.680E+02	6.344E+00	1.770E+03	1.189E-01
1989	4.707E+03	2.571E+06	1.728E+02	6.526E+00	1.821E+03	1.223E-01
1990	4.912E+03	2.683E+06	1.803E+02	6.810E+00	1.900E+03	1.277E-01
1991	5.080E+03	2.775E+06	1.865E+02	7.043E+00	1.965E+03	1.320E-01
1992	5.442E+03	2.973E+06	1.997E+02	7.544E+00	2.105E+03	1.414E-01
1993	5.742E+03	3.137E+06	2.108E+02	7.961E+00	2.221E+03	1.492E-01
1994	6.162E+03	3.366E+06	2.262E+02	8.542E+00	2.383E+03	1.601E-01
1995	6.719E+03	3.671E+06	2.466E+02	9.316E+00	2.599E+03	1.746E-01
1996	7.094E+03	3.876E+06	2.604E+02	9.836E+00	2.744E+03	1.844E-01
1997	7.437E+03	4.063E+06	2.730E+02	1.031E+01	2.877E+03	1.933E-01
1998	7.731E+03	4.223E+06	2.838E+02	1.072E+01	2.990E+03	2.009E-01
1999	8.083E+03	4.415E+06	2.967E+02	1.121E+01	3.126E+03	2.100E-01
2000	8.330E+03	4.550E+06	3.057E+02	1.155E+01	3.222E+03	2.165E-01
2001	8.964E+03	4.897E+06	3.290E+02	1.243E+01	3.467E+03	2.330E-01
2002	9.451E+03	5.163E+06	3.469E+02	1.310E+01	3.655E+03	2.456E-01
2003	9.925E+03	5.422E+06	3.643E+02	1.376E+01	3.839E+03	2.579E-01
2004	1.038E+04	5.673E+06	3.812E+02	1.440E+01	4.017E+03	2.699E-01
2005	1.077E+04	5.882E+06	3.952E+02	1.493E+01	4.164E+03	2.798E-01
2006	1.104E+04	6.031E+06	4.052E+02	1.531E+01	4.270E+03	2.869E-01
2007	1.084E+04	5.924E+06	3.980E+02	1.503E+01	4.194E+03	2.818E-01
2008	1.088E+04	5.944E+06	3.994E+02	1.509E+01	4.209E+03	2.828E-01
2009	1.083E+04	5.915E+06	3.975E+02	1.501E+01	4.188E+03	2.814E-01
2010	1.071E+04	5.852E+06	3.932E+02	1.485E+01	4.143E+03	2.784E-01
2011	1.072E+04	5.856E+06	3.935E+02	1.486E+01	4.146E+03	2.786E-01
2012	1.066E+04	5.825E+06	3.914E+02	1.478E+01	4.124E+03	2.771E-01
2013	1.103E+04	6.025E+06	4.048E+02	1.529E+01	4.266E+03	2.866E-01
2014	1.147E+04	6.268E+06	4.212E+02	1.591E+01	4.438E+03	2.982E-01
2015	1.203E+04	6.572E+06	4.416E+02	1.668E+01	4.653E+03	3.126E-01
2016	1.262E+04	6.894E+06	4.632E+02	1.749E+01	4.881E+03	3.279E-01
2017	1.328E+04	7.255E+06	4.875E+02	1.841E+01	5.137E+03	3.451E-01
2018	1.378E+04	7.528E+06	5.058E+02	1.910E+01	5.330E+03	3.581E-01
2019	1.446E+04	7.901E+06	5.309E+02	2.005E+01	5.594E+03	3.758E-01
2020	1.520E+04	8.302E+06	5.578E+02	2.107E+01	5.878E+03	3.949E-01
2021	1.608E+04	8.784E+06	5.902E+02	2.229E+01	6.219E+03	4.179E-01
2022	1.678E+04	9.164E+06	6.157E+02	2.326E+01	6.488E+03	4.360E-01
2023	1.743E+04	9.523E+06	6.399E+02	2.417E+01	6.742E+03	4.530E-01
2024	1.896E+04	1.036E+07	6.960E+02	2.629E+01	7.334E+03	4.927E-01
2025	2.036E+04	1.113E+07	7.475E+02	2.823E+01	7.877E+03	5.292E-01
2026	2.171E+04	1.186E+07	7.970E+02	3.010E+01	8.398E+03	5.643E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2027	2.301E+04	1.257E+07	8.446E+02	3.190E+01	8.899E+03	5.980E-01
2028	2.425E+04	1.325E+07	8.903E+02	3.363E+01	9.381E+03	6.303E-01
2029	2.545E+04	1.390E+07	9.342E+02	3.528E+01	9.844E+03	6.614E-01
2030	2.660E+04	1.453E+07	9.764E+02	3.688E+01	1.029E+04	6.913E-01
2031	2.770E+04	1.513E+07	1.017E+03	3.841E+01	1.072E+04	7.200E-01
2032	2.876E+04	1.571E+07	1.056E+03	3.988E+01	1.113E+04	7.475E-01
2033	2.978E+04	1.627E+07	1.093E+03	4.129E+01	1.152E+04	7.740E-01
2034	3.076E+04	1.681E+07	1.129E+03	4.265E+01	1.190E+04	7.995E-01
2035	2.956E+04	1.615E+07	1.085E+03	4.098E+01	1.143E+04	7.681E-01
2036	2.840E+04	1.551E+07	1.042E+03	3.937E+01	1.098E+04	7.380E-01
2037	2.728E+04	1.491E+07	1.002E+03	3.783E+01	1.055E+04	7.091E-01
2038	2.622E+04	1.432E+07	9.622E+02	3.634E+01	1.014E+04	6.813E-01
2039	2.519E+04	1.376E+07	9.245E+02	3.492E+01	9.742E+03	6.546E-01
2040	2.420E+04	1.322E+07	8.883E+02	3.355E+01	9.360E+03	6.289E-01
2041	2.325E+04	1.270E+07	8.534E+02	3.223E+01	8.993E+03	6.042E-01
2042	2.234E+04	1.220E+07	8.200E+02	3.097E+01	8.640E+03	5.805E-01
2043	2.146E+04	1.173E+07	7.878E+02	2.976E+01	8.301E+03	5.578E-01
2044	2.062E+04	1.127E+07	7.569E+02	2.859E+01	7.976E+03	5.359E-01
2045	1.981E+04	1.082E+07	7.272E+02	2.747E+01	7.663E+03	5.149E-01
2046	1.904E+04	1.040E+07	6.987E+02	2.639E+01	7.363E+03	4.947E-01
2047	1.829E+04	9.992E+06	6.713E+02	2.536E+01	7.074E+03	4.753E-01
2048	1.757E+04	9.600E+06	6.450E+02	2.436E+01	6.797E+03	4.567E-01
2049	1.688E+04	9.223E+06	6.197E+02	2.341E+01	6.530E+03	4.388E-01
2050	1.622E+04	8.862E+06	5.954E+02	2.249E+01	6.274E+03	4.216E-01
2051	1.559E+04	8.514E+06	5.721E+02	2.161E+01	6.028E+03	4.050E-01
2052	1.497E+04	8.180E+06	5.496E+02	2.076E+01	5.792E+03	3.891E-01
2053	1.439E+04	7.860E+06	5.281E+02	1.995E+01	5.565E+03	3.739E-01
2054	1.382E+04	7.551E+06	5.074E+02	1.916E+01	5.346E+03	3.592E-01
2055	1.328E+04	7.255E+06	4.875E+02	1.841E+01	5.137E+03	3.451E-01
2056	1.276E+04	6.971E+06	4.684E+02	1.769E+01	4.935E+03	3.316E-01
2057	1.226E+04	6.698E+06	4.500E+02	1.700E+01	4.742E+03	3.186E-01
2058	1.178E+04	6.435E+06	4.324E+02	1.633E+01	4.556E+03	3.061E-01
2059	1.132E+04	6.183E+06	4.154E+02	1.569E+01	4.377E+03	2.941E-01
2060	1.087E+04	5.940E+06	3.991E+02	1.508E+01	4.206E+03	2.826E-01
2061	1.045E+04	5.707E+06	3.835E+02	1.448E+01	4.041E+03	2.715E-01
2062	1.004E+04	5.484E+06	3.684E+02	1.392E+01	3.882E+03	2.609E-01
2063	9.644E+03	5.269E+06	3.540E+02	1.337E+01	3.730E+03	2.506E-01
2064	9.266E+03	5.062E+06	3.401E+02	1.285E+01	3.584E+03	2.408E-01
2065	8.903E+03	4.863E+06	3.268E+02	1.234E+01	3.443E+03	2.314E-01
2066	8.553E+03	4.673E+06	3.140E+02	1.186E+01	3.308E+03	2.223E-01
2067	8.218E+03	4.490E+06	3.017E+02	1.139E+01	3.179E+03	2.136E-01
2068	7.896E+03	4.313E+06	2.898E+02	1.095E+01	3.054E+03	2.052E-01
2069	7.586E+03	4.144E+06	2.785E+02	1.052E+01	2.934E+03	1.971E-01
2070	7.289E+03	3.982E+06	2.675E+02	1.011E+01	2.819E+03	1.894E-01
2071	7.003E+03	3.826E+06	2.570E+02	9.709E+00	2.709E+03	1.820E-01
2072	6.728E+03	3.676E+06	2.470E+02	9.328E+00	2.602E+03	1.749E-01
2073	6.465E+03	3.532E+06	2.373E+02	8.962E+00	2.500E+03	1.680E-01
2074	6.211E+03	3.393E+06	2.280E+02	8.611E+00	2.402E+03	1.614E-01
2075	5.968E+03	3.260E+06	2.190E+02	8.273E+00	2.308E+03	1.551E-01
2076	5.734E+03	3.132E+06	2.105E+02	7.949E+00	2.218E+03	1.490E-01
2077	5.509E+03	3.009E+06	2.022E+02	7.637E+00	2.131E+03	1.432E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2078	5.293E+03	2.891E+06	1.943E+02	7.338E+00	2.047E+03	1.375E-01
2079	5.085E+03	2.778E+06	1.867E+02	7.050E+00	1.967E+03	1.322E-01
2080	4.886E+03	2.669E+06	1.793E+02	6.774E+00	1.890E+03	1.270E-01
2081	4.694E+03	2.564E+06	1.723E+02	6.508E+00	1.816E+03	1.220E-01
2082	4.510E+03	2.464E+06	1.655E+02	6.253E+00	1.744E+03	1.172E-01
2083	4.333E+03	2.367E+06	1.591E+02	6.008E+00	1.676E+03	1.126E-01
2084	4.163E+03	2.274E+06	1.528E+02	5.772E+00	1.610E+03	1.082E-01
2085	4.000E+03	2.185E+06	1.468E+02	5.546E+00	1.547E+03	1.040E-01
2086	3.843E+03	2.100E+06	1.411E+02	5.328E+00	1.487E+03	9.988E-02
2087	3.693E+03	2.017E+06	1.355E+02	5.119E+00	1.428E+03	9.596E-02
2088	3.548E+03	1.938E+06	1.302E+02	4.919E+00	1.372E+03	9.220E-02
2089	3.409E+03	1.862E+06	1.251E+02	4.726E+00	1.318E+03	8.858E-02
2090	3.275E+03	1.789E+06	1.202E+02	4.541E+00	1.267E+03	8.511E-02
2091	3.147E+03	1.719E+06	1.155E+02	4.362E+00	1.217E+03	8.177E-02
2092	3.023E+03	1.652E+06	1.110E+02	4.191E+00	1.169E+03	7.857E-02
2093	2.905E+03	1.587E+06	1.066E+02	4.027E+00	1.123E+03	7.549E-02
2094	2.791E+03	1.525E+06	1.024E+02	3.869E+00	1.079E+03	7.253E-02
2095	2.681E+03	1.465E+06	9.842E+01	3.717E+00	1.037E+03	6.968E-02
2096	2.576E+03	1.407E+06	9.456E+01	3.572E+00	9.964E+02	6.695E-02
2097	2.475E+03	1.352E+06	9.086E+01	3.432E+00	9.574E+02	6.433E-02
2098	2.378E+03	1.299E+06	8.729E+01	3.297E+00	9.198E+02	6.180E-02
2099	2.285E+03	1.248E+06	8.387E+01	3.168E+00	8.838E+02	5.938E-02
2100	2.195E+03	1.199E+06	8.058E+01	3.044E+00	8.491E+02	5.705E-02
2101	2.109E+03	1.152E+06	7.742E+01	2.924E+00	8.158E+02	5.481E-02
2102	2.027E+03	1.107E+06	7.439E+01	2.810E+00	7.838E+02	5.267E-02
2103	1.947E+03	1.064E+06	7.147E+01	2.699E+00	7.531E+02	5.060E-02
2104	1.871E+03	1.022E+06	6.867E+01	2.594E+00	7.236E+02	4.862E-02
2105	1.797E+03	9.819E+05	6.597E+01	2.492E+00	6.952E+02	4.671E-02
2106	1.727E+03	9.434E+05	6.339E+01	2.394E+00	6.679E+02	4.488E-02
2107	1.659E+03	9.064E+05	6.090E+01	2.300E+00	6.417E+02	4.312E-02
2108	1.594E+03	8.709E+05	5.851E+01	2.210E+00	6.166E+02	4.143E-02
2109	1.532E+03	8.367E+05	5.622E+01	2.123E+00	5.924E+02	3.980E-02
2110	1.472E+03	8.039E+05	5.402E+01	2.040E+00	5.692E+02	3.824E-02
2111	1.414E+03	7.724E+05	5.190E+01	1.960E+00	5.469E+02	3.674E-02
2112	1.358E+03	7.421E+05	4.986E+01	1.883E+00	5.254E+02	3.530E-02
2113	1.305E+03	7.130E+05	4.791E+01	1.809E+00	5.048E+02	3.392E-02
2114	1.254E+03	6.851E+05	4.603E+01	1.739E+00	4.850E+02	3.259E-02
2115	1.205E+03	6.582E+05	4.422E+01	1.670E+00	4.660E+02	3.131E-02
2116	1.158E+03	6.324E+05	4.249E+01	1.605E+00	4.477E+02	3.008E-02
2117	1.112E+03	6.076E+05	4.082E+01	1.542E+00	4.302E+02	2.890E-02



Summary Report

Landfill Name or Identifier: Wexford West Landfill

Date: Wednesday, March 27, 2024

Description/Comments:

About LandGEM:

First-Order Decomposition Rate Equation:

$$Q_{CH_4} = \sum_{i=1}^n \sum_{j=0.1}^1 k L_o \left(\frac{M_i}{10} \right) e^{-k t_{ij}}$$

Where,

Q_{CH_4} = annual methane generation in the year of the calculation ($m^3/year$)

i = 1-year time increment

n = (year of the calculation) - (initial year of waste acceptance)

j = 0.1-year time increment

k = methane generation rate ($year^{-1}$)

L_o = potential methane generation capacity (m^3/Mg)

M_i = mass of waste accepted in the i^{th} year (Mg)

t_{ij} = age of the j^{th} section of waste mass M_i accepted in the i^{th} year (*decimal years*, e.g., 3.2 years)

LandGEM is based on a first-order decomposition rate equation for quantifying emissions from the decomposition of landfilled waste in municipal solid waste (MSW) landfills. The software provides a relatively simple approach to estimating landfill gas emissions. Model defaults are based on empirical data from U.S. landfills. Field test data can also be used in place of model defaults when available. Further guidance on EPA test methods, Clean Air Act (CAA) regulations, and other guidance regarding landfill gas emissions and control technology requirements can be found at <http://www.epa.gov/ttnatw01/landfill/landflpg.html>.

LandGEM is considered a screening tool — the better the input data, the better the estimates. Often, there are limitations with the available data regarding waste quantity and composition, variation in design and operating practices over time, and changes occurring over time that impact the emissions potential. Changes to landfill operation, such as operating under wet conditions through leachate recirculation or other liquid additions, will result in generating more gas at a faster rate. Defaults for estimating emissions for this type of operation are being developed to include in LandGEM along with defaults for conventional landfills (no leachate or liquid additions) for developing emission inventories and determining CAA applicability. Refer to the Web site identified above for future updates.

Input Review

LANDFILL CHARACTERISTICS

Landfill Open Year	2034	
Landfill Closure Year (with 80-year limit)	2073	
Actual Closure Year (without limit)	2073	
Have Model Calculate Closure Year?	No	
Waste Design Capacity	12,476,193	<i>short tons</i>

MODEL PARAMETERS

Methane Generation Rate, k	0.040	<i>year⁻¹</i>
Potential Methane Generation Capacity, L ₀	100	<i>m³/Mg</i>
NMOC Concentration	354	<i>ppmv as hexane</i>
Methane Content	50	<i>% by volume</i>

GASES / POLLUTANTS SELECTED

Gas / Pollutant #1:	Total landfill gas
Gas / Pollutant #2:	Methane
Gas / Pollutant #3:	Carbon dioxide
Gas / Pollutant #4:	NMOC

WASTE ACCEPTANCE RATES

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2034	283,550	311,905	0	0
2035	283,550	311,905	283,550	311,905
2036	283,550	311,905	567,100	623,810
2037	283,550	311,905	850,650	935,714
2038	283,550	311,905	1,134,199	1,247,619
2039	283,550	311,905	1,417,749	1,559,524
2040	283,550	311,905	1,701,299	1,871,429
2041	283,550	311,905	1,984,849	2,183,334
2042	283,550	311,905	2,268,399	2,495,239
2043	283,550	311,905	2,551,949	2,807,143
2044	283,550	311,905	2,835,498	3,119,048
2045	283,550	311,905	3,119,048	3,430,953
2046	283,550	311,905	3,402,598	3,742,858
2047	283,550	311,905	3,686,148	4,054,763
2048	283,550	311,905	3,969,698	4,366,668
2049	283,550	311,905	4,253,248	4,678,572
2050	283,550	311,905	4,536,797	4,990,477
2051	283,550	311,905	4,820,347	5,302,382
2052	283,550	311,905	5,103,897	5,614,287
2053	283,550	311,905	5,387,447	5,926,192
2054	283,550	311,905	5,670,997	6,238,097
2055	283,550	311,905	5,954,547	6,550,001
2056	283,550	311,905	6,238,097	6,861,906
2057	283,550	311,905	6,521,646	7,173,811
2058	283,550	311,905	6,805,196	7,485,716
2059	283,550	311,905	7,088,746	7,797,621
2060	283,550	311,905	7,372,296	8,109,525
2061	283,550	311,905	7,655,846	8,421,430
2062	283,550	311,905	7,939,396	8,733,335
2063	283,550	311,905	8,222,945	9,045,240
2064	283,550	311,905	8,506,495	9,357,145
2065	283,550	311,905	8,790,045	9,669,050
2066	283,550	311,905	9,073,595	9,980,954
2067	283,550	311,905	9,357,145	10,292,859
2068	283,550	311,905	9,640,695	10,604,764
2069	283,550	311,905	9,924,244	10,916,669
2070	283,550	311,905	10,207,794	11,228,574
2071	283,550	311,905	10,491,344	11,540,479
2072	283,550	311,905	10,774,894	11,852,383
2073	283,550	311,905	11,058,444	12,164,288

WASTE ACCEPTANCE RATES (Continued)

Year	Waste Accepted		Waste-In-Place	
	(Mg/year)	(short tons/year)	(Mg)	(short tons)
2074	0	0	11,341,994	12,476,193
2075	0	0	11,341,994	12,476,193
2076	0	0	11,341,994	12,476,193
2077	0	0	11,341,994	12,476,193
2078	0	0	11,341,994	12,476,193
2079	0	0	11,341,994	12,476,193
2080	0	0	11,341,994	12,476,193
2081	0	0	11,341,994	12,476,193
2082	0	0	11,341,994	12,476,193
2083	0	0	11,341,994	12,476,193
2084	0	0	11,341,994	12,476,193
2085	0	0	11,341,994	12,476,193
2086	0	0	11,341,994	12,476,193
2087	0	0	11,341,994	12,476,193
2088	0	0	11,341,994	12,476,193
2089	0	0	11,341,994	12,476,193
2090	0	0	11,341,994	12,476,193
2091	0	0	11,341,994	12,476,193
2092	0	0	11,341,994	12,476,193
2093	0	0	11,341,994	12,476,193
2094	0	0	11,341,994	12,476,193
2095	0	0	11,341,994	12,476,193
2096	0	0	11,341,994	12,476,193
2097	0	0	11,341,994	12,476,193
2098	0	0	11,341,994	12,476,193
2099	0	0	11,341,994	12,476,193
2100	0	0	11,341,994	12,476,193
2101	0	0	11,341,994	12,476,193
2102	0	0	11,341,994	12,476,193
2103	0	0	11,341,994	12,476,193
2104	0	0	11,341,994	12,476,193
2105	0	0	11,341,994	12,476,193
2106	0	0	11,341,994	12,476,193
2107	0	0	11,341,994	12,476,193
2108	0	0	11,341,994	12,476,193
2109	0	0	11,341,994	12,476,193
2110	0	0	11,341,994	12,476,193
2111	0	0	11,341,994	12,476,193
2112	0	0	11,341,994	12,476,193
2113	0	0	11,341,994	12,476,193

Pollutant Parameters

Gas / Pollutant Default Parameters:

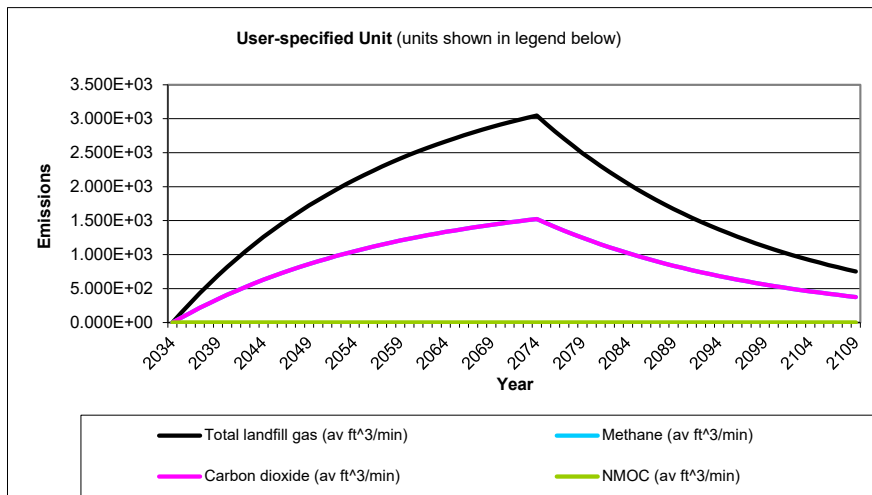
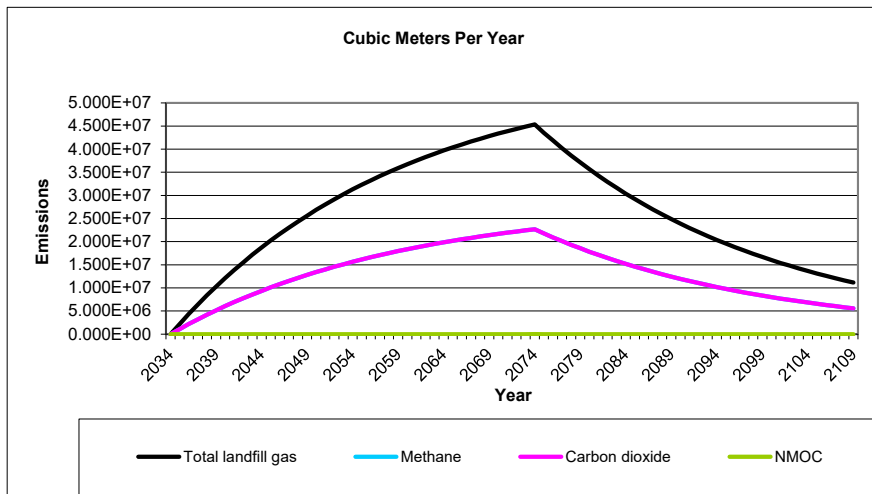
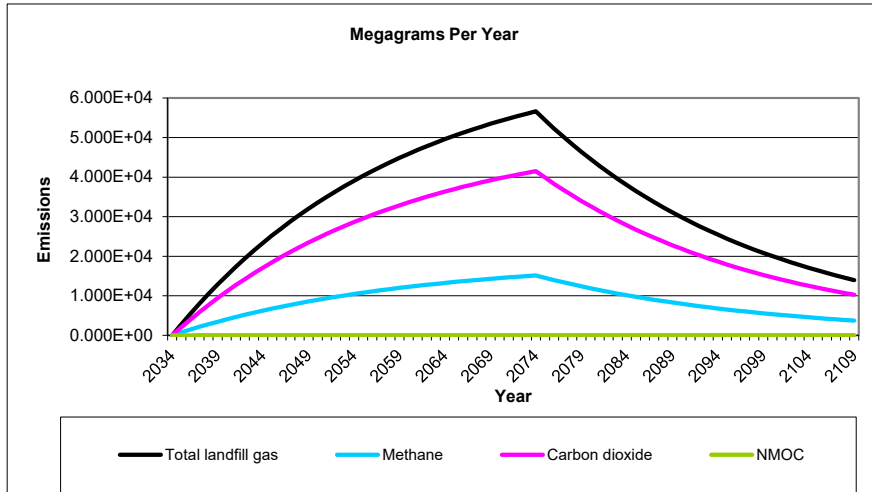
User-specified Pollutant Parameters:

	Compound	Concentration (ppmv)	Molecular Weight	Concentration (ppmv)	Molecular Weight
Gases	Total landfill gas		0.00		
	Methane		16.04		
	Carbon dioxide		44.01		
	NMOC	4,000	86.18		
Pollutants	1,1,1-Trichloroethane (methyl chloroform) - HAP	0.48	133.41		
	1,1,2,2- Tetrachloroethane - HAP/VOC	1.1	167.85		
	1,1-Dichloroethane (ethylidene dichloride) - HAP/VOC	2.4	98.97		
	1,1-Dichloroethene (vinylidene chloride) - HAP/VOC	0.20	96.94		
	1,2-Dichloroethane (ethylene dichloride) - HAP/VOC	0.41	98.96		
	1,2-Dichloropropane (propylene dichloride) - HAP/VOC	0.18	112.99		
	2-Propanol (isopropyl alcohol) - VOC	50	60.11		
	Acetone	7.0	58.08		
	Acrylonitrile - HAP/VOC	6.3	53.06		
	Benzene - No or Unknown Co-disposal - HAP/VOC	1.9	78.11		
	Benzene - Co-disposal - HAP/VOC	11	78.11		
	Bromodichloromethane - VOC	3.1	163.83		
	Butane - VOC	5.0	58.12		
	Carbon disulfide - HAP/VOC	0.58	76.13		
	Carbon monoxide	140	28.01		
	Carbon tetrachloride - HAP/VOC	4.0E-03	153.84		
	Carbonyl sulfide - HAP/VOC	0.49	60.07		
	Chlorobenzene - HAP/VOC	0.25	112.56		
	Chlorodifluoromethane	1.3	86.47		
	Chloroethane (ethyl chloride) - HAP/VOC	1.3	64.52		
	Chloroform - HAP/VOC	0.03	119.39		
	Chloromethane - VOC	1.2	50.49		
	Dichlorobenzene - (HAP for para isomer/VOC)	0.21	147		
	Dichlorodifluoromethane	16	120.91		
	Dichlorofluoromethane - VOC	2.6	102.92		
	Dichloromethane (methylene chloride) - HAP	14	84.94		
	Dimethyl sulfide (methyl sulfide) - VOC	7.8	62.13		
	Ethane	890	30.07		
	Ethanol - VOC	27	46.08		

Pollutant Parameters (Continued)

[illegible]

Graphs



Results

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2034	0	0	0	0	0	0
2035	2.782E+03	2.228E+06	1.497E+02	7.432E+02	1.114E+06	7.485E+01
2036	5.456E+03	4.369E+06	2.935E+02	1.457E+03	2.184E+06	1.468E+02
2037	8.024E+03	6.426E+06	4.317E+02	2.143E+03	3.213E+06	2.159E+02
2038	1.049E+04	8.402E+06	5.645E+02	2.803E+03	4.201E+06	2.823E+02
2039	1.286E+04	1.030E+07	6.921E+02	3.436E+03	5.150E+06	3.460E+02
2040	1.514E+04	1.212E+07	8.146E+02	4.044E+03	6.062E+06	4.073E+02
2041	1.733E+04	1.388E+07	9.324E+02	4.629E+03	6.939E+06	4.662E+02
2042	1.943E+04	1.556E+07	1.046E+03	5.191E+03	7.781E+06	5.228E+02
2043	2.145E+04	1.718E+07	1.154E+03	5.730E+03	8.590E+06	5.771E+02
2044	2.339E+04	1.873E+07	1.259E+03	6.249E+03	9.367E+06	6.294E+02
2045	2.526E+04	2.023E+07	1.359E+03	6.747E+03	1.011E+07	6.795E+02
2046	2.705E+04	2.166E+07	1.455E+03	7.226E+03	1.083E+07	7.277E+02
2047	2.877E+04	2.304E+07	1.548E+03	7.686E+03	1.152E+07	7.741E+02
2048	3.043E+04	2.437E+07	1.637E+03	8.128E+03	1.218E+07	8.186E+02
2049	3.202E+04	2.564E+07	1.723E+03	8.552E+03	1.282E+07	8.613E+02
2050	3.354E+04	2.686E+07	1.805E+03	8.960E+03	1.343E+07	9.024E+02
2051	3.501E+04	2.804E+07	1.884E+03	9.352E+03	1.402E+07	9.419E+02
2052	3.642E+04	2.916E+07	1.960E+03	9.729E+03	1.458E+07	9.798E+02
2053	3.778E+04	3.025E+07	2.032E+03	1.009E+04	1.512E+07	1.016E+03
2054	3.908E+04	3.129E+07	2.102E+03	1.044E+04	1.565E+07	1.051E+03
2055	4.033E+04	3.229E+07	2.170E+03	1.077E+04	1.615E+07	1.085E+03
2056	4.153E+04	3.325E+07	2.234E+03	1.109E+04	1.663E+07	1.117E+03
2057	4.268E+04	3.418E+07	2.296E+03	1.140E+04	1.709E+07	1.148E+03
2058	4.379E+04	3.507E+07	2.356E+03	1.170E+04	1.753E+07	1.178E+03
2059	4.486E+04	3.592E+07	2.413E+03	1.198E+04	1.796E+07	1.207E+03
2060	4.588E+04	3.674E+07	2.468E+03	1.226E+04	1.837E+07	1.234E+03
2061	4.686E+04	3.753E+07	2.521E+03	1.252E+04	1.876E+07	1.261E+03
2062	4.781E+04	3.828E+07	2.572E+03	1.277E+04	1.914E+07	1.286E+03
2063	4.872E+04	3.901E+07	2.621E+03	1.301E+04	1.951E+07	1.311E+03
2064	4.959E+04	3.971E+07	2.668E+03	1.325E+04	1.985E+07	1.334E+03
2065	5.043E+04	4.038E+07	2.713E+03	1.347E+04	2.019E+07	1.357E+03
2066	5.123E+04	4.102E+07	2.756E+03	1.368E+04	2.051E+07	1.378E+03
2067	5.201E+04	4.164E+07	2.798E+03	1.389E+04	2.082E+07	1.399E+03
2068	5.275E+04	4.224E+07	2.838E+03	1.409E+04	2.112E+07	1.419E+03
2069	5.346E+04	4.281E+07	2.876E+03	1.428E+04	2.141E+07	1.438E+03
2070	5.415E+04	4.336E+07	2.913E+03	1.446E+04	2.168E+07	1.457E+03
2071	5.481E+04	4.389E+07	2.949E+03	1.464E+04	2.194E+07	1.474E+03
2072	5.544E+04	4.440E+07	2.983E+03	1.481E+04	2.220E+07	1.491E+03
2073	5.605E+04	4.488E+07	3.016E+03	1.497E+04	2.244E+07	1.508E+03
2074	5.664E+04	4.535E+07	3.047E+03	1.513E+04	2.268E+07	1.524E+03
2075	5.441E+04	4.357E+07	2.928E+03	1.453E+04	2.179E+07	1.464E+03
2076	5.228E+04	4.186E+07	2.813E+03	1.396E+04	2.093E+07	1.406E+03
2077	5.023E+04	4.022E+07	2.703E+03	1.342E+04	2.011E+07	1.351E+03
2078	4.826E+04	3.865E+07	2.597E+03	1.289E+04	1.932E+07	1.298E+03
2079	4.637E+04	3.713E+07	2.495E+03	1.239E+04	1.857E+07	1.247E+03
2080	4.455E+04	3.567E+07	2.397E+03	1.190E+04	1.784E+07	1.198E+03
2081	4.280E+04	3.428E+07	2.303E+03	1.143E+04	1.714E+07	1.151E+03
2082	4.113E+04	3.293E+07	2.213E+03	1.099E+04	1.647E+07	1.106E+03
2083	3.951E+04	3.164E+07	2.126E+03	1.055E+04	1.582E+07	1.063E+03

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2084	3.796E+04	3.040E+07	2.043E+03	1.014E+04	1.520E+07	1.021E+03
2085	3.648E+04	2.921E+07	1.962E+03	9.743E+03	1.460E+07	9.812E+02
2086	3.505E+04	2.806E+07	1.886E+03	9.361E+03	1.403E+07	9.428E+02
2087	3.367E+04	2.696E+07	1.812E+03	8.994E+03	1.348E+07	9.058E+02
2088	3.235E+04	2.590E+07	1.741E+03	8.641E+03	1.295E+07	8.703E+02
2089	3.108E+04	2.489E+07	1.672E+03	8.302E+03	1.244E+07	8.361E+02
2090	2.986E+04	2.391E+07	1.607E+03	7.977E+03	1.196E+07	8.034E+02
2091	2.869E+04	2.298E+07	1.544E+03	7.664E+03	1.149E+07	7.719E+02
2092	2.757E+04	2.207E+07	1.483E+03	7.364E+03	1.104E+07	7.416E+02
2093	2.649E+04	2.121E+07	1.425E+03	7.075E+03	1.060E+07	7.125E+02
2094	2.545E+04	2.038E+07	1.369E+03	6.797E+03	1.019E+07	6.846E+02
2095	2.445E+04	1.958E+07	1.315E+03	6.531E+03	9.789E+06	6.577E+02
2096	2.349E+04	1.881E+07	1.264E+03	6.275E+03	9.405E+06	6.319E+02
2097	2.257E+04	1.807E+07	1.214E+03	6.029E+03	9.037E+06	6.072E+02
2098	2.169E+04	1.736E+07	1.167E+03	5.792E+03	8.682E+06	5.834E+02
2099	2.084E+04	1.668E+07	1.121E+03	5.565E+03	8.342E+06	5.605E+02
2100	2.002E+04	1.603E+07	1.077E+03	5.347E+03	8.015E+06	5.385E+02
2101	1.923E+04	1.540E+07	1.035E+03	5.137E+03	7.700E+06	5.174E+02
2102	1.848E+04	1.480E+07	9.942E+02	4.936E+03	7.399E+06	4.971E+02
2103	1.775E+04	1.422E+07	9.552E+02	4.742E+03	7.108E+06	4.776E+02
2104	1.706E+04	1.366E+07	9.178E+02	4.556E+03	6.830E+06	4.589E+02
2105	1.639E+04	1.312E+07	8.818E+02	4.378E+03	6.562E+06	4.409E+02
2106	1.575E+04	1.261E+07	8.472E+02	4.206E+03	6.305E+06	4.236E+02
2107	1.513E+04	1.211E+07	8.140E+02	4.041E+03	6.057E+06	4.070E+02
2108	1.454E+04	1.164E+07	7.821E+02	3.883E+03	5.820E+06	3.910E+02
2109	1.397E+04	1.118E+07	7.514E+02	3.731E+03	5.592E+06	3.757E+02
2110	1.342E+04	1.074E+07	7.219E+02	3.584E+03	5.372E+06	3.610E+02
2111	1.289E+04	1.032E+07	6.936E+02	3.444E+03	5.162E+06	3.468E+02
2112	1.239E+04	9.919E+06	6.664E+02	3.309E+03	4.959E+06	3.332E+02
2113	1.190E+04	9.530E+06	6.403E+02	3.179E+03	4.765E+06	3.202E+02
2114	1.143E+04	9.156E+06	6.152E+02	3.054E+03	4.578E+06	3.076E+02
2115	1.099E+04	8.797E+06	5.911E+02	2.935E+03	4.399E+06	2.955E+02
2116	1.056E+04	8.452E+06	5.679E+02	2.819E+03	4.226E+06	2.840E+02
2117	1.014E+04	8.121E+06	5.456E+02	2.709E+03	4.060E+06	2.728E+02
2118	9.744E+03	7.802E+06	5.242E+02	2.603E+03	3.901E+06	2.621E+02
2119	9.362E+03	7.496E+06	5.037E+02	2.501E+03	3.748E+06	2.518E+02
2120	8.995E+03	7.203E+06	4.839E+02	2.403E+03	3.601E+06	2.420E+02
2121	8.642E+03	6.920E+06	4.650E+02	2.308E+03	3.460E+06	2.325E+02
2122	8.303E+03	6.649E+06	4.467E+02	2.218E+03	3.324E+06	2.234E+02
2123	7.978E+03	6.388E+06	4.292E+02	2.131E+03	3.194E+06	2.146E+02
2124	7.665E+03	6.138E+06	4.124E+02	2.047E+03	3.069E+06	2.062E+02
2125	7.364E+03	5.897E+06	3.962E+02	1.967E+03	2.948E+06	1.981E+02
2126	7.075E+03	5.666E+06	3.807E+02	1.890E+03	2.833E+06	1.903E+02
2127	6.798E+03	5.444E+06	3.658E+02	1.816E+03	2.722E+06	1.829E+02
2128	6.531E+03	5.230E+06	3.514E+02	1.745E+03	2.615E+06	1.757E+02
2129	6.275E+03	5.025E+06	3.376E+02	1.676E+03	2.513E+06	1.688E+02
2130	6.029E+03	4.828E+06	3.244E+02	1.610E+03	2.414E+06	1.622E+02
2131	5.793E+03	4.639E+06	3.117E+02	1.547E+03	2.319E+06	1.558E+02
2132	5.566E+03	4.457E+06	2.995E+02	1.487E+03	2.228E+06	1.497E+02
2133	5.348E+03	4.282E+06	2.877E+02	1.428E+03	2.141E+06	1.439E+02
2134	5.138E+03	4.114E+06	2.764E+02	1.372E+03	2.057E+06	1.382E+02

Results (Continued)

Year	Total landfill gas			Methane		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2135	4.936E+03	3.953E+06	2.656E+02	1.319E+03	1.976E+06	1.328E+02
2136	4.743E+03	3.798E+06	2.552E+02	1.267E+03	1.899E+06	1.276E+02
2137	4.557E+03	3.649E+06	2.452E+02	1.217E+03	1.824E+06	1.226E+02
2138	4.378E+03	3.506E+06	2.356E+02	1.169E+03	1.753E+06	1.178E+02
2139	4.207E+03	3.368E+06	2.263E+02	1.124E+03	1.684E+06	1.132E+02
2140	4.042E+03	3.236E+06	2.174E+02	1.080E+03	1.618E+06	1.087E+02
2141	3.883E+03	3.109E+06	2.089E+02	1.037E+03	1.555E+06	1.045E+02
2142	3.731E+03	2.987E+06	2.007E+02	9.965E+02	1.494E+06	1.004E+02
2143	3.585E+03	2.870E+06	1.929E+02	9.575E+02	1.435E+06	9.643E+01
2144	3.444E+03	2.758E+06	1.853E+02	9.199E+02	1.379E+06	9.265E+01
2145	3.309E+03	2.650E+06	1.780E+02	8.839E+02	1.325E+06	8.902E+01
2146	3.179E+03	2.546E+06	1.710E+02	8.492E+02	1.273E+06	8.552E+01
2147	3.055E+03	2.446E+06	1.643E+02	8.159E+02	1.223E+06	8.217E+01
2148	2.935E+03	2.350E+06	1.579E+02	7.839E+02	1.175E+06	7.895E+01
2149	2.820E+03	2.258E+06	1.517E+02	7.532E+02	1.129E+06	7.585E+01
2150	2.709E+03	2.169E+06	1.458E+02	7.236E+02	1.085E+06	7.288E+01
2151	2.603E+03	2.084E+06	1.400E+02	6.953E+02	1.042E+06	7.002E+01
2152	2.501E+03	2.003E+06	1.346E+02	6.680E+02	1.001E+06	6.728E+01
2153	2.403E+03	1.924E+06	1.293E+02	6.418E+02	9.620E+05	6.464E+01
2154	2.309E+03	1.849E+06	1.242E+02	6.166E+02	9.243E+05	6.210E+01
2155	2.218E+03	1.776E+06	1.193E+02	5.925E+02	8.881E+05	5.967E+01
2156	2.131E+03	1.706E+06	1.147E+02	5.692E+02	8.532E+05	5.733E+01
2157	2.048E+03	1.640E+06	1.102E+02	5.469E+02	8.198E+05	5.508E+01
2158	1.967E+03	1.575E+06	1.058E+02	5.255E+02	7.876E+05	5.292E+01
2159	1.890E+03	1.514E+06	1.017E+02	5.049E+02	7.568E+05	5.085E+01
2160	1.816E+03	1.454E+06	9.771E+01	4.851E+02	7.271E+05	4.885E+01
2161	1.745E+03	1.397E+06	9.387E+01	4.661E+02	6.986E+05	4.694E+01
2162	1.676E+03	1.342E+06	9.019E+01	4.478E+02	6.712E+05	4.510E+01
2163	1.611E+03	1.290E+06	8.666E+01	4.302E+02	6.449E+05	4.333E+01
2164	1.547E+03	1.239E+06	8.326E+01	4.134E+02	6.196E+05	4.163E+01
2165	1.487E+03	1.191E+06	7.999E+01	3.971E+02	5.953E+05	4.000E+01
2166	1.429E+03	1.144E+06	7.686E+01	3.816E+02	5.719E+05	3.843E+01
2167	1.373E+03	1.099E+06	7.384E+01	3.666E+02	5.495E+05	3.692E+01
2168	1.319E+03	1.056E+06	7.095E+01	3.522E+02	5.280E+05	3.547E+01
2169	1.267E+03	1.015E+06	6.817E+01	3.384E+02	5.073E+05	3.408E+01
2170	1.217E+03	9.748E+05	6.549E+01	3.252E+02	4.874E+05	3.275E+01
2171	1.170E+03	9.365E+05	6.293E+01	3.124E+02	4.683E+05	3.146E+01
2172	1.124E+03	8.998E+05	6.046E+01	3.002E+02	4.499E+05	3.023E+01
2173	1.080E+03	8.645E+05	5.809E+01	2.884E+02	4.323E+05	2.904E+01
2174	1.037E+03	8.306E+05	5.581E+01	2.771E+02	4.153E+05	2.791E+01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2034	0	0	0	0	0	0
2035	2.039E+03	1.114E+06	7.485E+01	2.827E+00	7.887E+02	5.300E-02
2036	3.999E+03	2.184E+06	1.468E+02	5.544E+00	1.547E+03	1.039E-01
2037	5.881E+03	3.213E+06	2.159E+02	8.153E+00	2.275E+03	1.528E-01
2038	7.690E+03	4.201E+06	2.823E+02	1.066E+01	2.974E+03	1.998E-01
2039	9.427E+03	5.150E+06	3.460E+02	1.307E+01	3.646E+03	2.450E-01
2040	1.110E+04	6.062E+06	4.073E+02	1.538E+01	4.292E+03	2.884E-01
2041	1.270E+04	6.939E+06	4.662E+02	1.761E+01	4.913E+03	3.301E-01
2042	1.424E+04	7.781E+06	5.228E+02	1.975E+01	5.509E+03	3.701E-01
2043	1.572E+04	8.590E+06	5.771E+02	2.180E+01	6.081E+03	4.086E-01
2044	1.715E+04	9.367E+06	6.294E+02	2.377E+01	6.632E+03	4.456E-01
2045	1.851E+04	1.011E+07	6.795E+02	2.567E+01	7.160E+03	4.811E-01
2046	1.983E+04	1.083E+07	7.277E+02	2.749E+01	7.668E+03	5.152E-01
2047	2.109E+04	1.152E+07	7.741E+02	2.924E+01	8.156E+03	5.480E-01
2048	2.230E+04	1.218E+07	8.186E+02	3.092E+01	8.625E+03	5.795E-01
2049	2.347E+04	1.282E+07	8.613E+02	3.253E+01	9.076E+03	6.098E-01
2050	2.458E+04	1.343E+07	9.024E+02	3.408E+01	9.509E+03	6.389E-01
2051	2.566E+04	1.402E+07	9.419E+02	3.557E+01	9.925E+03	6.668E-01
2052	2.669E+04	1.458E+07	9.798E+02	3.701E+01	1.032E+04	6.937E-01
2053	2.769E+04	1.512E+07	1.016E+03	3.838E+01	1.071E+04	7.195E-01
2054	2.864E+04	1.565E+07	1.051E+03	3.971E+01	1.108E+04	7.443E-01
2055	2.956E+04	1.615E+07	1.085E+03	4.098E+01	1.143E+04	7.681E-01
2056	3.044E+04	1.663E+07	1.117E+03	4.220E+01	1.177E+04	7.910E-01
2057	3.128E+04	1.709E+07	1.148E+03	4.337E+01	1.210E+04	8.129E-01
2058	3.209E+04	1.753E+07	1.178E+03	4.450E+01	1.241E+04	8.341E-01
2059	3.288E+04	1.796E+07	1.207E+03	4.558E+01	1.272E+04	8.543E-01
2060	3.363E+04	1.837E+07	1.234E+03	4.662E+01	1.301E+04	8.738E-01
2061	3.435E+04	1.876E+07	1.261E+03	4.762E+01	1.328E+04	8.926E-01
2062	3.504E+04	1.914E+07	1.286E+03	4.858E+01	1.355E+04	9.106E-01
2063	3.570E+04	1.951E+07	1.311E+03	4.950E+01	1.381E+04	9.279E-01
2064	3.634E+04	1.985E+07	1.334E+03	5.039E+01	1.406E+04	9.445E-01
2065	3.696E+04	2.019E+07	1.357E+03	5.124E+01	1.429E+04	9.604E-01
2066	3.755E+04	2.051E+07	1.378E+03	5.206E+01	1.452E+04	9.758E-01
2067	3.811E+04	2.082E+07	1.399E+03	5.284E+01	1.474E+04	9.905E-01
2068	3.866E+04	2.112E+07	1.419E+03	5.360E+01	1.495E+04	1.005E+00
2069	3.918E+04	2.141E+07	1.438E+03	5.432E+01	1.516E+04	1.018E+00
2070	3.969E+04	2.168E+07	1.457E+03	5.502E+01	1.535E+04	1.031E+00
2071	4.017E+04	2.194E+07	1.474E+03	5.569E+01	1.554E+04	1.044E+00
2072	4.063E+04	2.220E+07	1.491E+03	5.633E+01	1.572E+04	1.056E+00
2073	4.108E+04	2.244E+07	1.508E+03	5.695E+01	1.589E+04	1.068E+00
2074	4.151E+04	2.268E+07	1.524E+03	5.755E+01	1.605E+04	1.079E+00
2075	3.988E+04	2.179E+07	1.464E+03	5.529E+01	1.542E+04	1.036E+00
2076	3.832E+04	2.093E+07	1.406E+03	5.312E+01	1.482E+04	9.958E-01
2077	3.681E+04	2.011E+07	1.351E+03	5.104E+01	1.424E+04	9.567E-01
2078	3.537E+04	1.932E+07	1.298E+03	4.904E+01	1.368E+04	9.192E-01
2079	3.398E+04	1.857E+07	1.247E+03	4.711E+01	1.314E+04	8.832E-01
2080	3.265E+04	1.784E+07	1.198E+03	4.527E+01	1.263E+04	8.485E-01
2081	3.137E+04	1.714E+07	1.151E+03	4.349E+01	1.213E+04	8.153E-01
2082	3.014E+04	1.647E+07	1.106E+03	4.179E+01	1.166E+04	7.833E-01
2083	2.896E+04	1.582E+07	1.063E+03	4.015E+01	1.120E+04	7.526E-01

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2084	2.782E+04	1.520E+07	1.021E+03	3.857E+01	1.076E+04	7.231E-01
2085	2.673E+04	1.460E+07	9.812E+02	3.706E+01	1.034E+04	6.947E-01
2086	2.568E+04	1.403E+07	9.428E+02	3.561E+01	9.934E+03	6.675E-01
2087	2.468E+04	1.348E+07	9.058E+02	3.421E+01	9.545E+03	6.413E-01
2088	2.371E+04	1.295E+07	8.703E+02	3.287E+01	9.170E+03	6.162E-01
2089	2.278E+04	1.244E+07	8.361E+02	3.158E+01	8.811E+03	5.920E-01
2090	2.189E+04	1.196E+07	8.034E+02	3.034E+01	8.465E+03	5.688E-01
2091	2.103E+04	1.149E+07	7.719E+02	2.915E+01	8.133E+03	5.465E-01
2092	2.020E+04	1.104E+07	7.416E+02	2.801E+01	7.814E+03	5.251E-01
2093	1.941E+04	1.060E+07	7.125E+02	2.691E+01	7.508E+03	5.045E-01
2094	1.865E+04	1.019E+07	6.846E+02	2.586E+01	7.214E+03	4.847E-01
2095	1.792E+04	9.789E+06	6.577E+02	2.484E+01	6.931E+03	4.657E-01
2096	1.722E+04	9.405E+06	6.319E+02	2.387E+01	6.659E+03	4.474E-01
2097	1.654E+04	9.037E+06	6.072E+02	2.293E+01	6.398E+03	4.299E-01
2098	1.589E+04	8.682E+06	5.834E+02	2.203E+01	6.147E+03	4.130E-01
2099	1.527E+04	8.342E+06	5.605E+02	2.117E+01	5.906E+03	3.968E-01
2100	1.467E+04	8.015E+06	5.385E+02	2.034E+01	5.674E+03	3.813E-01
2101	1.410E+04	7.700E+06	5.174E+02	1.954E+01	5.452E+03	3.663E-01
2102	1.354E+04	7.399E+06	4.971E+02	1.878E+01	5.238E+03	3.520E-01
2103	1.301E+04	7.108E+06	4.776E+02	1.804E+01	5.033E+03	3.382E-01
2104	1.250E+04	6.830E+06	4.589E+02	1.733E+01	4.835E+03	3.249E-01
2105	1.201E+04	6.562E+06	4.409E+02	1.665E+01	4.646E+03	3.122E-01
2106	1.154E+04	6.305E+06	4.236E+02	1.600E+01	4.464E+03	2.999E-01
2107	1.109E+04	6.057E+06	4.070E+02	1.537E+01	4.289E+03	2.882E-01
2108	1.065E+04	5.820E+06	3.910E+02	1.477E+01	4.120E+03	2.769E-01
2109	1.024E+04	5.592E+06	3.757E+02	1.419E+01	3.959E+03	2.660E-01
2110	9.834E+03	5.372E+06	3.610E+02	1.363E+01	3.804E+03	2.556E-01
2111	9.449E+03	5.162E+06	3.468E+02	1.310E+01	3.655E+03	2.455E-01
2112	9.078E+03	4.959E+06	3.332E+02	1.259E+01	3.511E+03	2.359E-01
2113	8.722E+03	4.765E+06	3.202E+02	1.209E+01	3.374E+03	2.267E-01
2114	8.380E+03	4.578E+06	3.076E+02	1.162E+01	3.241E+03	2.178E-01
2115	8.052E+03	4.399E+06	2.955E+02	1.116E+01	3.114E+03	2.092E-01
2116	7.736E+03	4.226E+06	2.840E+02	1.073E+01	2.992E+03	2.010E-01
2117	7.433E+03	4.060E+06	2.728E+02	1.030E+01	2.875E+03	1.932E-01
2118	7.141E+03	3.901E+06	2.621E+02	9.900E+00	2.762E+03	1.856E-01
2119	6.861E+03	3.748E+06	2.518E+02	9.512E+00	2.654E+03	1.783E-01
2120	6.592E+03	3.601E+06	2.420E+02	9.139E+00	2.550E+03	1.713E-01
2121	6.334E+03	3.460E+06	2.325E+02	8.781E+00	2.450E+03	1.646E-01
2122	6.085E+03	3.324E+06	2.234E+02	8.437E+00	2.354E+03	1.581E-01
2123	5.847E+03	3.194E+06	2.146E+02	8.106E+00	2.261E+03	1.519E-01
2124	5.617E+03	3.069E+06	2.062E+02	7.788E+00	2.173E+03	1.460E-01
2125	5.397E+03	2.948E+06	1.981E+02	7.483E+00	2.088E+03	1.403E-01
2126	5.186E+03	2.833E+06	1.903E+02	7.189E+00	2.006E+03	1.348E-01
2127	4.982E+03	2.722E+06	1.829E+02	6.907E+00	1.927E+03	1.295E-01
2128	4.787E+03	2.615E+06	1.757E+02	6.636E+00	1.851E+03	1.244E-01
2129	4.599E+03	2.513E+06	1.688E+02	6.376E+00	1.779E+03	1.195E-01
2130	4.419E+03	2.414E+06	1.622E+02	6.126E+00	1.709E+03	1.148E-01
2131	4.246E+03	2.319E+06	1.558E+02	5.886E+00	1.642E+03	1.103E-01
2132	4.079E+03	2.228E+06	1.497E+02	5.655E+00	1.578E+03	1.060E-01
2133	3.919E+03	2.141E+06	1.439E+02	5.434E+00	1.516E+03	1.018E-01
2134	3.765E+03	2.057E+06	1.382E+02	5.220E+00	1.456E+03	9.786E-02

Results (Continued)

Year	Carbon dioxide			NMOC		
	(Mg/year)	(m ³ /year)	(av ft ³ /min)	(Mg/year)	(m ³ /year)	(av ft ³ /min)
2135	3.618E+03	1.976E+06	1.328E+02	5.016E+00	1.399E+03	9.402E-02
2136	3.476E+03	1.899E+06	1.276E+02	4.819E+00	1.344E+03	9.033E-02
2137	3.340E+03	1.824E+06	1.226E+02	4.630E+00	1.292E+03	8.679E-02
2138	3.209E+03	1.753E+06	1.178E+02	4.449E+00	1.241E+03	8.339E-02
2139	3.083E+03	1.684E+06	1.132E+02	4.274E+00	1.192E+03	8.012E-02
2140	2.962E+03	1.618E+06	1.087E+02	4.107E+00	1.146E+03	7.698E-02
2141	2.846E+03	1.555E+06	1.045E+02	3.946E+00	1.101E+03	7.396E-02
2142	2.734E+03	1.494E+06	1.004E+02	3.791E+00	1.058E+03	7.106E-02
2143	2.627E+03	1.435E+06	9.643E+01	3.642E+00	1.016E+03	6.827E-02
2144	2.524E+03	1.379E+06	9.265E+01	3.499E+00	9.763E+02	6.559E-02
2145	2.425E+03	1.325E+06	8.902E+01	3.362E+00	9.380E+02	6.302E-02
2146	2.330E+03	1.273E+06	8.552E+01	3.230E+00	9.012E+02	6.055E-02
2147	2.239E+03	1.223E+06	8.217E+01	3.104E+00	8.659E+02	5.818E-02
2148	2.151E+03	1.175E+06	7.895E+01	2.982E+00	8.319E+02	5.590E-02
2149	2.067E+03	1.129E+06	7.585E+01	2.865E+00	7.993E+02	5.370E-02
2150	1.986E+03	1.085E+06	7.288E+01	2.753E+00	7.680E+02	5.160E-02
2151	1.908E+03	1.042E+06	7.002E+01	2.645E+00	7.378E+02	4.958E-02
2152	1.833E+03	1.001E+06	6.728E+01	2.541E+00	7.089E+02	4.763E-02
2153	1.761E+03	9.620E+05	6.464E+01	2.441E+00	6.811E+02	4.576E-02
2154	1.692E+03	9.243E+05	6.210E+01	2.346E+00	6.544E+02	4.397E-02
2155	1.626E+03	8.881E+05	5.967E+01	2.254E+00	6.287E+02	4.225E-02
2156	1.562E+03	8.532E+05	5.733E+01	2.165E+00	6.041E+02	4.059E-02
2157	1.501E+03	8.198E+05	5.508E+01	2.080E+00	5.804E+02	3.900E-02
2158	1.442E+03	7.876E+05	5.292E+01	1.999E+00	5.576E+02	3.747E-02
2159	1.385E+03	7.568E+05	5.085E+01	1.920E+00	5.358E+02	3.600E-02
2160	1.331E+03	7.271E+05	4.885E+01	1.845E+00	5.148E+02	3.459E-02
2161	1.279E+03	6.986E+05	4.694E+01	1.773E+00	4.946E+02	3.323E-02
2162	1.229E+03	6.712E+05	4.510E+01	1.703E+00	4.752E+02	3.193E-02
2163	1.180E+03	6.449E+05	4.333E+01	1.637E+00	4.566E+02	3.068E-02
2164	1.134E+03	6.196E+05	4.163E+01	1.572E+00	4.387E+02	2.947E-02
2165	1.090E+03	5.953E+05	4.000E+01	1.511E+00	4.215E+02	2.832E-02
2166	1.047E+03	5.719E+05	3.843E+01	1.451E+00	4.049E+02	2.721E-02
2167	1.006E+03	5.495E+05	3.692E+01	1.395E+00	3.891E+02	2.614E-02
2168	9.664E+02	5.280E+05	3.547E+01	1.340E+00	3.738E+02	2.512E-02
2169	9.286E+02	5.073E+05	3.408E+01	1.287E+00	3.591E+02	2.413E-02
2170	8.921E+02	4.874E+05	3.275E+01	1.237E+00	3.451E+02	2.318E-02
2171	8.572E+02	4.683E+05	3.146E+01	1.188E+00	3.315E+02	2.228E-02
2172	8.236E+02	4.499E+05	3.023E+01	1.142E+00	3.185E+02	2.140E-02
2173	7.913E+02	4.323E+05	2.904E+01	1.097E+00	3.060E+02	2.056E-02
2174	7.602E+02	4.153E+05	2.791E+01	1.054E+00	2.940E+02	1.976E-02

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4241336
DESCRIPTION: Vertical Well ROI Calculation CALC NO: SHEET 1 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 3/29/2024

Required: Determine the radius of influence (ROI) for vertical LFG extraction wells, using both EMCON and NSPS methods for the average flow rate condition.

Purpose: The ROI is calculated in the design of an active gas extraction system to properly locate extraction wells. The location of the extraction well will dictate the well depth, once the refuse depth is known. A detail of each well can then be created with respect to total well depth, depth to slotted pipe, applied vacuum and rate of extraction.

Although horizontal collectors are not modeled directly by this analysis, the effected flow patterns developed by horizontal collectors are analogous to those created by vertical extraction wells. The net result is that vertical well spacing criteria can be translated directly to horizontal collectors in a similar environment.

Method: The following methods were used to estimate the theoretical ROI for an LFG extraction well.

A) EMCON Method (from *Methane Generation and Recovery from Landfills*, EMCON, 1982, pg. 81)

$$Q_w = \frac{k \pi R^2 t D r}{C}$$

where: Q_w = individual extraction well LFG flow rate [L/s]
 k = conversion factor (1.157×10^{-8}) [(L/s)/(mL/day)]
 R = radius of influence [m]
 t = perforated pipe length [m]
 D = in-place refuse density [kg/m³]
 r = methane production rate [mL/kg/day]
 C = fractional methane concentration [-]

1. Noting that the methane production rate (r) divided by the fractional methane concentration (C) is equal to the LFG production rate (G), and solving for the ROI yields:

$$R = \left(\frac{Q_w}{k \pi t D G} \right)^{1/2}$$

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4241336
DESCRIPTION: Vertical Well ROI Calculation CALC NO: _____ SHEET 2 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 3/29/2024

2. Converting from metric to English units yields the following conversion factors (allowing input in English units):

Q_w (from cfm to L/s):

$$\frac{1 \text{ ft}^3}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times \frac{28.317 \text{ L}}{\text{ft}^3} = 0.47195$$

t (from ft to m):

$$1 \text{ ft} \times \frac{0.3048 \text{ m}}{\text{ft}} = 0.3048$$

D (from lb/cy to kg/m³):

$$\frac{1 \text{ lb}}{1 \text{ cy}} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} \times \frac{1 \text{ cy}}{27 \text{ ft}^3} \times \left(\frac{1 \text{ ft}}{0.3048 \text{ m}} \right)^3 = 0.5932$$

In addition, converting G from flow per volume to flow per mass equals:

G (from cfm/cy to mL/kg/day):

$$\begin{aligned} & \frac{1 \text{ ft}^3}{\text{min yd}^3} \times \left(\frac{1}{D} \times \frac{1 \text{ yd}^3}{1 \text{ lb}} \right) \times \frac{60 \text{ min}}{1 \text{ hr}} \times \frac{24 \text{ hr}}{1 \text{ day}} \times \frac{28.317 \text{ L}}{\text{ft}^3} \times \frac{1 \text{ lb}}{0.4536 \text{ kg}} \times \frac{1000 \text{ mL}}{\text{L}} \\ &= \frac{89,895,238}{D} \end{aligned}$$

Therefore:

$$R = \left[\frac{0.47195 Q_w}{1.157 \times 10^{-8} \pi (0.3048 t) (0.5932 D) \left(\frac{89,895,238}{D} \times G \right)} \right]^{1/2}$$

Where: R is in meters

Q_w is in cfm

t is in feet

D is in lb/cy

G is in cfm/cy

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4241336
DESCRIPTION: Vertical Well ROI Calculation CALC NO: SHEET 3 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 3/29/2024

Then:

$$R = 0.8938 \left(\frac{Q_w}{t G} \right)^{1/2}$$

Converting results from meters to feet (1 ft = 0.3048 m):

$$0.3048R = 0.8938 \left(\frac{Q_w}{t G} \right)^{1/2}$$

$$R = 2.932 \left(\frac{Q_w}{t G} \right)^{1/2}$$

B) NSPS Method (from EPA *NSPS Bid*, 1991, pg. G-1)

$$R = \left(\frac{Q_w DC}{\pi L \rho Q_{gen} \eta} \right)^{1/2}$$

Where: ρ = in-place density of refuse [kg/m³]
 Q_w = LFG flow rate per well [m³/s]
DC = design capacity of landfill [kg]
L = perforated pipe length [m]
 Q_{gen} = peak LFG generation rate [m³/s]
 η = system collection efficiency [%]
R = radius of influence [m]

1. Converting from metric to English units yields the following conversion factors (allowing input in English units):

Q_w (from cfm to m³/s):

$$\frac{1 \text{ ft}^3}{1 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ sec}} \times \left(\frac{0.3048 \text{ m}}{\text{ft}} \right)^3 = 4.719 \times 10^{-4}$$

DC (from tons to kg):

$$1 \text{ ton} \times \frac{2,000 \text{ lb}}{1 \text{ ton}} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} = 907.2$$

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4241336
DESCRIPTION: Vertical Well ROI Calculation CALC NO: _____ SHEET 4 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 3/29/2024

L (from ft to m):

$$1 \text{ ft} \times \frac{0.3048 \text{ m}}{\text{ft}} = 0.3048$$

ρ (from lb/cy to kg/m³):

$$\frac{1 \text{ lb}}{\text{yd}^3} \times \frac{0.4536 \text{ kg}}{1 \text{ lb}} \times \frac{1 \text{ yd}^3}{27 \text{ ft}^3} \times \left(\frac{1 \text{ ft}}{0.3048 \text{ m}} \right)^3 = 0.5933$$

Converting results from meters to feet (1 ft = 0.3048 m):

$$0.3048R = \left[\frac{(4.719 \times 10^{-4}) Q_w (907.2) DC}{\pi (0.3048L) (0.5933\rho) (4.719 \times 10^{-4} Q_{\text{gen}}) \eta} \right]^{1/2}$$

$$R = \frac{1}{0.3048} \left[\frac{(4.719 \times 10^{-4}) Q_w (907.2) DC}{\pi (0.3048L) (0.5933\rho) (4.719 \times 10^{-4} Q_{\text{gen}}) \eta} \right]^{1/2}$$

Example Calculation:

Estimate the ROI for the following average LFG flow rate condition at the XYZ Landfill

Given:

Design Capacity	=	2.5 x 10 ⁷ tons
Collection efficiency	=	70%
Average LFG generation rate	=	2,469 scfm
In-place refuse density	=	1,400 lb/cy
Depth of well	=	100 ft
Well flow rate	=	26 scfm

COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4241336
DESCRIPTION: Vertical Well ROI Calculation CALC NO: _____ SHEET 5 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 3/29/2024

A) EMCON Method

$$R = 2.932 \left(\frac{26 \text{ scfm}}{(100 \text{ ft}) \left(\frac{2,469 \text{ scfm}}{25,000,000 \text{ tons}} \right) \left(\frac{1 \text{ ton}}{2,000 \text{ lb}} \right) \left(\frac{1,400 \text{ lb}}{1 \text{ cy}} \right)} \right)^{1/2}$$

$$\underline{\underline{R = 180 \text{ ft}}}$$

B) NSPS Method

$$R = \frac{1}{0.3048} \left[\frac{(4.719 \times 10^{-4})(26 \text{ scfm})(907.2)(25,000,000 \text{ tons})}{\pi (0.3048)(100 \text{ ft})(0.5933)(1,400 \text{ lb / cy})(4.719 \times 10^{-4})(2,469 \text{ scfm})(0.70)} \right]^{1/2}$$

$$\underline{\underline{R = 215 \text{ ft}}}$$

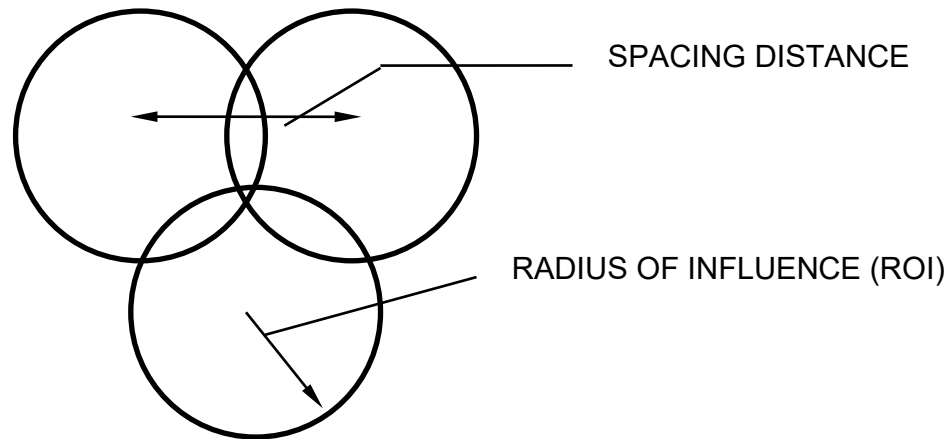
COMPUTATION SHEET

PROJECT TITLE: Wexford County Landfill PROJECT NO: 209-4241336
DESCRIPTION: Vertical Well ROI Calculation CALC NO: SHEET 6 OF 6
PREPARED BY: JPS CHECKED BY: JPS APPROVED BY: KM DATE: 3/29/2024

Well Spacing

Utilizing the more conservative (EMCON) value for ROI of $R = 180$ ft for this extraction well, and applying the criteria for well spacing of:

$$\text{SPACING} = \sqrt{3} \text{ ROI}$$



$$\underline{\text{SPACING} = 312 \text{ ft}}$$

Site Specific Calculations:

ROI calculations for the facility are shown on the following spreadsheet. These calculations were performed for a series of typical well depths, ranging from 40 feet to 100 feet and will be utilized as the basis for determining well placement across the selected area of the Project.

**Landfill Gas Extraction Well Radius of Influence
Based on Volume of Affected Refuse
Wexford County Landfill - East Hill**

Well Number ^(c)	Well Depth (ft)	LFG Flow Rate Condition	LFG Flow Rate (scfm)	Radius of Influence (ft) ^{(a)(b)}	
				NSPS	EMCON
A	40	Low	10	107	101
		Average	14	125	119
		High	20	151	143
B	60	Low	20	123	117
		Average	25	138	131
		High	30	151	143
C	80	Low	30	131	124
		Average	40	150	143
		High	50	169	160
D	100	Low	50	151	143
		Average	60	165	157
		High	70	179	170

Degradable Waste Design Tonnage	8,525,645 tons	LFG Generation Rate	2,258 scfm
Total East Hill Waste Design Capacity	10,783,807 cy	Refuse Density	1,581 lb/cy
Total Landfill Waste Design Capacity	23,260,000 cy	NSPS Collection Efficiency	90 %

ROI of Perimeter Wells =<	125 feet	Well Spacing =>	217 feet
ROI of Interior Wells =<	150 feet	Well Spacing =>	260 feet

Notes:

- a) Radius of influence based on the estimated capacity of the facility and the anticipated LFG generation rate.
b) Calculations assume 20 feet of solid well casing from ground surface to start of perforations.
c) Well number is general for the typical depth shown.

**Landfill Gas Extraction Well Radius of Influence
Based on Volume of Affected Refuse
Wexford County Landfill - West Hill**

Well Number ^(c)	Well Depth (ft)	LFG Flow Rate Condition	LFG Flow Rate (scfm)	Radius of Influence (ft) ^{(a)(b)}	
				NSPS	EMCON
A	40	Low	10	99	94
		Average	16	125	119
		High	20	140	133
B	60	Low	20	114	108
		Average	25	128	121
		High	30	140	133
C	80	Low	30	121	115
		Average	46	150	142
		High	50	156	148
D	100	Low	50	140	133
		Average	60	153	145
		High	70	165	157

Degradable Waste Design Tonnage	12,476,193 tons	LFG Generation Rate	3,047 scfm
Total East Hill Waste Design Capacity	12,476,193 cy	Refuse Density	2,000 lb/cy
Total Landfill Waste Design Capacity	23,260,000 cy	NSPS Collection Efficiency	90 %

ROI of Perimeter Wells =<	125 feet	Well Spacing =>	217 feet
ROI of Interior Wells =<	150 feet	Well Spacing =>	260 feet

Notes:

- a) Radius of influence based on the estimated capacity of the facility and the anticipated LFG generation rate.
b) Calculations assume 20 feet of solid well casing from ground surface to start of perforations.
c) Well number is general for the typical depth shown.

```

* * * * * K Y G A S * * * * *
*
* Gas Network Analysis Software (KYGAS)
*
* CopyRighted by KYPIPE LLC (www.kypipe.com)
* Version: 11.112 01-27-2023
* Company: TETRAWIN Serial #: 580297
* Interface: KYnetic
* Licensed for Pipe2022
*
* * * * *

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```

INPUT DATA FILE NAME FOR THIS SIMULATION =
c:\users\JARRET~1.DUT\DOCUME~1\WVWC
-2~2.KYP\wvwc-202.DAT
OUTPUT DATA FILE NAME FOR THIS SIMULATION =
c:\users\JARRET~1.DUT\DOCUME~1\WVWC
-2~2.KYP\wvwc-202.OT2

```

```

DATE FOR THIS COMPUTER RUN      : 2-22-2024
START TIME FOR THIS COMPUTER RUN : 16: 0:46:25

```

SUMMARY OF DISTRIBUTION SYSTEM CHARACTERISTICS:

```

NUMBER OF PIPES      = 133
NUMBER OF JUNCTION NODES = 129

UNITS SPECIFIED      = ENGLISH

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A CONSTANT DENSITY FLUID IS SPECIFIED - DENSITY = .07POUNDS/CUBIC
FOOT
ABSOLUTE VISCOSITY      = .280E-06 POUND SECONDS/SQUARE
FOOT

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USER SPEC. FLOW UNITS (USFU)      = SCF/MIN
USER SPEC. PRESSURE UNITS(USPU)   = INCHES OF WATER (GAUGE)

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----- SIMULATION DESCRIPTION -----

----- SUMMARY OF PIPE NETWORK GEOMETRIC AND OPERATING DATA -----

PIPE NAME	NODE #1	NODE #2	LENGTH (FT.)	DIAM. (IN.)	ROUGHNESS (MILLIFEET)	SUM-M FACT.	PUMP ID	ELEVATION CHANGE
--------------	------------	------------	-----------------	----------------	--------------------------	----------------	------------	---------------------

1	2	J-1	3.0	7.5	.100	1.8	0	.0
2	3	4	60.2	5.8	.100	1.8	0	.0
3	3	5	62.5	5.8	.100	1.8	0	.0
4	2	6	248.2	7.5	.100	1.8	0	.0
5	7	J-56	157.4	5.8	.100	1.8	0	.0
7	10	26	15.6	5.8	.100	1.8	0	.0
8	13	J-43	24.8	5.8	.100	1.8	0	.0
9	15	J-88	13.8	5.8	.100	1.8	0	.0
10	J-88	98	78.6	15.8	.100	1.8	0	.0
12	20	21	.5	3.9	.100	1.8	0	.0
13	20	J-19	10.7	5.8	.100	1.8	0	.0
15	J-19	26	111.4	15.8	.100	1.8	0	.0
17	J-45	J-40	177.2	15.8	.100	1.8	0	.0
18	J-29	51	225.3	5.8	.100	3.6	0	.0
19	31	J-29	63.5	5.8	.100	1.8	0	.0
20	J-30	J-32	80.9	15.8	.100	1.8	0	.0
22	J-34	J-35	157.1	7.5	.100	1.8	0	.0
23	37	J-32	94.5	5.8	.100	1.8	0	.0
24	39	J-40	71.2	5.8	.100	1.8	0	.0
25	9	J-8	16.2	5.8	.100	1.8	0	.0
26	42	J-41	119.4	5.8	.100	1.8	0	.0
27	44	J-41	27.1	5.8	.100	1.8	0	.0
28	46	J-38	40.7	5.8	.100	1.8	0	.0
29	48	J-34	41.2	5.8	.100	1.8	0	.0
31	52	J-36	127.4	5.8	.100	1.8	0	.0
34	55	J-27	25.8	5.8	.100	1.8	0	.0
36	J-60	33	15.9	15.8	.100	1.8	0	.0
38	61	62	12.8	5.8	.100	1.8	0	.0
41	J-67	J-93	190.2	11.2	.100	1.8	0	.0
42	69	70	18.1	5.8	.100	1.8	0	.0
44	73	51	148.5	5.8	.100	1.8	0	.0
45	J-77	J-92	3.9	15.8	.100	1.8	0	.0
72	69	J-14	29.1	15.8	.100	1.8	0	.0
80	31	106	116.1	5.8	.100	1.8	0	.0
82	109	62	176.5	5.8	.100	1.8	0	.0
83	110	J-64	157.2	5.8	.100	1.8	0	.0
84	112	J-89	162.5	5.8	.100	1.8	0	.0
85	113	J-66	175.2	5.8	.100	1.8	0	.0
87	115	76	182.2	5.8	.100	1.8	0	.0
88	116	J-75	177.6	5.8	.100	1.8	0	.0
90	120	J-142	161.7	5.8	.100	1.8	0	.0
93	126	J-143	23.3	5.8	.100	1.8	0	.0
94	128	J-145	28.2	5.8	.100	1.8	0	.0
96	132	J-149	182.0	5.8	.100	1.8	0	.0
97	134	J-148	116.4	5.8	.100	1.8	0	.0
100	J-100	J-133	86.8	15.8	.100	1.8	0	.0
P-101	J-140	9	7.8	5.8	.100	1.8	0	.0
P-102	J-90	J-64	215.0	5.8	.100	1.8	0	.0
P-103	J-66	J-68	123.9	5.8	.100	1.8	0	.0
P-104	J-67	J-92	57.0	11.2	.100	1.8	0	.0
P-105	J-93	J-94	185.6	11.2	.100	1.8	0	.0
P-106	J-94	J-95	168.3	11.2	.100	1.8	0	.0
P-107	J-95	J-97	166.8	11.2	.100	1.8	0	.0

P-108	J-96	J-99	170.4	11.2	.100	1.8	0	.0
P-109	108	J-97	220.0	5.8	.100	1.8	0	.0
P-11	J-18	J-47	20.0	15.8	.100	1.8	0	.0
P-111	J-100	J-99	4.0	11.2	.100	1.8	0	.0
P-112	J-77	76	44.0	5.8	.100	1.8	0	.0
P-113	J-103	J-77	92.9	15.8	.100	1.8	0	.0
P-114	J-102	J-101	184.1	11.2	.100	1.8	0	.0
P-115	J-103	117	32.0	11.2	.100	1.8	0	.0
P-116	117	J-101	184.0	11.2	.100	1.8	0	.0
P-117	J-107	J-146	38.1	11.2	.100	1.8	0	.0
P-118	J-111	J-114	73.1	11.2	.100	1.8	0	.0
P-119	J-114	J-107	187.0	11.2	.100	1.8	0	.0
P-120	J-119	J-111	178.5	11.2	.100	1.8	0	.0
P-121	J-130	J-119	182.7	11.2	.100	1.8	0	.0
P-122	J-141	J-130	179.9	11.2	.100	1.8	0	.0
P-123	J-118	J-141	37.0	11.2	.100	1.8	0	.0
P-124	J-121	J-103	94.4	15.8	.100	1.8	0	.0
P-125	J-121	J-142	18.0	5.8	.100	1.8	0	.0
P-126	J-123	122	3.0	5.8	.100	1.8	0	.0
P-127	J-125	124	42.0	5.8	.100	1.8	0	.0
P-128	J-127	J-125	151.2	15.8	.100	1.8	0	.0
P-129	J-127	J-143	64.0	5.8	.100	1.8	0	.0
P-130	J-129	J-127	99.4	15.8	.100	1.8	0	.0
P-131	J-145	J-144	26.6	5.8	.100	1.8	0	.0
P-132	J-129	J-144	68.0	5.8	.100	1.8	0	.0
P-133	J-146	J-147	39.6	11.2	.100	1.8	0	.0
P-134	J-131	J-129	95.7	15.8	.100	1.8	0	.0
P-135	J-147	J-131	74.0	11.2	.100	1.8	0	.0
P-136	J-137	136	178.0	5.8	.100	1.8	0	.0
P-137	J-135	J-137	92.5	15.8	.100	1.8	0	.0
P-138	J-135	J-148	174.0	5.8	.100	1.8	0	.0
P-139	J-139	138	38.0	5.8	.100	1.8	0	.0
P-14	J-19	J-18	184.5	15.8	.100	1.8	0	.0
P-140	J-149	J-150	177.8	5.8	.100	1.8	0	.0
P-141	J-150	J-151	174.3	5.8	.100	1.8	0	.0
P-142	J-133	J-118	88.1	15.8	.100	1.8	0	.0
P-143	J-133	J-151	20.0	7.5	.100	1.8	0	.0
P-21	J-32	J-28	35.6	15.8	.100	1.8	0	.0
P-30	58	J-30	58.4	5.8	.100	1.8	0	.0
P-32	J-28	J-47	60.5	15.8	.100	1.8	0	.0
P-33	J-40	J-27	86.8	15.8	.100	1.8	0	.0
P-35	J-30	33	17.5	15.8	.100	1.8	0	.0
P-37	J-29	33	163.7	5.8	.100	1.8	0	.0
P-39	J-90	J-91	173.3	7.5	.100	1.8	0	.0
P-40	J-65	J-92	37.0	15.8	.100	1.8	0	.0
P-43	R-1	J-18	194.4	15.8	.100	1.8	0	.0
P-46	J-75	117	187.0	5.8	.100	1.8	0	.0
P-47	J-60	J-100	45.7	15.8	.100	1.8	0	.0
P-48	J-34	J-28	10.7	7.5	.100	1.8	0	.0
P-49	J-35	J-38	40.3	7.5	.100	1.8	0	.0
P-50	J-36	J-35	90.2	5.8	.100	1.8	0	.0
P-51	J-38	61	163.5	7.5	.100	1.8	0	.0
P-52	J-47	72	170.0	5.8	.100	1.8	0	.0
P-6	J-8	26	157.1	15.8	.100	1.8	0	.0

P-61	J-45	J-8	275.1	15.8	.100	1.8	0	.0
P-62	J-41	J-45	35.1	5.8	.100	1.8	0	.0
P-63	J-27	J-43	94.3	15.8	.100	1.8	0	.0
P-64	J-1	J-17	81.1	7.5	.100	1.8	0	.0
P-65	J-17	J-43	62.5	15.8	.100	1.8	0	.0
P-66	J-1	3	75.0	5.8	.100	1.8	0	.0
P-67	J-12	2	118.3	5.8	.100	1.8	0	.0
P-68	J-54	J-12	142.7	5.8	.100	1.8	0	.0
P-69	J-56	J-54	160.9	5.8	.100	1.8	0	.0
P-70	J-88	J-17	11.3	15.8	.100	1.8	0	.0
P-71	J-16	J-65	31.6	15.8	.100	1.8	0	.0
P-73	J-14	J-16	4.6	15.8	.100	1.8	0	.0
P-74	J-14	98	28.0	11.2	.100	1.8	0	.0
P-75	J-89	J-90	177.3	5.8	.100	1.8	0	.0
P-76	J-16	J-91	60.3	7.5	.100	1.8	0	.0
P-77	J-65	J-63	92.0	7.5	.100	1.8	0	.0
P-79	J-60	104	15.0	5.8	.100	1.8	0	.0
P-81	J-97	J-96	192.0	11.2	.100	1.8	0	.0
P-86	J-68	J-63	51.1	7.5	.100	1.8	0	.0
P-89	J-118	J-139	109.8	15.8	.100	1.8	0	.0
P-91	J-123	J-121	92.0	15.8	.100	1.8	0	.0
P-92	J-125	J-123	90.2	15.8	.100	1.8	0	.0
P-95	J-114	J-102	107.8	11.2	.100	1.8	0	.0
P-98	J-137	J-131	108.8	15.8	.100	1.8	0	.0
P-99	J-139	J-135	86.0	15.8	.100	1.8	0	.0

JUNCTION NAME	NODE TITLE	ELEV	DEMAND (USFU)	FPN PRESSURE
2		.00	.00	
3		.00	.00	
4		.00	-27.54	
5		.00	-27.54	
6		.00	-27.54	
7		.00	-27.54	
9		.00	-27.54	
10		.00	-27.54	
13		.00	-27.54	
15		.00	-27.54	
20		.00	.00	
21		.00	-27.54	
26		.00	.00	
31		.00	-27.54	
33		.00	.00	
37		.00	-27.54	
39		.00	-27.54	
42		.00	-27.54	
44		.00	-27.54	
46		.00	-27.54	
48		.00	-27.54	
51		.00	-27.54	
52		.00	-27.54	
55		.00	-27.54	

58	.00	-27.54
61	.00	.00
62	.00	-27.54
69	.00	.00
70	.00	-27.54
72	.00	-27.54
73	.00	-27.54
76	.00	-27.54
98	.00	.00
104	.00	-27.54
106	.00	-27.54
108	.00	-27.54
109	.00	-27.54
110	.00	-27.54
112	.00	-27.54
113	.00	-27.54
115	.00	-27.54
116	.00	-27.54
117	.00	-27.54
120	.00	-27.54
122	.00	-27.54
124	.00	-27.54
126	.00	-27.54
128	.00	-27.54
132	.00	-27.54
134	.00	-27.54
136	.00	-27.54
138	.00	-27.54
J-1	.00	.00
J-100	.00	.00
J-101	.00	-27.54
J-102	.00	-27.54
J-103	.00	.00
J-107	.00	-27.54
J-111	.00	-27.54
J-114	.00	.00
J-118	.00	.00
J-119	.00	-27.54
J-12	.00	-27.54
J-121	.00	.00
J-123	.00	.00
J-125	.00	.00
J-127	.00	.00
J-129	.00	.00
J-130	.00	-27.54
J-131	.00	.00
J-133	.00	.00
J-135	.00	-27.54
J-137	.00	-27.54
J-139	.00	.00
J-14	.00	.00
J-140	.00	-27.54
J-141	.00	-27.54
J-142	.00	-27.54

J-143	.00	-27.54	
J-144	.00	-27.54	
J-145	.00	-27.54	
J-146	.00	-27.54	
J-147	.00	-27.54	
J-148	.00	-27.54	
J-149	.00	-27.54	
J-150	.00	-27.54	
J-151	.00	-27.54	
J-16	.00	.00	
J-17	.00	.00	
J-18	.00	.00	
J-19	.00	.00	
J-27	.00	.00	
J-28	.00	.00	
J-29	.00	.00	
J-30	.00	.00	
J-32	.00	.00	
J-34	.00	.00	
J-35	.00	.00	
J-36	.00	-27.54	
J-38	.00	.00	
J-40	.00	.00	
J-41	.00	.00	
J-43	.00	.00	
J-45	.00	.00	
J-47	.00	.00	
J-54	.00	-27.54	
J-56	.00	-27.54	
J-60	.00	.00	
J-63	.00	-27.54	
J-64	.00	-27.54	
J-65	.00	.00	
J-66	.00	-27.54	
J-67	.00	-27.54	
J-68	.00	.00	
J-75	.00	-27.54	
J-77	.00	.00	
J-8	.00	.00	
J-88	.00	.00	
J-89	.00	-27.54	
J-90	.00	-27.54	
J-91	.00	-27.54	
J-92	.00	.00	
J-93	.00	-27.54	
J-94	.00	-27.54	
J-95	.00	-27.54	
J-96	.00	-27.54	
J-97	.00	-27.54	
J-99	.00	-27.54	
R-1	.00	.00	-60.00

=====

=====

Set = 0

===== RESULTS FOR THIS SIMULATION FOLLOW
=====

Solution was obtained in 17 trials
Flow Accuracy = .1692E-02[< .500E-02]
RV Accuracy = .0000E+00[< .100E-02]

PIPE	NODE	NODE	FLOW	LOSS	VELOCITY	DENSITY	
FRICTION AREA							
NO.	#1	#2	(USFU)	(USPU)	(FT/S)	(#/CF)	FACTOR
RATIO							

.0225	1	2	J-1	137.700	.03	7.95	.0710
.0311	2	3	4	-27.540	.01	2.70	.0710
.0311	3	3	5	-27.540	.01	2.70	.0710
.0333	4	2	6	-27.540	.01	1.59	.0710
.0311	5	7	J-56	27.540	.02	2.70	.0710
.0311	7	10	26	27.540	.00	2.70	.0710
.0311	8	13	J-43	27.540	.01	2.70	.0710
.0311	9	15	J-88	27.540	.00	2.70	.0710
.0200	10	J-88	98	-469.692	.02	6.23	.0710
.0284	12	20	21	-27.540	.01	5.84	.0710
.0311	13	20	J-19	27.540	.00	2.70	.0710
.0176	15	J-19	26	-910.332	.10	12.07	.0710
.0181	17	J-45	J-40	-772.632	.09	10.24	.0710
.0262	18	J-29	51	-55.080	.10	5.39	.0710
.0262	19	31	J-29	55.080	.03	5.39	.0710

.0170	20	J-30	J-32	1100.088	.13	14.59	.0710
.0225	22	J-34	J-35	-137.700	.10	7.95	.0710
.0311	23	37	J-32	27.540	.01	2.70	.0710
.0311	24	39	J-40	27.540	.01	2.70	.0710
.0262	25	9	J-8	55.080	.02	5.39	.0710
.0311	26	42	J-41	27.540	.01	2.70	.0710
.0311	27	44	J-41	27.540	.01	2.70	.0710
.0311	28	46	J-38	27.540	.01	2.70	.0710
.0311	29	48	J-34	27.540	.01	2.70	.0710
.0311	31	52	J-36	27.540	.02	2.70	.0710
.0311	34	55	J-27	27.540	.01	2.70	.0710
.0174	36	J-60	33	962.388	.07	12.76	.0710
.0262	38	61	62	-55.080	.02	5.39	.0710
.0300	41	J-67	J-93	59.222	.00	1.57	.0710
.0311	42	69	70	-27.540	.00	2.70	.0710
.0311	44	73	51	27.540	.02	2.70	.0710
.0234	45	J-77	J-92	225.974	.00	3.00	.0710
.0414	72	69	J-14	27.540	.00	.37	.0710
.0311	80	31	106	-27.540	.01	2.70	.0710
.0311	82	109	62	27.540	.02	2.70	.0710
.0311	83	110	J-64	27.540	.02	2.70	.0710
.0311	84	112	J-89	27.540	.02	2.70	.0710
.0311	85	113	J-66	27.540	.02	2.70	.0710
.0311	87	115	76	27.540	.02	2.70	.0710
.0311	88	116	J-75	27.540	.02	2.70	.0710
.0311	90	120	J-142	27.540	.02	2.70	.0710
.0311	93	126	J-143	27.540	.01	2.70	.0710

.0311	94	128	J-145	27.540	.01	2.70	.0710
.0311	96	132	J-149	27.540	.02	2.70	.0710
.0311	97	134	J-148	27.540	.01	2.70	.0710
.0185	100	J-100	J-133	-682.846	.05	9.05	.0710
.0311	P-101	J-140	9	27.540	.00	2.70	.0710
.0262	P-102	J-90	J-64	-55.080	.08	5.39	.0710
.0262	P-103	J-66	J-68	55.080	.05	5.39	.0710
.0356	P-104	J-67	J-92	-31.682	.00	.84	.0710
.0272	P-105	J-93	J-94	86.762	.01	2.29	.0710
.0254	P-106	J-94	J-95	114.302	.01	3.02	.0710
.0242	P-107	J-95	J-97	141.842	.02	3.75	.0710
.0218	P-108	J-96	J-99	224.462	.04	5.93	.0710
.0311	P-109	108	J-97	27.540	.02	2.70	.0710
.0165	P-11	J-18	J-47	-1320.408	.13	17.51	.0710
.0213	P-111	J-100	J-99	-252.002	.02	6.66	.0710
.0262	P-112	J-77	76	-55.080	.03	5.39	.0710
.0250	P-113	J-103	J-77	170.894	.00	2.27	.0710
.0389	P-114	J-102	J-101	-24.077	.00	.64	.0710
.0273	P-115	J-103	117	-86.083	.00	2.27	.0710
.0047	P-116	117	J-101	-3.463	.00	.09	.0710
.0458	P-117	J-107	J-146	14.504	.00	.38	.0710
.0294	P-118	J-111	J-114	-64.652	.00	1.71	.0710
.0470	P-119	J-114	J-107	-13.036	.00	.34	.0710
.0268	P-120	J-119	J-111	-92.192	.01	2.44	.0710
.0252	P-121	J-130	J-119	-119.732	.01	3.16	.0710
.0240	P-122	J-141	J-130	-147.272	.02	3.89	.0710
.0231	P-123	J-118	J-141	-174.812	.01	4.62	.0710

P-124	J-121	J-103	84.811	.00	1.12	.0710
.0298						
P-125	J-121	J-142	-55.080	.02	5.39	.0710
.0262						
P-126	J-123	122	-27.540	.00	2.70	.0710
.0311						
P-127	J-125	124	-27.540	.01	2.70	.0710
.0311						
P-128	J-127	J-125	-25.349	.00	.34	.0710
.0426						
P-129	J-127	J-143	-55.080	.03	5.39	.0710
.0262						
P-130	J-129	J-127	-80.429	.00	1.07	.0710
.0303						
P-131	J-145	J-144	55.080	.02	5.39	.0710
.0262						
P-132	J-129	J-144	-82.620	.07	8.09	.0710
.0239						
P-133	J-146	J-147	42.044	.00	1.11	.0710
.0330						
P-134	J-131	J-129	-163.049	.00	2.16	.0710
.0253						
P-135	J-147	J-131	69.584	.00	1.84	.0710
.0288						
P-136	J-137	136	-27.540	.02	2.70	.0710
.0311						
P-137	J-135	J-137	-287.714	.01	3.81	.0710
.0222						
P-138	J-135	J-148	-55.080	.07	5.39	.0710
.0262						
P-139	J-139	138	-27.540	.01	2.70	.0710
.0311						
P-14	J-19	J-18	937.872	.14	12.44	.0710
.0175						
P-140	J-149	J-150	55.080	.07	5.39	.0710
.0262						
P-141	J-150	J-151	82.620	.14	8.09	.0710
.0239						
P-142	J-133	J-118	-572.686	.04	7.59	.0710
.0192						
P-143	J-133	J-151	-110.160	.02	6.36	.0710
.0236						
P-21	J-32	J-28	1127.628	.11	14.95	.0710
.0169						
P-30	58	J-30	27.540	.01	2.70	.0710
.0311						
P-32	J-28	J-47	1292.868	.16	17.14	.0710
.0165						
P-33	J-40	J-27	-745.092	.06	9.88	.0710
.0182						
P-35	J-30	33	-1072.548	.09	14.22	.0710
.0171						
P-37	J-29	33	110.160	.23	10.78	.0710
.0225						

.0225	P-39	J-90	J-91	137.700	.11	7.95	.0710
.0243	P-40	J-65	J-92	-194.292	.00	2.58	.0710
.0151	P-43	R-1	J-18	-2258.280	.77	29.94	.0710
.0262	P-46	J-75	117	55.080	.07	5.39	.0710
.0175	P-47	J-60	J-100	-934.848	.08	12.40	.0710
.0216	P-48	J-34	J-28	165.240	.04	9.54	.0710
.0252	P-49	J-35	J-38	-82.620	.02	4.77	.0710
.0262	P-50	J-36	J-35	55.080	.04	5.39	.0710
.0278	P-51	J-38	61	-55.080	.02	3.18	.0710
.0311	P-52	J-47	72	-27.540	.02	2.70	.0710
.0177	P-6	J-8	26	882.792	.11	11.71	.0710
.0179	P-61	J-45	J-8	827.712	.14	10.98	.0710
.0262	P-62	J-41	J-45	55.080	.02	5.39	.0710
.0184	P-63	J-27	J-43	-717.552	.06	9.51	.0710
.0210	P-64	J-1	J-17	192.780	.12	11.13	.0710
.0185	P-65	J-17	J-43	690.012	.05	9.15	.0710
.0262	P-66	J-1	3	-55.080	.04	5.39	.0710
.0225	P-67	J-12	2	110.160	.18	10.78	.0710
.0239	P-68	J-54	J-12	82.620	.12	8.09	.0710
.0262	P-69	J-56	J-54	55.080	.06	5.39	.0710
.0197	P-70	J-88	J-17	497.232	.02	6.59	.0710
.0224	P-71	J-16	J-65	-276.912	.01	3.67	.0710
.0202	P-73	J-14	J-16	-442.152	.01	5.86	.0710
.0188	P-74	J-14	98	469.692	.08	12.41	.0710
.0262	P-75	J-89	J-90	55.080	.07	5.39	.0710
.0216	P-76	J-16	J-91	-165.240	.07	9.54	.0710
.0252	P-77	J-65	J-63	-82.620	.03	4.77	.0710

.0311	P-79	J-60	104	-27.540	.00	2.70	.0710
.0225	P-81	J-97	J-96	196.922	.04	5.20	.0710
.0278	P-86	J-68	J-63	55.080	.01	3.18	.0710
.0207	P-89	J-118	J-139	-397.874	.02	5.28	.0710
.0403	P-91	J-123	J-121	29.731	.00	.39	.0710
.0411	P-92	J-125	J-123	2.191	.00	.03	.0710
.0312	P-95	J-114	J-102	-51.617	.00	1.36	.0710
.0233	P-98	J-137	J-131	-232.634	.01	3.08	.0710
.0210	P-99	J-139	J-135	-370.334	.02	4.91	.0710
.0381	R-1	R-1	R-1	-2258.280	.00	.01	.0710

JUNCTION DENSITY	NODE	DEMAND	PRESSURE	PRESSURE	PRESSURE	
NAME	TITLE	(USFU)	(USPU)	(PSIA)	(PSIG)	
#/CF						

.071	2	.00	-58.33	12.59	-2.10	
.071	3	.00	-58.32	12.59	-2.10	
.071	4	-27.54	-58.31	12.59	-2.10	
.071	5	-27.54	-58.31	12.59	-2.10	
.071	6	-27.54	-58.32	12.59	-2.10	
.071	7	-27.54	-57.94	12.60	-2.09	
.071	9	-27.54	-58.86	12.57	-2.12	
.071	10	-27.54	-58.99	12.57	-2.13	
.071	13	-27.54	-58.51	12.58	-2.11	
.071	15	-27.54	-58.45	12.59	-2.11	
.071	20	.00	-59.09	12.56	-2.13	

.071	21	-27.54	-59.07	12.56	-2.13
.071	26	.00	-58.99	12.57	-2.13
.071	31	-27.54	-58.35	12.59	-2.11
.071	33	.00	-58.62	12.58	-2.12
.071	37	-27.54	-58.82	12.57	-2.12
.071	39	-27.54	-58.63	12.58	-2.12
.071	42	-27.54	-58.70	12.58	-2.12
.071	44	-27.54	-58.71	12.58	-2.12
.071	46	-27.54	-58.77	12.57	-2.12
.071	48	-27.54	-58.89	12.57	-2.13
.071	51	-27.54	-58.29	12.59	-2.10
.071	52	-27.54	-58.74	12.58	-2.12
.071	55	-27.54	-58.57	12.58	-2.11
.071	58	-27.54	-58.69	12.58	-2.12
.071	61	.00	-58.76	12.58	-2.12
.071	62	-27.54	-58.75	12.58	-2.12
.071	69	.00	-58.35	12.59	-2.11
.071	70	-27.54	-58.35	12.59	-2.11
.071	72	-27.54	-59.08	12.56	-2.13
.071	73	-27.54	-58.27	12.59	-2.10
.071	76	-27.54	-58.30	12.59	-2.10
.071	98	.00	-58.43	12.59	-2.11
.071	104	-27.54	-58.54	12.58	-2.11
.071	106	-27.54	-58.34	12.59	-2.11
.071	108	-27.54	-58.34	12.59	-2.11
.071	109	-27.54	-58.73	12.58	-2.12
.071	110	-27.54	-58.05	12.60	-2.10

	112	-27.54	-58.06	12.60	-2.10
.071					
	113	-27.54	-58.22	12.59	-2.10
.071					
	115	-27.54	-58.28	12.59	-2.10
.071					
	116	-27.54	-58.22	12.59	-2.10
.071					
	117	-27.54	-58.32	12.59	-2.10
.071					
	120	-27.54	-58.28	12.59	-2.10
.071					
	122	-27.54	-58.31	12.59	-2.10
.071					
	124	-27.54	-58.31	12.59	-2.10
.071					
	126	-27.54	-58.28	12.59	-2.10
.071					
	128	-27.54	-58.22	12.59	-2.10
.071					
	132	-27.54	-58.16	12.60	-2.10
.071					
	134	-27.54	-58.26	12.59	-2.10
.071					
	136	-27.54	-58.31	12.59	-2.10
.071					
	138	-27.54	-58.35	12.59	-2.11
.071					
	J-1	.00	-58.35	12.59	-2.11
.071					
	J-100	.00	-58.47	12.59	-2.11
.071					
	J-101	-27.54	-58.32	12.59	-2.10
.071					
	J-102	-27.54	-58.32	12.59	-2.10
.071					
	J-103	.00	-58.32	12.59	-2.10
.071					
	J-107	-27.54	-58.32	12.59	-2.10
.071					
	J-111	-27.54	-58.32	12.59	-2.10
.071					
	J-114	.00	-58.32	12.59	-2.10
.071					
	J-118	.00	-58.38	12.59	-2.11
.071					
	J-119	-27.54	-58.33	12.59	-2.11
.071					
	J-12	-27.54	-58.15	12.60	-2.10
.071					
	J-121	.00	-58.32	12.59	-2.10
.071					
	J-123	.00	-58.32	12.59	-2.10
.071					

	J-125	.00	-58.32	12.59	-2.10
.071	J-127	.00	-58.32	12.59	-2.10
.071	J-129	.00	-58.32	12.59	-2.10
.071	J-130	-27.54	-58.34	12.59	-2.11
.071	J-131	.00	-58.32	12.59	-2.10
.071	J-133	.00	-58.41	12.59	-2.11
.071	J-135	-27.54	-58.34	12.59	-2.11
.071	J-137	-27.54	-58.33	12.59	-2.11
.071	J-139	.00	-58.36	12.59	-2.11
.071	J-14	.00	-58.35	12.59	-2.11
.071	J-140	-27.54	-58.86	12.57	-2.12
.071	J-141	-27.54	-58.36	12.59	-2.11
.071	J-142	-27.54	-58.30	12.59	-2.10
.071	J-143	-27.54	-58.28	12.59	-2.10
.071	J-144	-27.54	-58.25	12.59	-2.10
.071	J-145	-27.54	-58.23	12.59	-2.10
.071	J-146	-27.54	-58.32	12.59	-2.10
.071	J-147	-27.54	-58.32	12.59	-2.10
.071	J-148	-27.54	-58.27	12.59	-2.10
.071	J-149	-27.54	-58.18	12.60	-2.10
.071	J-150	-27.54	-58.25	12.59	-2.10
.071	J-151	-27.54	-58.39	12.59	-2.11
.071	J-16	.00	-58.34	12.59	-2.11
.071	J-17	.00	-58.47	12.59	-2.11
.071	J-18	.00	-59.23	12.56	-2.14
.071	J-19	.00	-59.09	12.56	-2.13
.071	J-27	.00	-58.58	12.58	-2.11
.071					

	J-28	.00	-58.94	12.57	-2.13
.071	J-29	.00	-58.38	12.59	-2.11
.071	J-30	.00	-58.70	12.58	-2.12
.071	J-32	.00	-58.83	12.57	-2.12
.071	J-34	.00	-58.90	12.57	-2.13
.071	J-35	.00	-58.80	12.57	-2.12
.071	J-36	-27.54	-58.76	12.58	-2.12
.071	J-38	.00	-58.78	12.57	-2.12
.071	J-40	.00	-58.64	12.58	-2.12
.071	J-41	.00	-58.71	12.58	-2.12
.071	J-43	.00	-58.52	12.58	-2.11
.071	J-45	.00	-58.73	12.58	-2.12
.071	J-47	.00	-59.10	12.56	-2.13
.071	J-54	-27.54	-58.02	12.60	-2.09
.071	J-56	-27.54	-57.96	12.60	-2.09
.071	J-60	.00	-58.55	12.58	-2.11
.071	J-63	-27.54	-58.30	12.59	-2.10
.071	J-64	-27.54	-58.07	12.60	-2.10
.071	J-65	.00	-58.33	12.59	-2.11
.071	J-66	-27.54	-58.24	12.59	-2.10
.071	J-67	-27.54	-58.33	12.59	-2.11
.071	J-68	.00	-58.29	12.59	-2.10
.071	J-75	-27.54	-58.24	12.59	-2.10
.071	J-77	.00	-58.32	12.59	-2.10
.071	J-8	.00	-58.88	12.57	-2.12
.071	J-88	.00	-58.45	12.59	-2.11
.071	J-89	-27.54	-58.08	12.60	-2.10
.071					

.071	J-90	-27.54	-58.15	12.60	-2.10
.071	J-91	-27.54	-58.26	12.59	-2.10
.071	J-92	.00	-58.33	12.59	-2.10
.071	J-93	-27.54	-58.33	12.59	-2.11
.071	J-94	-27.54	-58.34	12.59	-2.11
.071	J-95	-27.54	-58.35	12.59	-2.11
.071	J-96	-27.54	-58.41	12.59	-2.11
.071	J-97	-27.54	-58.37	12.59	-2.11
.071	J-99	-27.54	-58.45	12.59	-2.11
.071	R-1	.00	-60.00	12.53	-2.17

* This designates the use of default density in a low pressure region

THE NET SYSTEM DEMAND (USFU) = -2258.281

SUMMARY OF INFLOWS (+) AND OUTFLOWS (-) :

NAME	FLOW (USFU)	FPN TITLE
R-1	-2258.2	R-1

SUMMARY OF MINIMUM AND MAXIMUM VELOCITIES (FT/S)

	MINIMUM		MAXIMUM
R-1	.01	P-43	29.94
P-92	.03	P-11	17.51
P-116	.09	P-32	17.14
P-128	.34	P-21	14.95
P-119	.34	20	14.59

SUMMARY OF MINIMUM AND MAXIMUM LOSS/1000. (PSI)

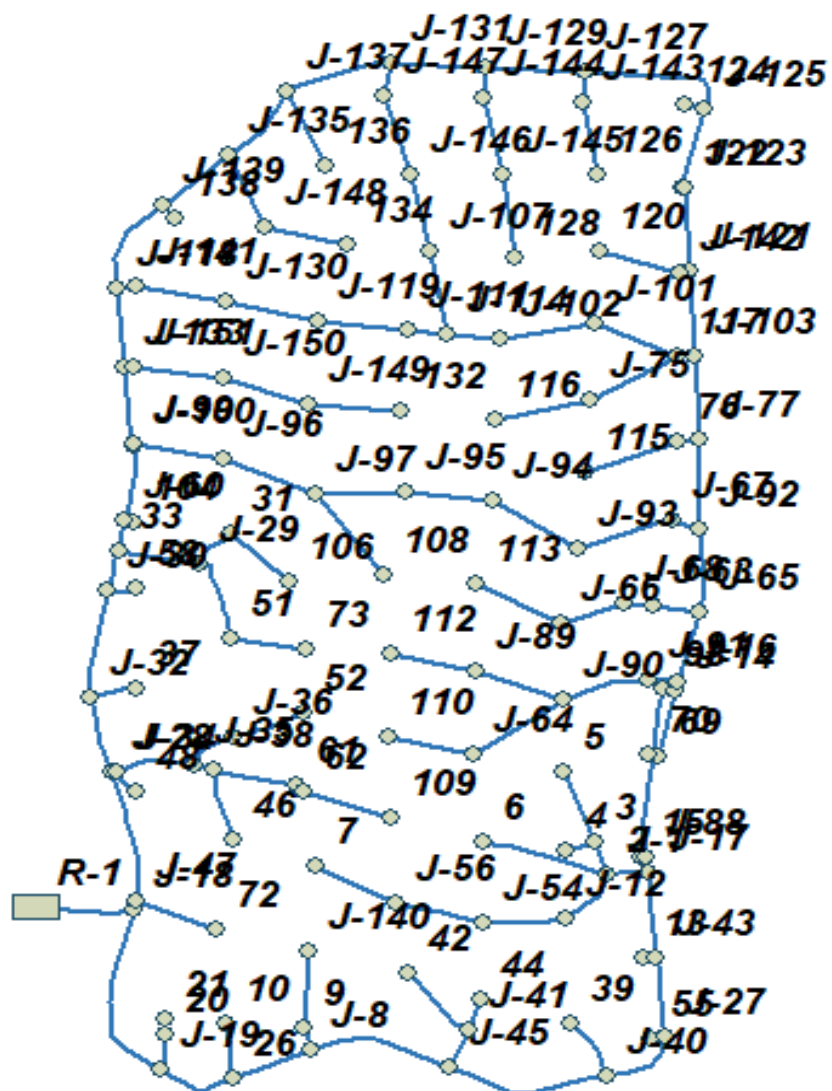
	MINIMUM		MAXIMUM
R-1	.00	P-43	.08
P-92	.00	P-67	.04
P-116	.00	P-37	.04

P-128	.00	P-64	.03
72	.00	P-11	.03

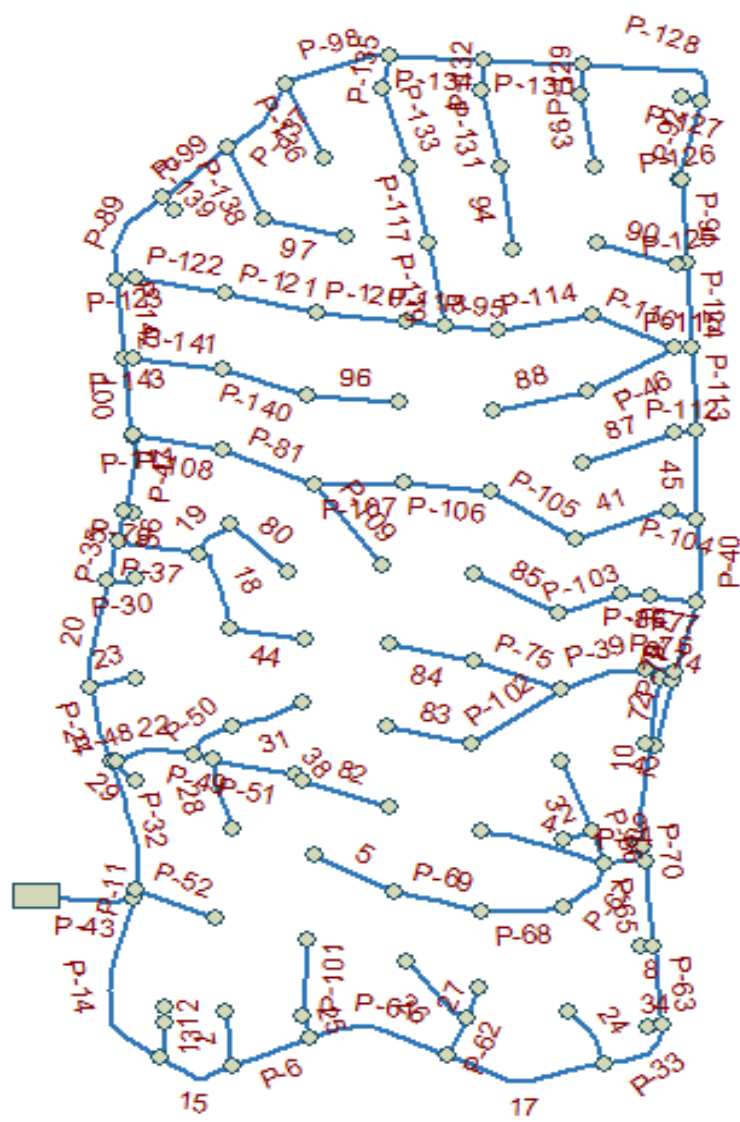
SUMMARY OF MINIMUM,AND,MAXIMUM PRESSURES (USPU)

	MINIMUM		MAXIMUM
R-1	-60.00	7	-57.94
J-18	-59.23	J-56	-57.96
J-47	-59.10	J-54	-58.02
J-19	-59.09	110	-58.05
20	-59.09	112	-58.06

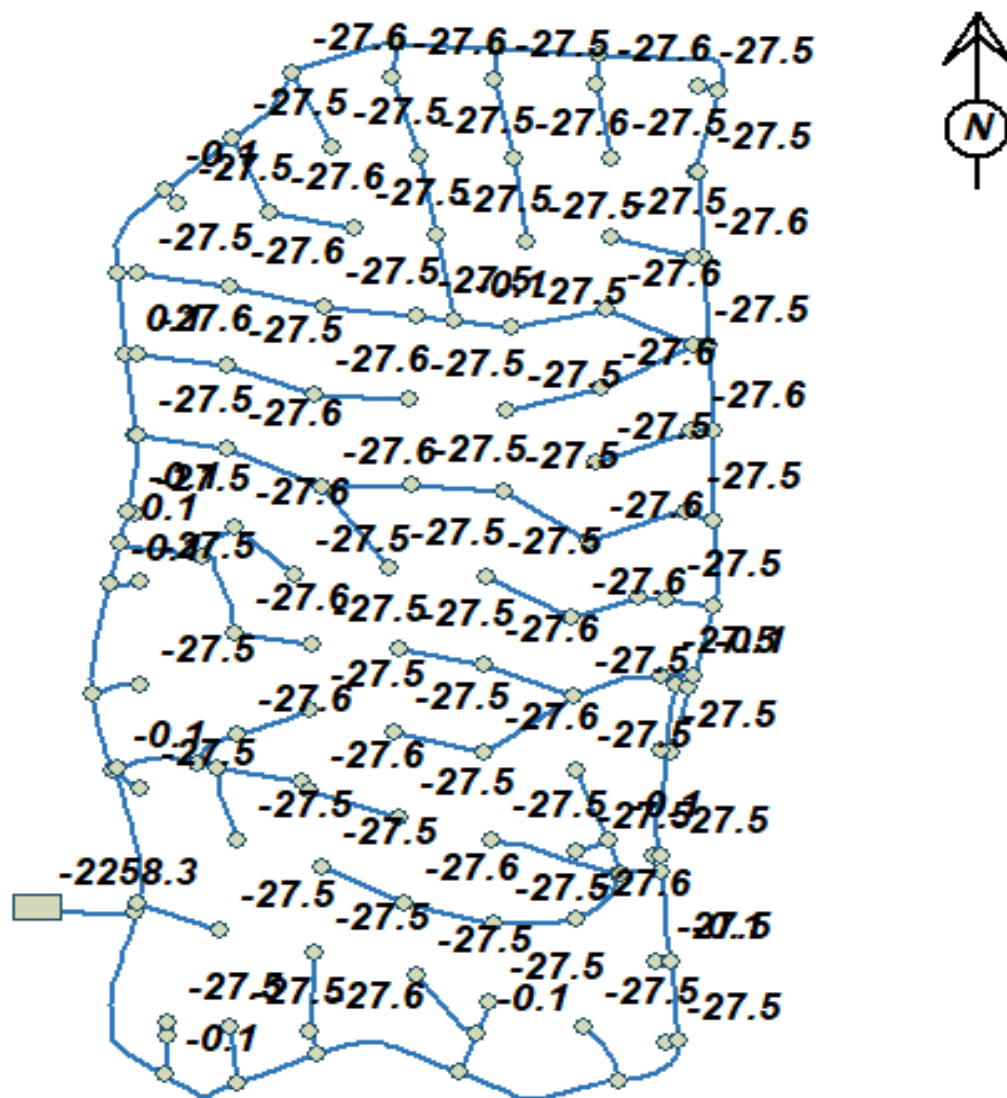
Wexford County Landfill (East) – Final Development
 Landfill Gas Collection and Control System
 Junction Name



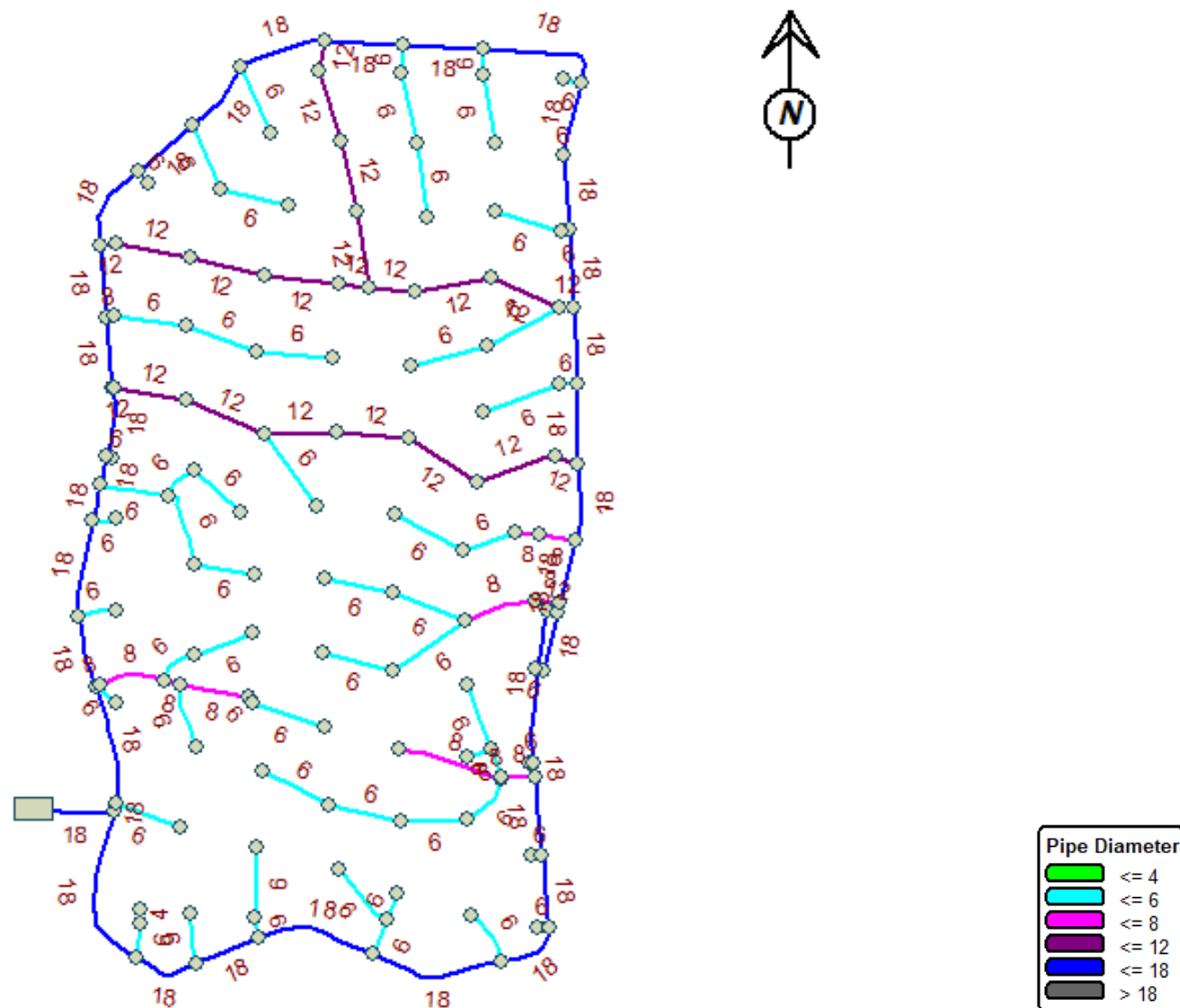
Wexford County Landfill (East) – Final Development
 Landfill Gas Collection and Control System
 Pipe Name



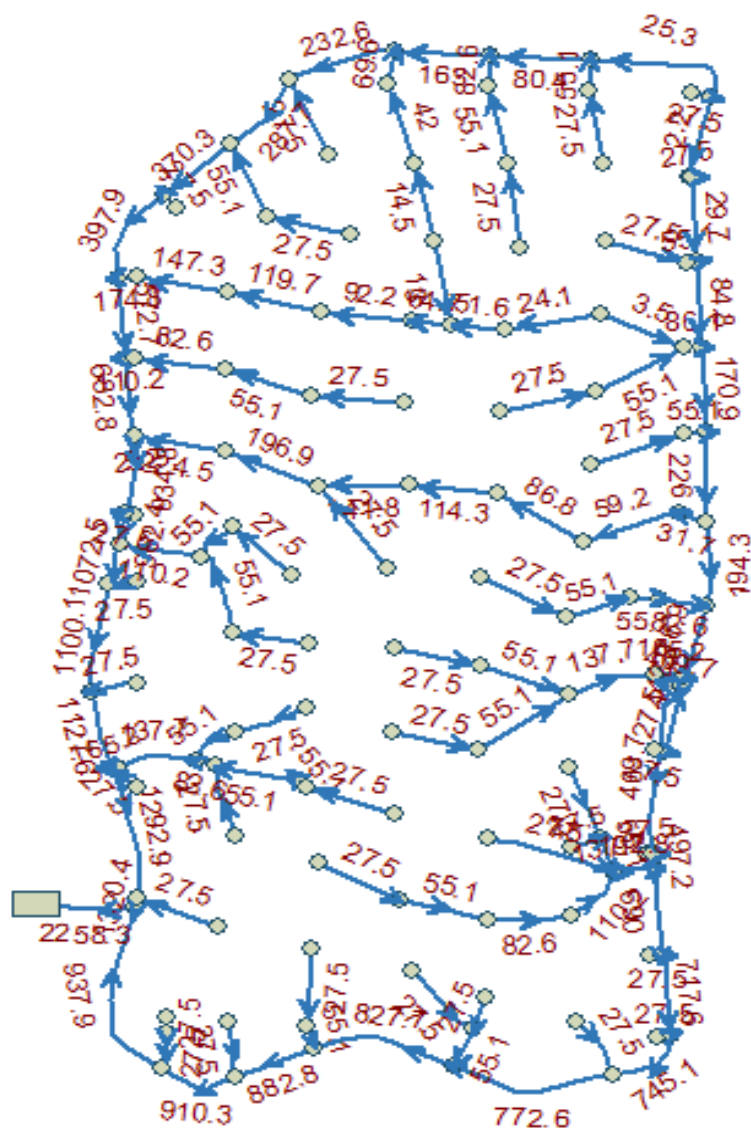
Wexford County Landfill (East) – Final Development
Landfill Gas Collection and Control System
Junction LFG Demand (scfm)



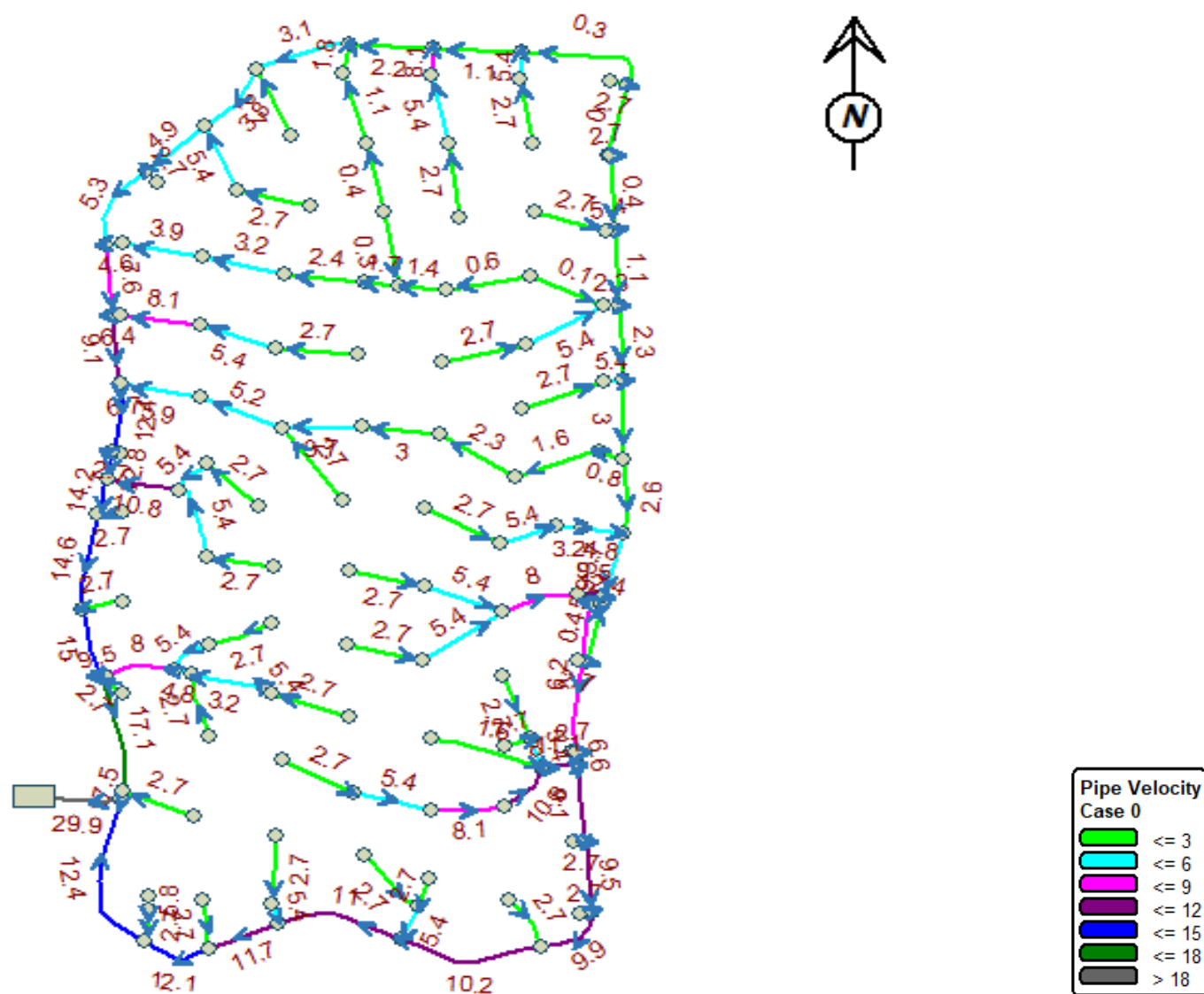
Wexford County Landfill (East) – Final Development
 Landfill Gas Collection and Control System
 Pipe Diameter (in.)



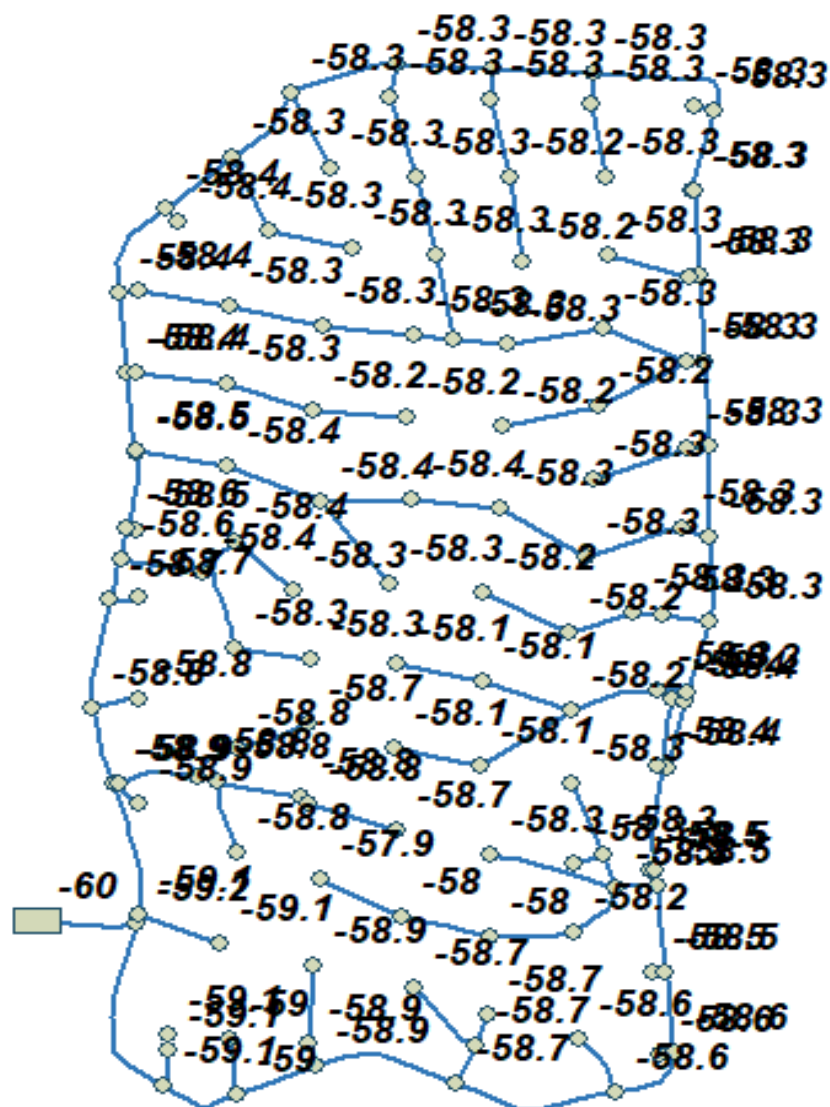
Wexford County Landfill (East) – Final Development
 Landfill Gas Collection and Control System
 Pipe Flow (scfm)



Wexford County Landfill (East) – Final Development
 Landfill Gas Collection and Control System
 Pipe Velocity (fps)



Wexford County Landfill (East) – Final Development
Landfill Gas Collection and Control System
Junction Available Vacuum (in. w.c.)



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* * * * * K Y G A S * * * * *
*
* Gas Network Analysis Software (KYGAS)
*
* CopyRighted by KYPIPE LLC (www.kypipe.com)
* Version: 11.112 01-27-2023
* Company: TETRAWIN Serial #: 580297
* Interface: KYnetic
* Licensed for Pipe2022
*
* * * * *

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DATE FOR THIS COMPUTER RUN      : 2-22-2024
START TIME FOR THIS COMPUTER RUN : 15:51:22: 1

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SUMMARY OF DISTRIBUTION SYSTEM CHARACTERISTICS:

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NUMBER OF PIPES      = 97
NUMBER OF JUNCTION NODES = 94

UNITS SPECIFIED      = ENGLISH

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A CONSTANT DENSITY FLUID IS SPECIFIED - DENSITY = .07POUNDS/CUBIC
FOOT
ABSOLUTE VISCOSITY      = .280E-06 POUND SECONDS/SQUARE
FOOT

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USER SPEC. FLOW UNITS (USFU)      = SCF/MIN
USER SPEC. PRESSURE UNITS(USPU)   = INCHES OF WATER (GAUGE)

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----- SIMULATION DESCRIPTION -----

----- SUMMARY OF PIPE NETWORK GEOMETRIC AND OPERATING DATA -----

PIPE NAME	NODE #1	NODE #2	LENGTH (FT.)	DIAM. (IN.)	ROUGHNESS (MILLIFEET)	SUM-M FACT.	PUMP ID	ELEVATION CHANGE
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6	11	J-91	126.2	5.8	.100	1.8	0	.0
8	15	J-89	150.8	5.8	.100	1.8	0	.0
13	25	J-88	159.3	5.8	.100	1.8	0	.0
14	27	J-87	163.2	5.8	.100	1.8	0	.0
15	29	J-59	167.6	5.8	.100	1.8	0	.0
18	35	J-47	30.3	5.8	.100	1.8	0	.0
19	37	J-55	198.4	5.8	.100	1.8	0	.0
20	39	J-84	158.8	5.8	.100	1.8	0	.0
21	41	J-68	169.7	5.8	.100	1.8	0	.0
23	45	J-82	158.0	5.8	.100	1.8	0	.0
24	J-32	J-34	147.8	15.8	.100	1.8	0	.0
32	38	J-77	176.5	7.5	.100	1.8	0	.0
P-1	J-2	J-6	152.8	15.8	.100	1.8	0	.0
P-10	J-19	J-40	86.5	15.8	.100	1.8	0	.0
P-11	J-34	J-36	83.0	15.8	.100	1.8	0	.0
P-12	J-23	J-12	82.9	15.8	.100	1.8	0	.0
P-16	J-32	J-19	81.2	15.8	.100	1.8	0	.0
P-17	J-34	33	46.0	5.8	.100	1.8	0	.0
P-2	J-4	J-2	87.4	15.8	.100	1.8	0	.0
P-22	J-43	J-61	88.5	15.8	.100	1.8	0	.0
P-25	J-50	J-81	212.4	11.2	.100	1.8	0	.0
P-26	J-50	49	178.0	5.8	.100	1.8	0	.0
P-27	J-53	J-21	48.5	15.8	.100	1.8	0	.0
P-28	J-55	38	161.3	5.8	.100	1.8	0	.0
P-29	J-57	J-10	56.3	15.8	.100	1.8	0	.0
P-3	J-6	J-8	83.5	15.8	.100	1.8	0	.0
P-30	J-48	J-30	16.5	15.8	.100	1.8	0	.0
P-31	J-60	J-43	87.6	15.8	.100	1.8	0	.0
P-33	J-62	J-63	89.3	15.8	.100	1.8	0	.0
P-34	J-63	J-64	89.3	15.8	.100	1.8	0	.0
P-35	J-64	J-65	89.3	15.8	.100	1.8	0	.0
P-36	J-65	J-66	89.3	15.8	.100	1.8	0	.0
P-37	J-66	J-67	89.3	15.8	.100	1.8	0	.0
P-38	J-67	J-60	94.9	15.8	.100	1.8	0	.0
P-39	J-68	J-69	158.9	5.8	.100	1.8	0	.0
P-4	J-8	J-57	75.5	15.8	.100	1.8	0	.0
P-40	J-69	42	167.9	5.8	.100	1.8	0	.0
P-41	J-60	42	72.0	7.5	.100	1.8	0	.0
P-42	J-70	J-71	148.8	11.2	.100	1.8	0	.0
P-43	J-71	J-72	159.6	11.2	.100	1.8	0	.0
P-44	J-72	J-73	171.2	11.2	.100	1.8	0	.0
P-45	J-73	J-74	165.9	11.2	.100	1.8	0	.0
P-46	J-74	J-75	166.3	11.2	.100	1.8	0	.0
P-47	J-75	J-76	163.6	11.2	.100	1.8	0	.0
P-48	J-44	J-46	96.6	15.8	.100	1.8	0	.0
P-49	J-43	J-70	85.0	11.2	.100	1.8	0	.0
P-5	J-10	J-14	84.5	15.8	.100	1.8	0	.0
P-50	J-61	J-22	83.8	15.8	.100	1.8	0	.0
P-51	J-55	54	180.0	5.8	.100	1.8	0	.0
P-52	J-61	J-77	38.0	7.5	.100	1.8	0	.0
P-53	J-22	J-48	67.4	15.8	.100	1.8	0	.0
P-54	J-79	J-78	135.0	11.2	.100	1.8	0	.0
P-55	J-80	J-79	133.3	11.2	.100	1.8	0	.0

P-56	J-22	J-78	21.0	11.2	.100	1.8	0	.0
P-57	J-81	J-80	136.8	11.2	.100	1.8	0	.0
P-58	J-50	51	162.0	5.8	.100	1.8	0	.0
P-59	J-21	J-50	133.0	11.2	.100	1.8	0	.0
P-60	J-82	J-83	168.4	5.8	.100	1.8	0	.0
P-61	J-46	J-62	89.2	15.8	.100	1.8	0	.0
P-62	J-46	J-83	66.0	5.8	.100	1.8	0	.0
P-63	J-76	J-44	51.0	11.2	.100	1.8	0	.0
P-64	J-40	J-44	73.3	15.8	.100	1.8	0	.0
P-65	J-40	J-93	54.0	5.8	.100	1.8	0	.0
P-66	J-85	20	174.9	5.8	.100	1.8	0	.0
P-67	J-86	J-85	171.3	5.8	.100	1.8	0	.0
P-68	J-19	J-86	19.0	5.8	.100	1.8	0	.0
P-69	J-32	31	28.0	5.8	.100	1.8	0	.0
P-7	J-14	J-18	59.1	15.8	.100	1.8	0	.0
P-70	J-36	J-21	45.5	15.8	.100	1.8	0	.0
P-71	J-36	J-47	8.0	5.8	.100	1.8	0	.0
P-72	J-48	R-1	184.0	15.8	.100	1.8	0	.0
P-73	J-30	J-28	86.3	15.8	.100	1.8	0	.0
P-74	J-30	J-58	2.0	5.8	.100	1.8	0	.0
P-75	J-59	J-58	171.4	5.8	.100	1.8	0	.0
P-76	J-28	J-26	82.6	15.8	.100	1.8	0	.0
P-77	J-28	J-87	172.0	5.8	.100	1.8	0	.0
P-78	52	J-53	164.0	5.8	.100	1.8	0	.0
P-79	J-26	J-23	97.3	15.8	.100	1.8	0	.0
P-80	J-26	J-88	43.0	5.8	.100	1.8	0	.0
P-81	J-16	J-53	92.7	15.8	.100	1.8	0	.0
P-82	J-16	J-89	15.0	5.8	.100	1.8	0	.0
P-83	J-90	24	148.4	5.8	.100	1.8	0	.0
P-84	J-23	J-90	64.0	5.8	.100	1.8	0	.0
P-85	J-12	J-92	84.4	15.8	.100	1.8	0	.0
P-86	J-12	J-91	23.0	5.8	.100	1.8	0	.0
P-87	J-18	17	99.0	5.8	.100	1.8	0	.0
P-88	J-14	13	84.0	5.8	.100	1.8	0	.0
P-89	J-10	9	71.0	5.8	.100	1.8	0	.0
P-9	J-18	J-16	85.0	15.8	.100	1.8	0	.0
P-90	J-92	J-4	83.6	15.8	.100	1.8	0	.0
P-91	J-57	56	12.0	5.8	.100	1.8	0	.0
P-92	J-4	3	20.0	5.8	.100	1.8	0	.0
P-93	J-8	7	6.0	5.8	.100	1.8	0	.0
P-94	J-6	5	18.0	5.8	.100	1.8	0	.0
P-95	J-2	1	40.0	5.8	.100	1.8	0	.0
P-96	J-93	J-84	155.0	5.8	.100	1.8	0	.0

JUNCTION NAME	NODE TITLE	ELEV	DEMAND (USFU)	FPN PRESSURE
1		.00	-45.48	
3		.00	-45.48	
5		.00	-45.48	
7		.00	-45.48	
9		.00	-45.48	
11		.00	-45.48	

13	.00	-45.48
15	.00	-45.48
17	.00	-45.48
20	.00	-45.48
24	.00	-45.48
25	.00	-45.48
27	.00	-45.48
29	.00	-45.48
31	.00	-45.48
33	.00	-45.48
35	.00	-45.48
37	.00	-45.48
38	.00	-45.48
39	.00	-45.48
41	.00	-45.48
42	.00	-45.48
45	.00	-45.48
49	.00	-45.48
51	.00	-45.48
52	.00	-45.48
54	.00	-45.48
56	.00	-45.48
J-10	.00	.00
J-12	.00	.00
J-14	.00	.00
J-16	.00	.00
J-18	.00	.00
J-19	.00	.00
J-2	.00	.00
J-21	.00	.00
J-22	.00	.00
J-23	.00	.00
J-26	.00	.00
J-28	.00	-45.48
J-30	.00	.00
J-32	.00	.00
J-34	.00	.00
J-36	.00	.00
J-4	.00	.00
J-40	.00	.00
J-43	.00	.00
J-44	.00	.00
J-46	.00	.00
J-47	.00	-45.48
J-48	.00	.00
J-50	.00	-45.48
J-53	.00	-45.48
J-55	.00	-45.48
J-57	.00	.00
J-58	.00	-45.48
J-59	.00	-45.48
J-6	.00	.00
J-60	.00	.00
J-61	.00	.00

J-62	.00	-45.48	
J-63	.00	-45.48	
J-64	.00	-45.48	
J-65	.00	-45.48	
J-66	.00	-45.48	
J-67	.00	-45.48	
J-68	.00	-45.48	
J-69	.00	-45.48	
J-70	.00	-45.48	
J-71	.00	-45.48	
J-72	.00	-45.48	
J-73	.00	-45.48	
J-74	.00	-45.48	
J-75	.00	-45.48	
J-76	.00	-45.48	
J-77	.00	-45.48	
J-78	.00	-45.48	
J-79	.00	-45.48	
J-8	.00	.00	
J-80	.00	-45.48	
J-81	.00	-45.48	
J-82	.00	-45.48	
J-83	.00	-45.48	
J-84	.00	-45.48	
J-85	.00	-45.48	
J-86	.00	-45.48	
J-87	.00	-45.48	
J-88	.00	-45.48	
J-89	.00	-45.48	
J-90	.00	-45.48	
J-91	.00	-45.48	
J-92	.00	-45.48	
J-93	.00	-45.48	
R-1	.00	.00	-60.00

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Set = 0

===== RESULTS FOR THIS SIMULATION FOLLOW

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Solution was obtained in 14 trials

Flow Accuracy = .1371E-02[< .500E-02]

RV Accuracy = .0000E+00[< .100E-02]

PIPE FRICTION RATIO	AREA NO.	NODE #1	NODE #2	FLOW (USFU)	LOSS (USPU)	VELOCITY (FT/S)	DENSITY (#/CF)	FACTOR
.0274	6	11	J-91	45.480	.04	4.45	.0710	
.0274	8	15	J-89	45.480	.04	4.45	.0710	
.0274	13	25	J-88	45.480	.05	4.45	.0710	
.0274	14	27	J-87	45.480	.05	4.45	.0710	
.0274	15	29	J-59	45.480	.05	4.45	.0710	
.0274	18	35	J-47	45.480	.01	4.45	.0710	
.0274	19	37	J-55	45.480	.05	4.45	.0710	
.0274	20	39	J-84	45.480	.05	4.45	.0710	
.0274	21	41	J-68	45.480	.05	4.45	.0710	
.0274	23	45	J-82	45.480	.05	4.45	.0710	
.0258	24	J-32	J-34	150.619	.00	2.00	.0710	
.0212	32	38	J-77	181.920	.18	10.51	.0710	
.0192	P-1	J-2	J-6	-569.786	.05	7.56	.0710	
.0395	P-10	J-19	J-40	31.301	.00	.42	.0710	
.0242	P-11	J-34	J-36	196.099	.00	2.60	.0710	
.0180	P-12	J-23	J-12	-797.186	.07	10.57	.0710	
.0283	P-16	J-32	J-19	-105.139	.00	1.39	.0710	
.0274	P-17	J-34	33	-45.480	.02	4.45	.0710	
.0189	P-2	J-4	J-2	-615.266	.04	8.16	.0710	
.0170	P-22	J-43	J-61	1077.341	.13	14.29	.0710	
.0204	P-25	J-50	J-81	308.514	.09	8.15	.0710	
.0274	P-26	J-50	49	-45.480	.05	4.45	.0710	
.0276	P-27	J-53	J-21	-114.986	.00	1.52	.0710	

	P-28	J-55	38	136.440	.34	13.35	.0710
.0215							
	P-29	J-57	J-10	-433.346	.02	5.75	.0710
.0203							
	P-3	J-6	J-8	-524.306	.03	6.95	.0710
.0195							
	P-30	J-48	J-30	-1251.986	.12	16.60	.0710
.0166							
	P-31	J-60	J-43	749.752	.06	9.94	.0710
.0182							
	P-33	J-62	J-63	340.432	.01	4.51	.0710
.0214							
	P-34	J-63	J-64	385.912	.02	5.12	.0710
.0208							
	P-35	J-64	J-65	431.392	.02	5.72	.0710
.0203							
	P-36	J-65	J-66	476.872	.03	6.32	.0710
.0199							
	P-37	J-66	J-67	522.352	.03	6.93	.0710
.0195							
	P-38	J-67	J-60	567.832	.04	7.53	.0710
.0192							
	P-39	J-68	J-69	90.960	.16	8.90	.0710
.0234							
	P-4	J-8	J-57	-478.826	.03	6.35	.0710
.0199							
	P-40	J-69	42	136.440	.35	13.35	.0710
.0215							
	P-41	J-60	42	-181.920	.10	10.51	.0710
.0212							
	P-42	J-70	J-71	-282.109	.06	7.46	.0710
.0208							
	P-43	J-71	J-72	-236.629	.05	6.25	.0710
.0216							
	P-44	J-72	J-73	-191.149	.03	5.05	.0710
.0226							
	P-45	J-73	J-74	-145.669	.02	3.85	.0710
.0240							
	P-46	J-74	J-75	-100.189	.01	2.65	.0710
.0263							
	P-47	J-75	J-76	-54.709	.00	1.45	.0710
.0306							
	P-48	J-44	J-46	158.512	.00	2.10	.0710
.0255							
	P-49	J-43	J-70	-327.589	.06	8.66	.0710
.0202							
	P-5	J-10	J-14	-387.866	.02	5.14	.0710
.0208							
	P-50	J-61	J-22	1304.741	.18	17.30	.0710
.0165							
	P-51	J-55	54	-45.480	.05	4.45	.0710
.0274							
	P-52	J-61	J-77	-227.400	.11	13.13	.0710
.0203							

	P-53	J-22	J-48	1795.174	.31	23.80	.0710
.0157							
	P-54	J-79	J-78	444.954	.13	11.76	.0710
.0190							
	P-55	J-80	J-79	399.474	.11	10.56	.0710
.0194							
	P-56	J-22	J-78	-490.434	.08	12.96	.0710
.0187							
	P-57	J-81	J-80	353.993	.09	9.35	.0710
.0199							
	P-58	J-50	51	-45.480	.05	4.45	.0710
.0274							
	P-59	J-21	J-50	172.074	.02	4.55	.0710
.0232							
	P-60	J-82	J-83	90.960	.17	8.90	.0710
.0234							
	P-61	J-46	J-62	294.952	.01	3.91	.0710
.0221							
	P-62	J-46	J-83	-136.440	.18	13.35	.0710
.0215							
	P-63	J-76	J-44	-9.229	.00	.24	.0710
.0536							
	P-64	J-40	J-44	167.741	.00	2.22	.0710
.0251							
	P-65	J-40	J-93	-136.440	.16	13.35	.0710
.0215							
	P-66	J-85	20	-45.480	.05	4.45	.0710
.0274							
	P-67	J-86	J-85	-90.960	.17	8.90	.0710
.0234							
	P-68	J-19	J-86	-136.440	.10	13.35	.0710
.0215							
	P-69	J-32	31	-45.480	.01	4.45	.0710
.0274							
	P-7	J-14	J-18	-342.386	.01	4.54	.0710
.0214							
	P-70	J-36	J-21	287.059	.01	3.81	.0710
.0222							
	P-71	J-36	J-47	-90.960	.04	8.90	.0710
.0234							
	P-72	J-48	R-1	3047.160	1.33	40.40	.0710
.0145							
	P-73	J-30	J-28	-1115.546	.14	14.79	.0710
.0169							
	P-74	J-30	J-58	-136.440	.07	13.35	.0710
.0215							
	P-75	J-59	J-58	90.960	.17	8.90	.0710
.0234							
	P-76	J-28	J-26	-979.106	.10	12.98	.0710
.0173							
	P-77	J-28	J-87	-90.960	.17	8.90	.0710
.0234							
	P-78	52	J-53	45.480	.05	4.45	.0710
.0274							

.0177	P-79	J-26	J-23	-888.146	.09	11.78	.0710
.0234	P-80	J-26	J-88	-90.960	.07	8.90	.0710
.0239	P-81	J-16	J-53	-205.946	.01	2.73	.0710
.0234	P-82	J-16	J-89	-90.960	.04	8.90	.0710
.0274	P-83	J-90	24	-45.480	.04	4.45	.0710
.0234	P-84	J-23	J-90	-90.960	.08	8.90	.0710
.0184	P-85	J-12	J-92	-706.226	.06	9.36	.0710
.0234	P-86	J-12	J-91	-90.960	.05	8.90	.0710
.0274	P-87	J-18	17	-45.480	.03	4.45	.0710
.0274	P-88	J-14	13	-45.480	.03	4.45	.0710
.0274	P-89	J-10	9	-45.480	.02	4.45	.0710
.0220	P-9	J-18	J-16	-296.906	.01	3.94	.0710
.0187	P-90	J-92	J-4	-660.746	.05	8.76	.0710
.0274	P-91	J-57	56	-45.480	.01	4.45	.0710
.0274	P-92	J-4	3	-45.480	.01	4.45	.0710
.0274	P-93	J-8	7	-45.480	.01	4.45	.0710
.0274	P-94	J-6	5	-45.480	.01	4.45	.0710
.0274	P-95	J-2	1	-45.480	.02	4.45	.0710
.0234	P-96	J-93	J-84	-90.960	.16	8.90	.0710
.0349	R-1	R-1	R-1	-3047.160	.00	.01	.0710

JUNCTION DENSITY	NODE	DEMAND	PRESSURE	PRESSURE	PRESSURE
NAME #/CF	TITLE	(USFU)	(USPU)	(PSIA)	(PSIG)
1		-45.48	-57.99	12.60	-2.09

.071	3	-45.48	-58.04	12.60	-2.09
.071	5	-45.48	-57.95	12.60	-2.09
.071	7	-45.48	-57.92	12.61	-2.09
.071	9	-45.48	-57.86	12.61	-2.09
.071	11	-45.48	-58.07	12.60	-2.10
.071	13	-45.48	-57.84	12.61	-2.09
.071	15	-45.48	-57.76	12.61	-2.08
.071	17	-45.48	-57.82	12.61	-2.09
.071	20	-45.48	-57.50	12.62	-2.08
.071	24	-45.48	-58.10	12.60	-2.10
.071	25	-45.48	-58.21	12.60	-2.10
.071	27	-45.48	-58.20	12.60	-2.10
.071	29	-45.48	-58.27	12.59	-2.10
.071	31	-45.48	-57.81	12.61	-2.09
.071	33	-45.48	-57.81	12.61	-2.09
.071	35	-45.48	-57.78	12.61	-2.09
.071	37	-45.48	-57.49	12.62	-2.07
.071	38	-45.48	-57.89	12.61	-2.09
.071	39	-45.48	-57.46	12.62	-2.07
.071	41	-45.48	-57.33	12.63	-2.07
.071	42	-45.48	-57.89	12.61	-2.09
.071	45	-45.48	-57.43	12.62	-2.07
.071	49	-45.48	-57.81	12.61	-2.09
.071	51	-45.48	-57.81	12.61	-2.09
.071	52	-45.48	-57.79	12.61	-2.09
.071	54	-45.48	-57.50	12.62	-2.07
.071	56	-45.48	-57.89	12.61	-2.09

.071	J-10	.00	-57.88	12.61	-2.09
.071	J-12	.00	-58.15	12.60	-2.10
.071	J-14	.00	-57.87	12.61	-2.09
.071	J-16	.00	-57.84	12.61	-2.09
.071	J-18	.00	-57.85	12.61	-2.09
.071	J-19	.00	-57.82	12.61	-2.09
.071	J-2	.00	-58.01	12.60	-2.09
.071	J-21	.00	-57.84	12.61	-2.09
.071	J-22	.00	-58.36	12.59	-2.11
.071	J-23	.00	-58.22	12.59	-2.10
.071	J-26	.00	-58.32	12.59	-2.10
.071	J-28	-45.48	-58.42	12.59	-2.11
.071	J-30	.00	-58.55	12.58	-2.11
.071	J-32	.00	-57.82	12.61	-2.09
.071	J-34	.00	-57.82	12.61	-2.09
.071	J-36	.00	-57.83	12.61	-2.09
.071	J-4	.00	-58.05	12.60	-2.10
.071	J-40	.00	-57.82	12.61	-2.09
.071	J-43	.00	-58.05	12.60	-2.10
.071	J-44	.00	-57.82	12.61	-2.09
.071	J-46	.00	-57.83	12.61	-2.09
.071	J-47	-45.48	-57.79	12.61	-2.09
.071	J-48	.00	-58.67	12.58	-2.12
.071	J-50	-45.48	-57.86	12.61	-2.09
.071	J-53	-45.48	-57.84	12.61	-2.09
.071	J-55	-45.48	-57.55	12.62	-2.08
.071	J-57	.00	-57.90	12.61	-2.09

.071	J-58	-45.48	-58.48	12.59	-2.11
.071	J-59	-45.48	-58.31	12.59	-2.10
.071	J-6	.00	-57.96	12.60	-2.09
.071	J-60	.00	-57.99	12.60	-2.09
.071	J-61	.00	-58.18	12.60	-2.10
.071	J-62	-45.48	-57.84	12.61	-2.09
.071	J-63	-45.48	-57.85	12.61	-2.09
.071	J-64	-45.48	-57.87	12.61	-2.09
.071	J-65	-45.48	-57.89	12.61	-2.09
.071	J-66	-45.48	-57.92	12.61	-2.09
.071	J-67	-45.48	-57.95	12.60	-2.09
.071	J-68	-45.48	-57.38	12.63	-2.07
.071	J-69	-45.48	-57.54	12.62	-2.08
.071	J-70	-45.48	-57.99	12.60	-2.09
.071	J-71	-45.48	-57.93	12.61	-2.09
.071	J-72	-45.48	-57.89	12.61	-2.09
.071	J-73	-45.48	-57.85	12.61	-2.09
.071	J-74	-45.48	-57.83	12.61	-2.09
.071	J-75	-45.48	-57.82	12.61	-2.09
.071	J-76	-45.48	-57.82	12.61	-2.09
.071	J-77	-45.48	-58.07	12.60	-2.10
.071	J-78	-45.48	-58.28	12.59	-2.10
.071	J-79	-45.48	-58.15	12.60	-2.10
.071	J-8	.00	-57.93	12.61	-2.09
.071	J-80	-45.48	-58.04	12.60	-2.09
.071	J-81	-45.48	-57.95	12.60	-2.09
.071	J-82	-45.48	-57.48	12.62	-2.07

.071	J-83	-45.48	-57.65	12.62	-2.08
.071	J-84	-45.48	-57.50	12.62	-2.08
.071	J-85	-45.48	-57.55	12.62	-2.08
.071	J-86	-45.48	-57.72	12.61	-2.08
.071	J-87	-45.48	-58.25	12.59	-2.10
.071	J-88	-45.48	-58.25	12.59	-2.10
.071	J-89	-45.48	-57.80	12.61	-2.09
.071	J-90	-45.48	-58.14	12.60	-2.10
.071	J-91	-45.48	-58.11	12.60	-2.10
.071	J-92	-45.48	-58.10	12.60	-2.10
.071	J-93	-45.48	-57.66	12.62	-2.08
.071	R-1	.00	-60.00	12.53	-2.17

* This designates the use of default density in a low pressure region

THE NET SYSTEM DEMAND (USFU) = -3047.159

SUMMARY OF INFLOWS(+) .AND. OUTFLOWS(-) :

NAME	FLOW (USFU)	FPN TITLE
R-1	-3047.1	R-1

SUMMARY OF MINIMUM.AND.MAXIMUM VELOCITIES (FT/S)

	MINIMUM		MAXIMUM
R-1	.01	P-72	40.40
P-63	.24	P-53	23.80
P-10	.42	P-50	17.30
P-16	1.39	P-30	16.60
P-47	1.45	P-73	14.79

SUMMARY OF MINIMUM.AND.MAXIMUM LOSS/1000. (PSI)

MINIMUM	MAXIMUM
---------	---------

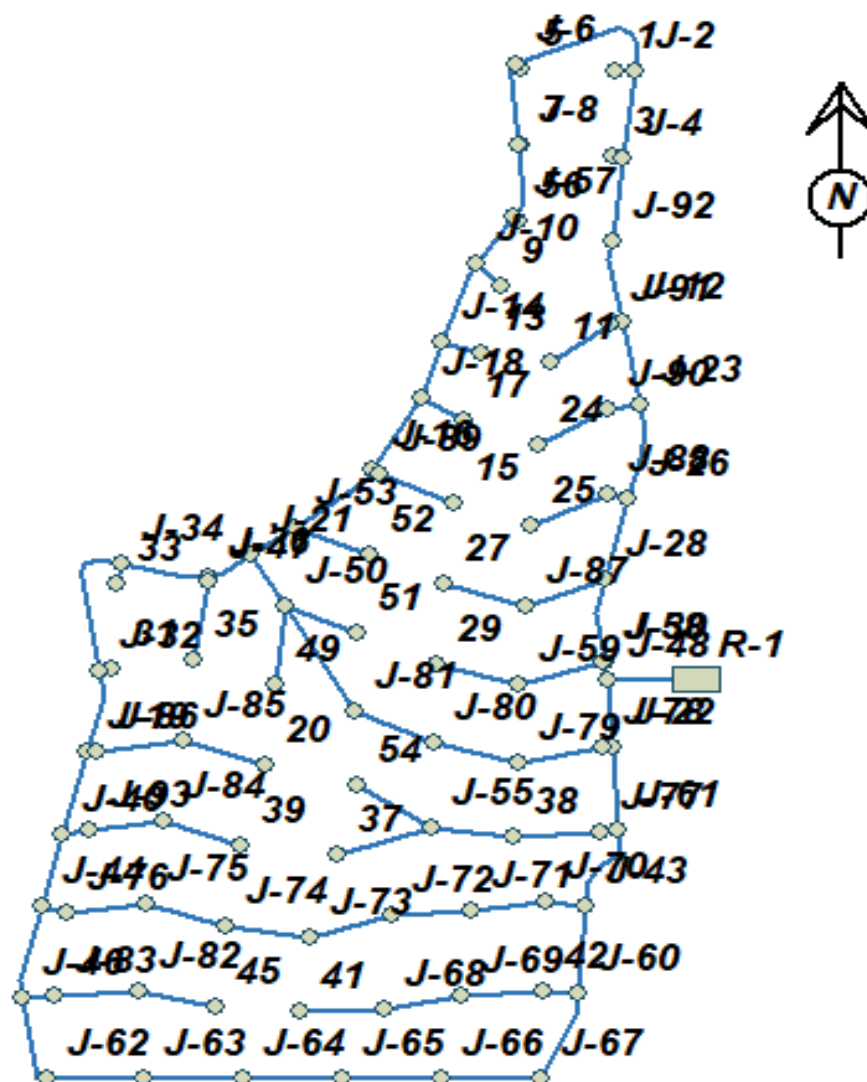
R-1	.00	P-72	.14
P-63	.00	P-74	.06
P-10	.00	P-68	.06
P-16	.00	P-53	.05
P-27	.00	P-52	.04

SUMMARY OF MINIMUM.AND.MAXIMUM PRESSURES (USPU)

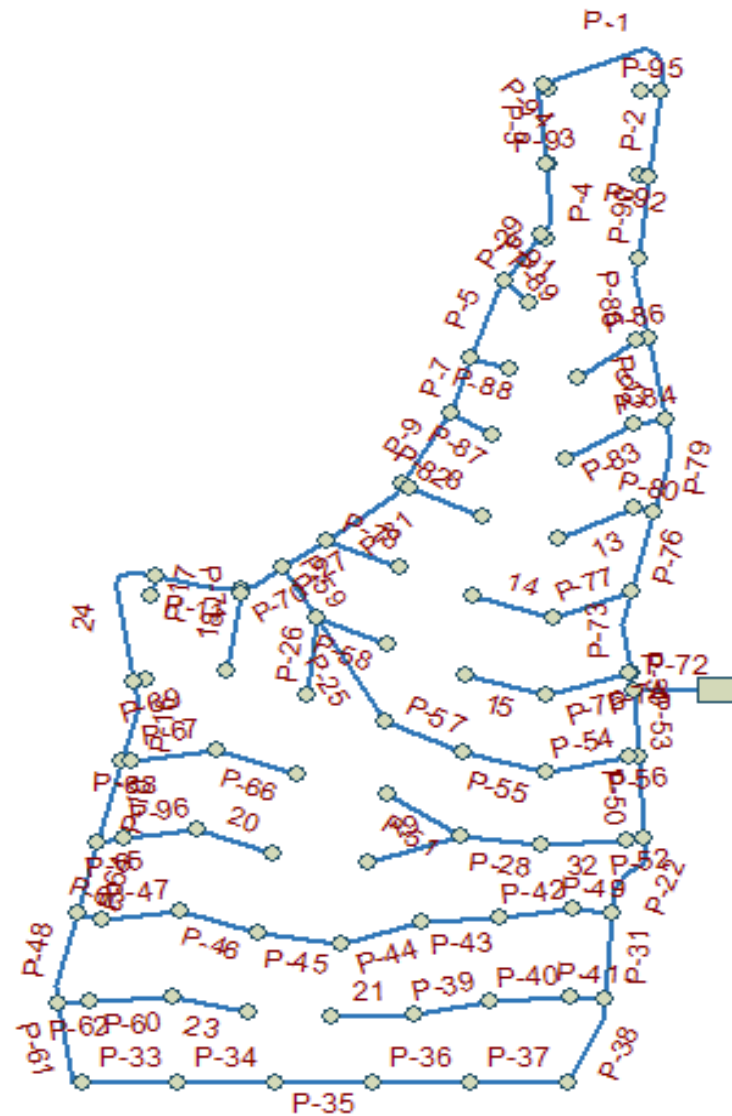
	MINIMUM		MAXIMUM

R-1	-60.00	41	-57.33
J-48	-58.67	J-68	-57.38
J-30	-58.55	45	-57.43
J-58	-58.48	39	-57.46
J-28	-58.42	J-82	-57.48

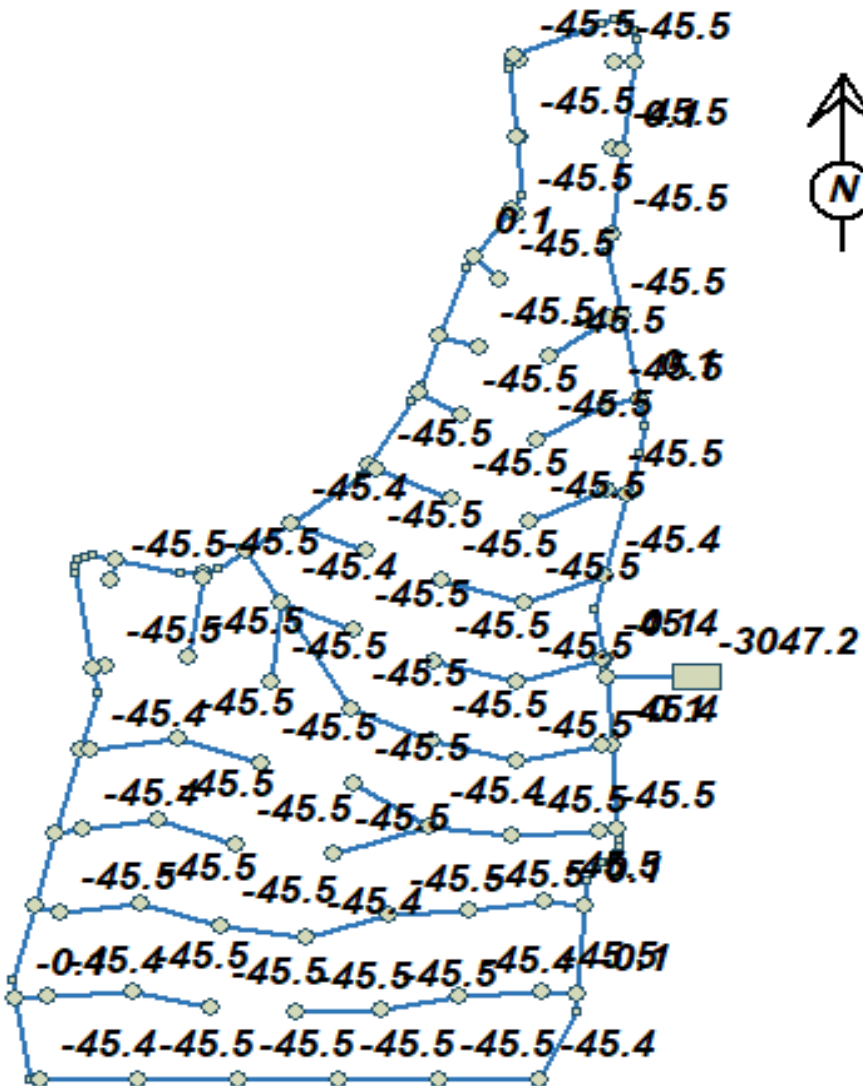
Wexford County Landfill (West) – Final Development
 Landfill Gas Collection and Control System
 Junction Name



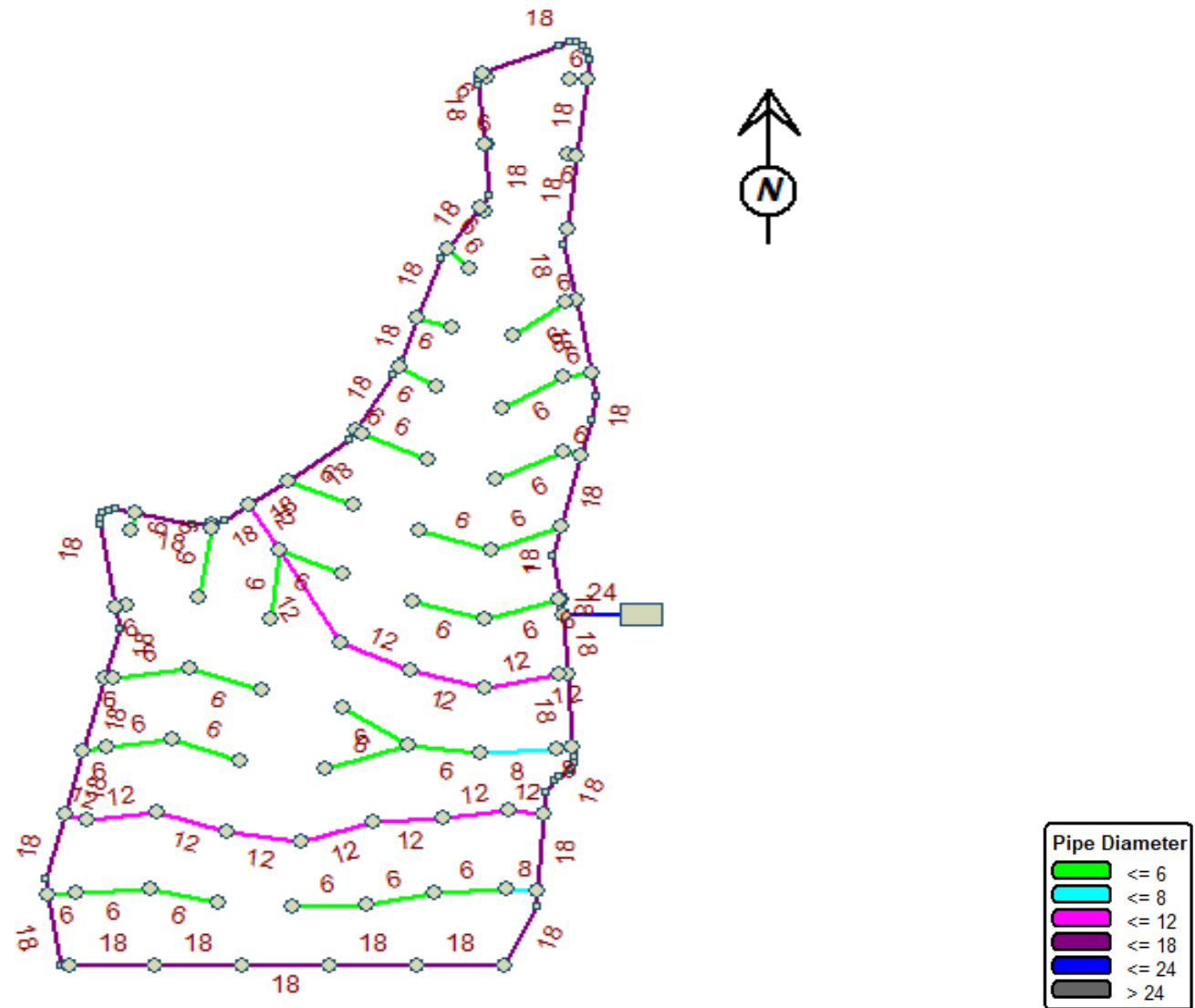
Wexford County Landfill (West) – Final Development
 Landfill Gas Collection and Control System
 Pipe Name



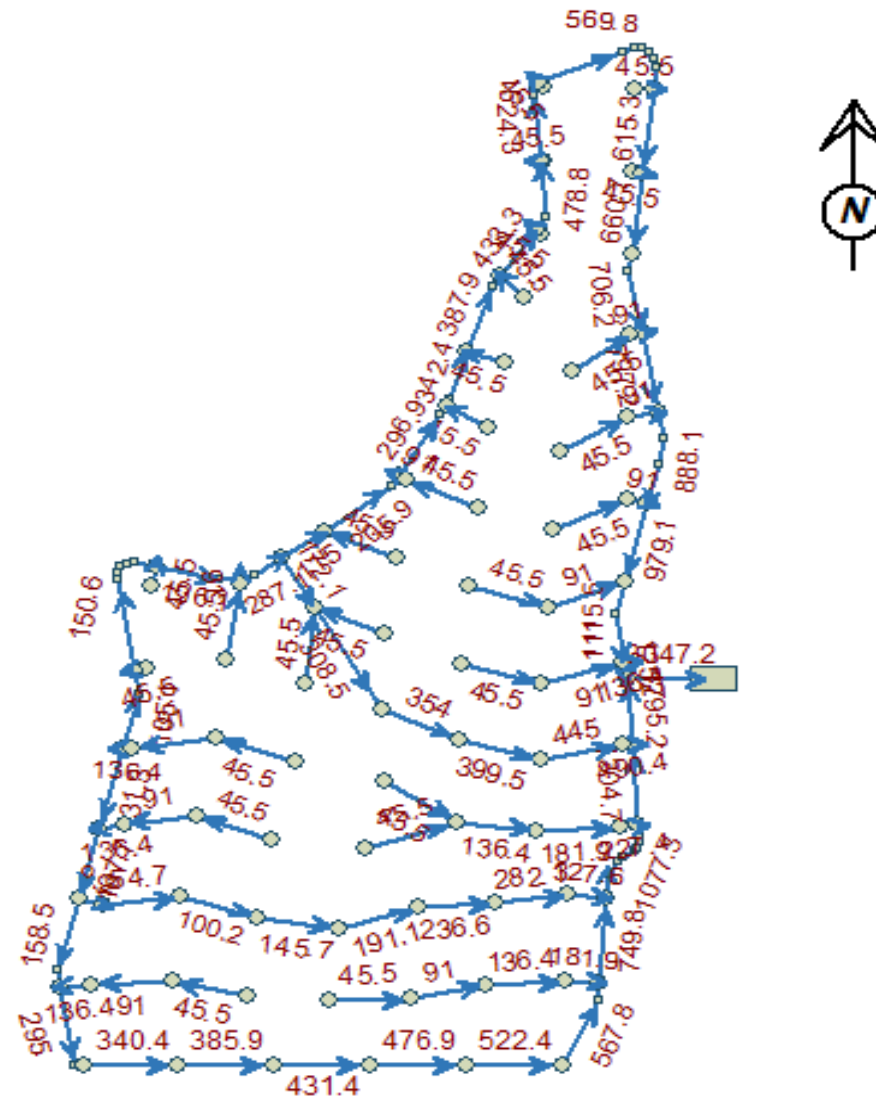
Wexford County Landfill (West) – Final Development
Landfill Gas Collection and Control System
Junction LFG Demand (scfm)



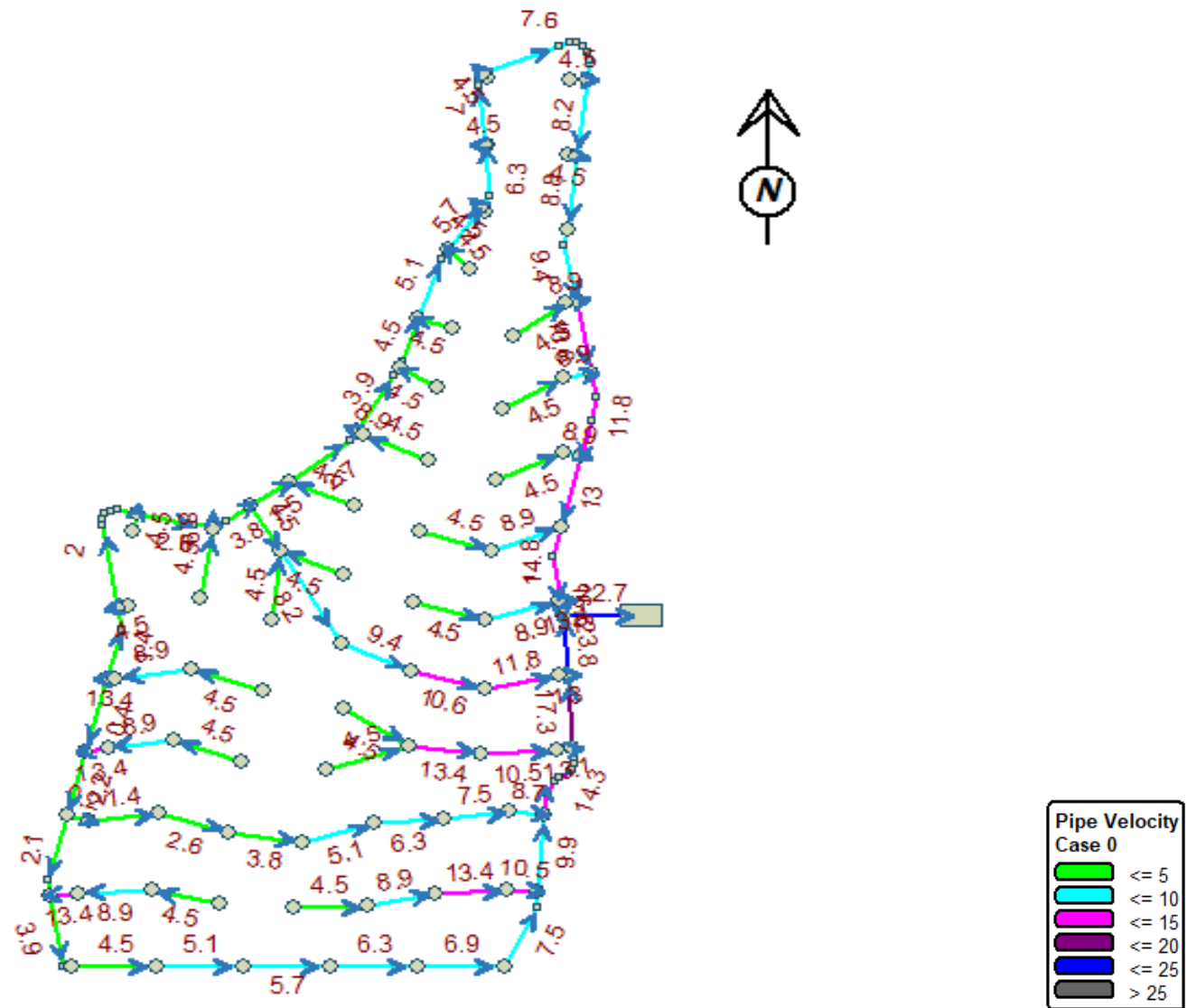
Wexford County Landfill (West) – Final Development
Landfill Gas Collection and Control System
Pipe Diameter (in.)



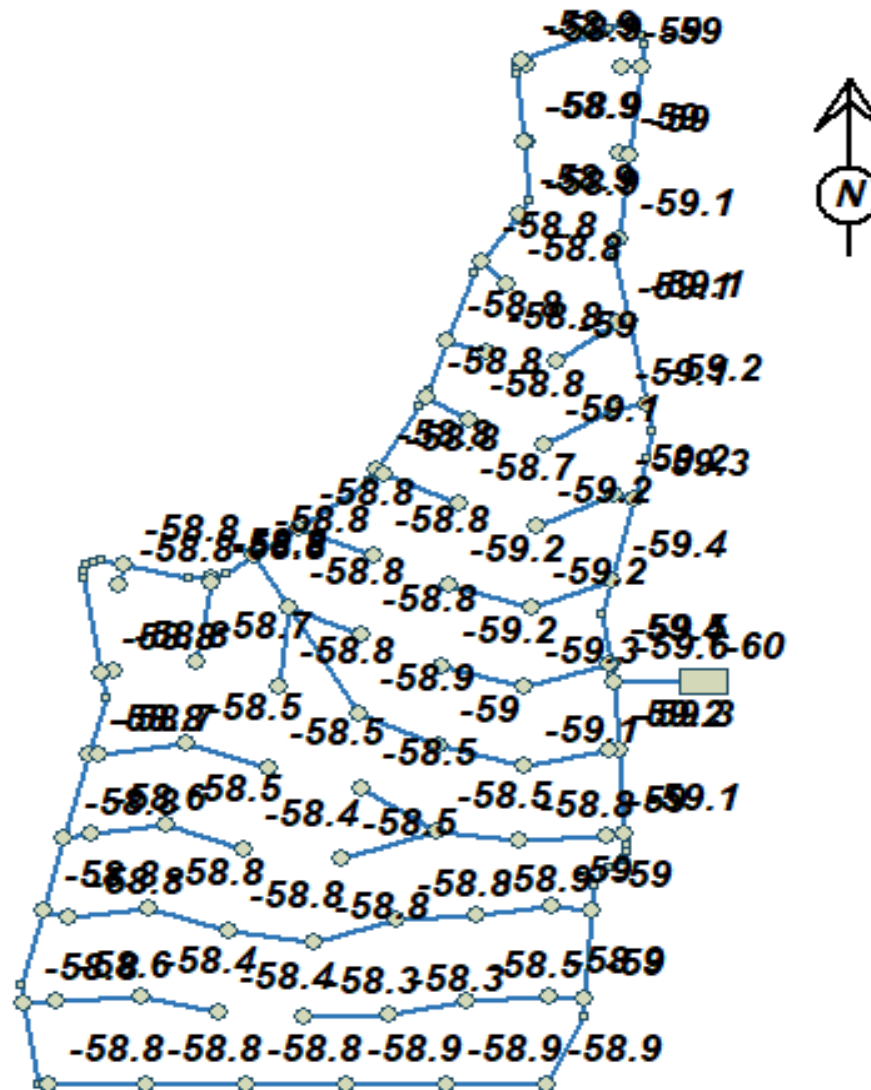
Wexford County Landfill (West) – Final Development
Landfill Gas Collection and Control System
Pipe Flow (scfm)



Wexford County Landfill (West) – Final Development
Landfill Gas Collection and Control System
Pipe Velocity (fps)



Wexford County Landfill (West) – Final Development
Landfill Gas Collection and Control System
Junction Available Vacuum (in. w.c.)



Appendix B

Current GCCS System As-builts on Effective Date of March 29, 2023

Appendix C

GCCS System Design – Final Grades



- LEGEND**
- PROPERTY BOUNDARY
 - EXISTING SOLID WASTE BOUNDARY
 - EXISTING PHASE BOUNDARY
 - 9000 --- PROPOSED 10' CONTOUR
 - 18" --- EXISTING LANDFILL GAS HEADER
 - 6" --- EXISTING LANDFILL GAS HORIZONTAL COLLECTOR
 - FW-3 --- EXISTING 3" FORCEMAIN
 - FW-36 --- EXISTING 3"6" DUAL CONTAINED FORCEMAIN LINE
 - FW-40 --- EXISTING 4"0" DUAL CONTAINED FORCEMAIN LINE
 - A-2 --- EXISTING 2" AIRLINE
 - ⊕ GW- --- EXISTING LFG EXTRACTION WELL
 - ⊕ GW- --- EXISTING LFG EXTRACTION WELL WITH CASSON
 - ⊕ RW- --- EXISTING REMOTE WELLHEAD
 - ⊕ --- EXISTING HEADER ACCESS RISER
 - ⊕ --- EXISTING BLIND FLANGE
 - ⊕ --- EXISTING FLANGE CONNECTION
 - ⊕ --- EXISTING REDUCER FITTING
 - ⊕ DL-3 --- EXISTING CONDENSATE DRIPLEG
 - ⊕ MH --- EXISTING MANHOLE
 - ⊕ --- EXISTING FORCEMAIN ISOLATION VALVE
 - ⊕ --- EXISTING AIRLINE ISOLATION VALVE
 - ⊕ FCO --- EXISTING FORCEMAIN CLEANOUT
 - ⊕ --- EXISTING FLANGE CONNECTION
 - ⊕ CS --- EXISTING CONDENSATE SUMP
 - ⊕ --- EXISTING FINAL COVER AREA
 - 6" --- PROPOSED LANDFILL GAS HEADER/LATERAL
 - FW --- PROPOSED FORCEMAIN LINE
 - A --- PROPOSED AIR LINE
 - ⊕ GW- --- PROPOSED LFG EXTRACTION WELL
 - ⊕ --- PROPOSED REMOTE WELLHEAD
 - ⊕ --- PROPOSED HEADER ISOLATION VALVE
 - ⊕ --- PROPOSED FLANGE CONNECTION
 - ⊕ CS --- PROPOSED CONDENSATE SUMP
 - ⊕ --- PROPOSED HEADER ACCESS RISER
 - ⊕ --- PROPOSED HEADER HIGH POINT

- NOTES:**
1. TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED "WC TOW GRADES".
 2. AS-BUILT GAS COLLECTION AND CONTROL SYSTEM DESIGN TAKEN FROM AXIN LAND SURVEYS, INC. IN DRAWING TITLED "GAS LINES 2016 WEXFORD COUNTY LANDFILL" DATED OCTOBER 24, 2016 AND FROM GOSLING ZUBAK ENGINEERING SCIENCES, INC. IN DRAWING TITLED "LANDFILL GAS COLLECTION SYSTEM WEXFORD COUNTY LANDFILL GFL ENVIRONMENTAL" DATED FEBRUARY 3, 2022.
 3. LOCATION OF PROPOSED LANDFILL GAS COLLECTION AND CONTROL SYSTEM COMPONENTS IS APPROXIMATE AND MAY VARY TO MEET FIELD CONDITIONS AT THE TIME OF CONSTRUCTION. ALL HORIZONTAL AND VERTICAL DATUM TO BE VERIFIED PRIOR TO CONSTRUCTION.



REV	DATE	DESCRIPTION	DESIGNED BY	CHECKED BY	DATE
1	5/28/2024	ISSUED FOR PERMIT	JHD	TAB	
2		REVISED	JHD	TAB	



GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
GCCS DESIGN PLAN - EAST

SHEET NO.
2A

PROJECT NO.
209-0241336



- NOTES:
1. TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED 'WC TOW GRADES'.
 2. AS-BUILT GAS COLLECTION AND CONTROL SYSTEM DESIGN TAKEN FROM AXIN LAND SURVEYS, INC. IN DRAWING TITLED 'GAS LINES 2016 WEXFORD COUNTY LANDFILL' DATED OCTOBER 24, 2016 AND FROM GOSLING COSBURG ENGINEERING SCIENCES, INC. IN DRAWING TITLED 'LANDFILL GAS COLLECTION SYSTEM WEXFORD COUNTY LANDFILL GFL ENVIRONMENTAL' DATED FEBRUARY 3, 2022.
 3. LOCATION OF PROPOSED LANDFILL GAS COLLECTION AND CONTROL SYSTEM COMPONENTS IS APPROXIMATE AND MAY VARY TO MEET FIELD CONDITIONS AT THE TIME OF CONSTRUCTION. ALL HORIZONTAL AND VERTICAL DATUM TO BE VERIFIED PRIOR TO CONSTRUCTION.



REV.	DATE	DESCRIPTION	DESIGNED BY	CHECKED BY	DATE
1	07/28/2024	ISSUED FOR PERMIT	JED	TAB	
2		REVISION	JED	TAB	

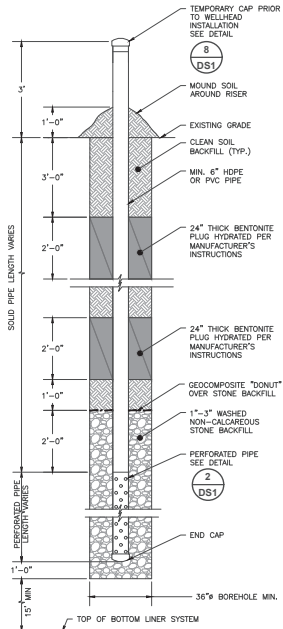


GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

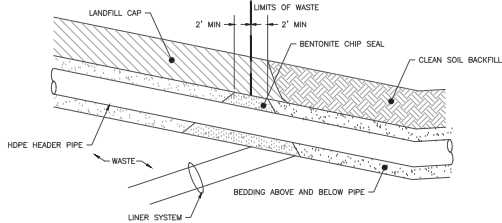
2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
GCCS DESIGN PLAN - WEST

SHEET NO.
2B

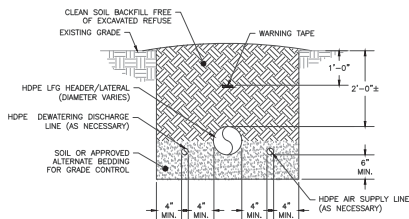
PROJECT NO.
209-4241336



EXTRACTION WELL
DETAIL 1
SCALE: NOT TO SCALE
DS1

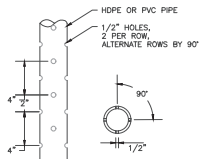


ANTI-SEEP COLLAR
DETAIL 6
SCALE: NOT TO SCALE
DS1



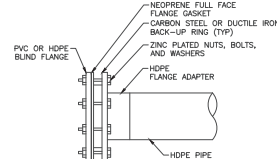
HEADER AND LATERAL PIPELINE TRENCH
DETAIL 9
SCALE: NOT TO SCALE
DS1

- NOTES:**
- WARNING TAPE SHALL BE MIN. 3" WIDE AND IMPRINTED WITH "GAS LINE BURIED BELOW".
 - ALL HEADER AND LATERAL SHALL BE INSTALLED AT MIN. 3% SLOPE UNLESS APPROVED IN ADVANCE BY ENGINEER.
 - THE NUMBER AND TYPES OF PIPES INSTALLED IN THE TRENCH MAY VARY.

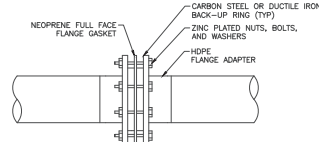


PERFORATED PIPE
DETAIL 2
SCALE: NOT TO SCALE
DS1

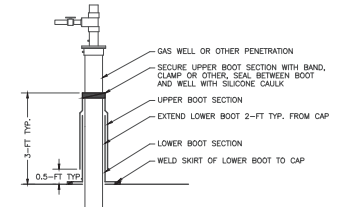
- NOTES:**
- PERFORATIONS SPACED 90° APART HORIZONTALLY.
 - PERFORATIONS SPACED 4" APART VERTICALLY.
 - 90° AND 270° ROWS STAGGERED 2" BELOW 0' AND 180° ROWS.
 - ALTERNATE PERFORATED SIZES AND PATTERNS MAY BE UTILIZED.
 - ALTERNATE MATERIALS MAY BE UTILIZED TO ACCOMMODATE FIELD CONDITIONS.



BLIND FLANGE ASSEMBLY
DETAIL 3
SCALE: NOT TO SCALE
DS1

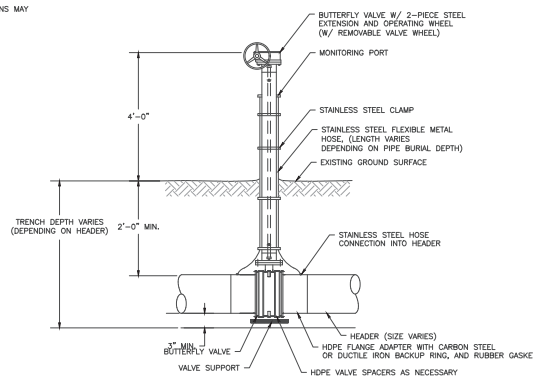


FLANGE CONNECTION
DETAIL 4
SCALE: NOT TO SCALE
DS1

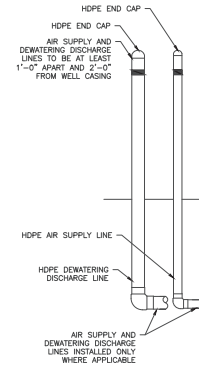


EXTRACTION WELL AND VERTICAL PIPE PENETRATION BOOT
DETAIL 5
SCALE: NOT TO SCALE
DS1

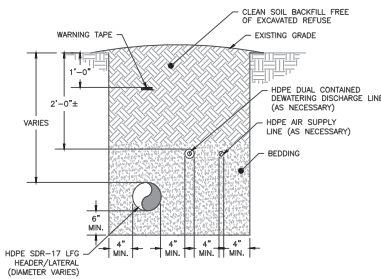
- NOTE:**
- WELL SEAL IS TO BE INSTALLED ON WELLS AND VERTICAL PIPE PENETRATIONS IN FINAL CAPPED AREAS.



ISOLATION VALVE
DETAIL 7
SCALE: NOT TO SCALE
DS1



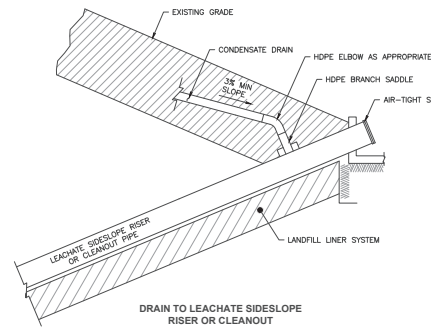
WELLHEAD
DETAIL 8
SCALE: NOT TO SCALE
DS1



HEADER/LATERAL TRENCH
DETAIL 10
SCALE: NOT TO SCALE
DS1

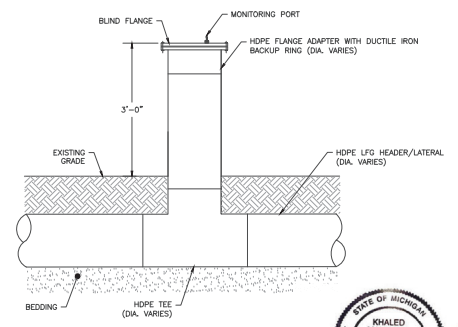
- NOTES:**
- WARNING TAPE SHALL BE MIN. 3" WIDE AND IMPRINTED WITH "GAS LINE BURIED BELOW".
 - ALL HEADER AND LATERAL SHALL BE INSTALLED AT MIN. 0.5% SLOPE UNLESS APPROVED IN ADVANCE BY ENGINEER.
 - THE NUMBER AND TYPES OF PIPES INSTALLED IN THE TRENCH MAY VARY.
 - DEWATERING LINES SHALL BE DUAL CONTAINED WHEN INSTALLED OUTSIDE LIMITS OF WASTE.

NOTE:
LAYOUT OF COMPONENTS AND DETAILS OF CONSTRUCTION ARE TYPICAL AND MAY VARY TO ACCOMMODATE FIELD CONDITIONS AND AVAILABLE TECHNOLOGY AT THE TIME OF CONSTRUCTION.



DRAIN TO LEACHATE SIDESLOPE RISER OR CLEANOUT
DETAIL 11
SCALE: NOT TO SCALE
DS1

- NOTES:**
- FIELD FIT CONDENSATE CONNECTION TO EXISTING LEACHATE RISER. RISER SIZE VARIES (6" TO 18" TYPICAL).
 - MAINTAIN MINIMUM 3% SLOPE INTO CLEANOUT RISER CONNECTION.



HEADER ACCESS RISER
DETAIL 12
SCALE: NOT TO SCALE
DS1

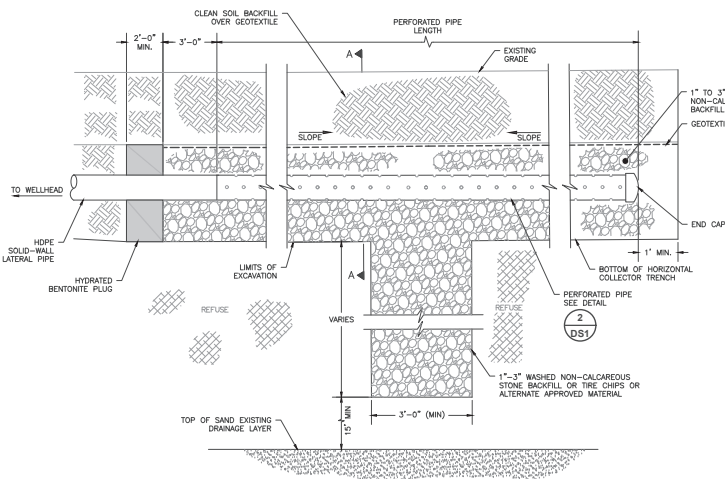
DATE OF MODIFICATION
KHALED MAHMOOD
ENGINEER
48110
PROJECT NO. 209-041136

REV.	DATE	DESCRIPTION	DESIGNED BY	CHECKED BY	DATE
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2			JSD	TAB	



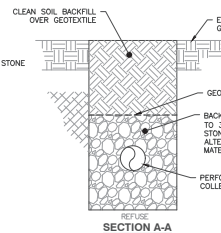
GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN
2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
DETAILS

SHEET NO.
DS1
PROJECT NO.
209-041136

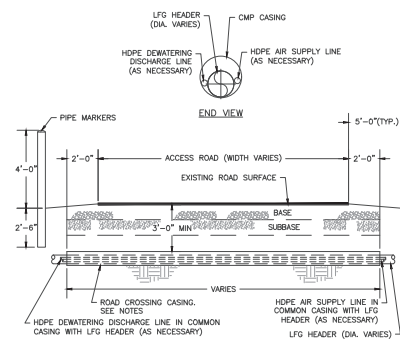


INTERIM HORIZONTAL COLLECTOR

DETAIL 1
SCALE: NOT TO SCALE DS2



SECTION A-A

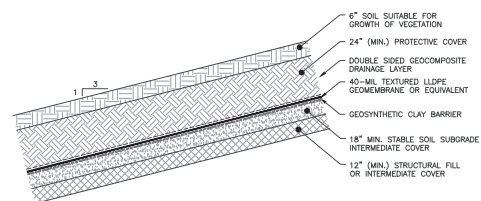


TRENCHED ROAD CROSSING

DETAIL 2
SCALE: NOT TO SCALE DS2

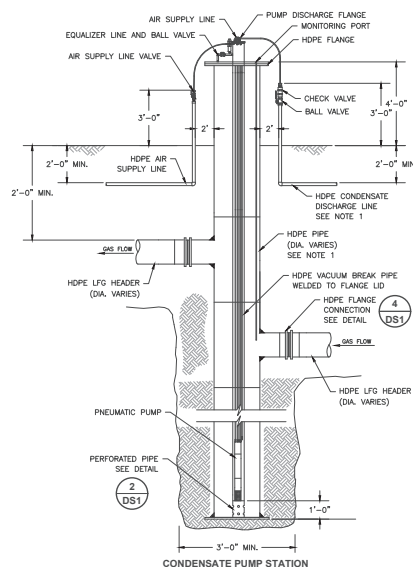
NOTES:

1. ROAD SHALL BE RESTORED TO MATCH ORIGINAL CONDITIONS.
2. CMP CASING A MINIMUM OF 5" GREATER THAN THE CUMULATIVE DIAMETER OF THE PIPES ENCASED.
3. NUMBER AND TYPES OF PIPES ENCASED IN ROAD CROSSING VARIES.



TYPICAL FINAL COVER

DETAIL 4
SCALE: NOT TO SCALE DS2

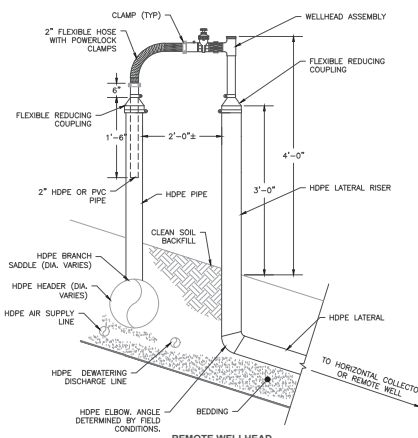


CONDENSATE PUMP STATION

DETAIL 5
SCALE: NOT TO SCALE DS2

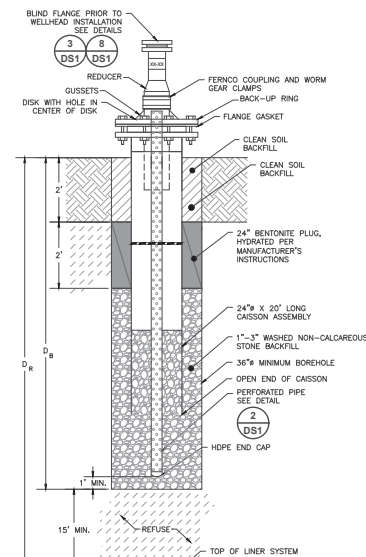
NOTES:

1. CONDENSATE PUMP STATIONS AND CONDENSATE DISCHARGE LINES INSTALLED OUTSIDE OF LIMITS OF WASTE SHALL BE DUAL CONTAINED.



REMOTE WELLHEAD

DETAIL 6
SCALE: NOT TO SCALE DS2

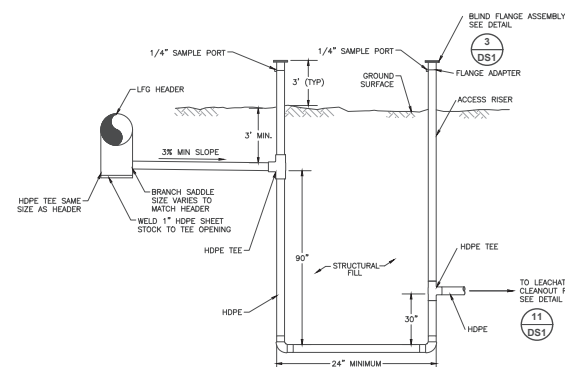


LFG EXTRACTION WELL WITH CAISSON

DETAIL 3
SCALE: NOT TO SCALE DS2

NOTES:

1. D_h = DEPTH OF REFUSE
2. D_b = DEPTH OF BORING
3. MINIMUM 15 FEET SEPARATION BETWEEN BASE OF BORING AND TOP OF LINER SYSTEM.



CONDENSATE DRIPLEG

DETAIL 7
SCALE: NOT TO SCALE DS2

NOTES:

1. DRIPLEG DESIGNED FOR 60" W.G. APPLIED VACUUM.
2. FILL DRIPLEG WITH CLEAN WATER PRIOR TO OPERATION.



NOTE:
LAYOUT OF COMPONENTS AND DETAILS OF CONSTRUCTION ARE TYPICAL AND MAY VARY TO ACCOMMODATE FIELD CONDITIONS AND AVAILABLE TECHNOLOGY AT THE TIME OF CONSTRUCTION.

REV	DATE	DESCRIPTION	DESIGNED BY	CHECKED BY	DATE
1	05/28/2024	ISSUED FOR PERMIT	JSD	TAB	
2			JSD	TAB	



GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
DETAILS

SHEET NO. DS2
PROJECT NO. 209-0241336

Appendix D

Surface Monitoring Design Plan Map



- LEGEND**
- PROPERTY BOUNDARY
 - EXISTING SOLID WASTE BOUNDARY
 - EXISTING PHASE BOUNDARY
 - PROPOSED 10' CONTOUR
 - PROPOSED 2' CONTOUR
 - EXISTING LANDFILL GAS HEADER
 - EXISTING LANDFILL GAS HORIZONTAL COLLECTOR
 - EXISTING FORCEMAIN
 - EXISTING 3" FORCEMAIN
 - EXISTING 37.5" DUAL CONTAINED FORCEMAIN LINE
 - EXISTING 47.5" DUAL CONTAINED FORCEMAIN LINE
 - EXISTING AIRLINE
 - EXISTING 2" AIRLINE
 - EXISTING LFG EXTRACTION WELL
 - EXISTING LFG EXTRACTION WELL WITH CAISSON
 - EXISTING REMOTE WELLHEAD
 - EXISTING ISOLATION VALVE
 - EXISTING BLIND FLANGE
 - EXISTING FLANGE CONNECTION
 - EXISTING REDUCER FITTING
 - EXISTING CONDENSATE DRIPLEG
 - EXISTING MANHOLE
 - EXISTING FORCEMAIN ISOLATION VALVE
 - EXISTING AIRLINE ISOLATION VALVE
 - EXISTING FORCEMAIN CLEANOUT
 - EXISTING FLANGE CONNECTION
 - EXISTING CONDENSATE SUMP
 - EXISTING FINAL COVER AREA
 - PROPOSED SURFACE EMISSIONS MONITORING PATHWAY

- NOTES:**
1. TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED "WC TOW GRADES".
 2. SURFACE EMISSIONS MONITORING FOR METHANE TO BE PERFORMED IN ACCORDANCE WITH PART 115, SECTION 115124, DESCRIBED IN SECTION 6.0 OF THE PART 115 PLAN FOR WEXFORD COUNTY LANDFILL.
 3. SURFACE EMISSIONS MONITORING SERPENTINE PATHWAY REPRESENTS FINAL GRADE CONDITIONS. INTERIM PATHWAYS WILL BE MONITORED BASED UPON THE SEQUENCE AND EXTENT OF WASTE PLACEMENT AND GOC'S DEVELOPMENT.
 4. TYPICAL SPACING FOR SERPENTINE PATHWAY WILL BE NO MORE THAN 30 METERS (100 FT). SOME LOCAL VARIATIONS MAY OCCUR TO ACCOMMODATE LANDFILL PHASING AND FIELD CONDITIONS AT THE TIME OF MONITORING.
 5. IN ADDITION TO SURFACE EMISSIONS MONITORING FOR METHANE ALONG THE SERPENTINE PATHWAY, SURFACE EMISSIONS MONITORING FOR METHANE SHALL ALSO BE CONDUCTED AT EACH COVER PENETRATION AND WHERE VISUAL OBSERVATION INDICATES ELEVATED CONCENTRATION OF LANDFILL GAS (SPECIFIED IN SECTION 6.1 OF THE PART 115 PLAN FOR WEXFORD COUNTY LANDFILL).
 6. THE FACILITY MAY EXCLUDE DANGEROUS AREAS, SUCH AS ROADS, THE ACTIVE FILL AREA, TRUCK TRAFFIC AREAS, AREAS WITH SNOW OR ICE COVER AND SLOPES STEEPER THAN OR EQUAL TO 3:1 FROM SURFACE TESTING.

REV	DATE	DESCRIPTION	DRAWN BY	CHECKED BY	DATE
1	05/28/2024	ISSUED FOR PERMITTING	JRC	TAB	05/28/2024
2		REVISED FOR COMMENTS	JRC	TAB	

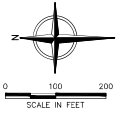


GFL ENVIRONMENTAL
WEXFORD COUNTY LANDFILL
WEXFORD COUNTY, MICHIGAN

2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN
SURFACE EMISSIONS MONITORING PLAN

SHEET NO.
SEM A

PROJECT NO.
209-424134



LEGEND

- PROPERTY BOUNDARY
- EXISTING SOLID WASTE BOUNDARY
- EXISTING PHASE BOUNDARY
- PROPOSED 10' CONTOUR
- PROPOSED 2' CONTOUR
- EXISTING LANDFILL GAS HEADER
- EXISTING LANDFILL GAS HEADER/LATERAL
- EXISTING FORCEMAIN
- EXISTING AIRLINE
- EXISTING LFG EXTRACTION WELL
- EXISTING REMOTE WELLHEAD
- EXISTING HEADER ISOLATION VALVE
- EXISTING FLANGE CONNECTION
- EXISTING CONDENSATE SUMP
- EXISTING HEADER ACCESS RISER
- EXISTING HEADER HIGH POINT
- PROPOSED SURFACE EMISSIONS MONITORING PATHWAY

NOTES:

1. TOP OF WASTE GRADES PROVIDED BY WEAVER CONSULTANTS GROUP, INC. IN FILE TITLED "WC TOW GRADES".
2. SURFACE EMISSIONS MONITORING FOR METHANE TO BE PERFORMED IN ACCORDANCE WITH PART 115 SECTION 115.1.5.4 DESCRIBED IN SECTION 6.0 OF THE PART 115 PLAN FOR WEXFORD COUNTY LANDFILL.
3. SURFACE EMISSIONS MONITORING SERPENTINE PATHWAY REPRESENTS FINAL GRADE CONDITIONS, INTERIM PATHWAYS WILL BE MONITORED BASED UPON THE SEQUENCE AND EXTENT OF WASTE PLACEMENT AND SOCS DEVELOPMENT.
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6. THE FACILITY MAY EXCLUDE DANGEROUS AREAS, SUCH AS ROADS, THE ACTIVE FILL AREA, TRUCK TRAFFIC AREAS, AREAS WITH SNOW OR ICE COVER AND SLOPES STEEPER THAN OR EQUAL TO 3:1 FROM SURFACE TESTING.

START

END

REV	DATE	DESCRIPTION	OWN BY	DES BY	CHECK BY	APP BY
1	07/28/2024	DESIGNED BY JRD				
2		DRAWN BY JRD				
3		CHECKED BY JRD				
4		APPROVED BY JRD				



GFL ENVIRONMENTAL WEXFORD COUNTY LANDFILL WEXFORD COUNTY, MICHIGAN	SHEET NO. SEM B
2024 LANDFILL GAS MONITORING AND MANAGEMENT PLAN SURFACE EMISSIONS MONITORING PLAN	PROJECT NO. 204-KM1336